UPSC CSE 2027 · GS Foundation

GEOGRAPHY

Geomorphology · Climatology · Oceanography · Indian geography · Human geography · Economic geography

Handwritten revision notes

Built from the Sarrthi IAS GS Foundation Geography handouts (pictures and the mind-map pages are in the textbook edition).
Bracket trees · diagrams · mnemonics · PYQ tags · current affairs
Right column and left binding margin are kept blank for your own use.

Personal study edition · for printing

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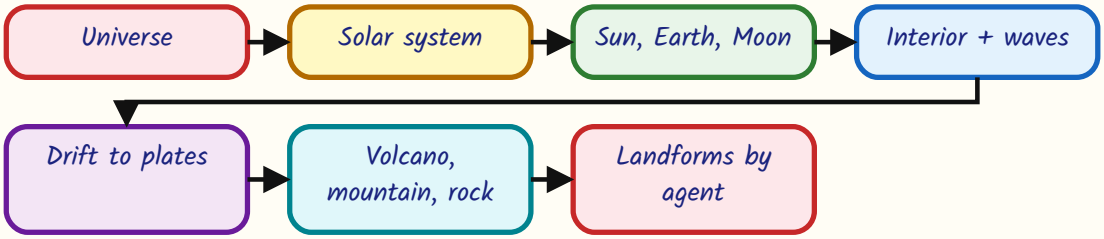
1 Geomorphology (universe to landforms) 2
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GEOMORPHOLOGY

PYQ Prelims 2025, 2024 ×3, 2023, 2022, 2019, 2018, 2015, 2014, 2013 ×2, 2012 ×2, 2011 + 4 with no year · Mains 2022 (primary rocks, 10m), 2017 (Juno mission, 10m)

✎ *From the Big Bang to the Sun, Earth, its interior, plates, rocks and every landform: 19 Prelims PYQs make this a HIGH chapter.*

At a glance: the Geomorphology route



Hot clusters

→ Earth–Sun geometry (pole star, day length, 21 June, time zones)
→ solar storm effects
→ P vs S waves
→ weathering by rain
→ drift evidences
→ volcanic products
→ chalk vs clay. No teacher ink: priority comes from handout PYQ counts only.

1 · UNIVERSE: ORIGIN THEORIES

Theory	By	Core idea	Drawback
Gaseous	Kant	Scattered primordial matter collided → heat → solid→liquid→gas = nebula; rotation → bulge; rings separated; residual centre = Sun, rings = planets; same process → moons	Source of primordial matter unexplained; collisions can't give rotation; how moons formed?
Nebular	Laplace, 1796 (modified Kant)	Faster spin → rings detach, each ring condenses into gas mass, cools → planets; remaining part = Sun	Source of nebula unexplained; Sun should spin fastest but doesn't
Planetesimal	Chamberlin–Moulton	Star passed close to young Sun → tidal bulges, solar prominences → ejected material; spiral arms; cooled → planetesimals + larger protoplanets → planets + moons; comets and asteroids = leftover debris	–
Accretion	Otto Schmidt 1944, Viktor Safronov 1969	Nebula → dust grains → boulders → planetesimals → planets (constant accretion)	–
Big Bang	Georges Lemaitre, 1920s	Universe began from a single primordial atom; violent explosion; dust scattered; denser regions pulled together over millions of years → gas clouds, stars, galaxies	Validated 1972

KLS — Kant = Kollision · Laplace = Loses heat (cools, shrinks, spins faster) · Schmidt–Safronov = Small Steps (dust → planet)

Proofs of Big Bang

→ Hubble's law (Hubble–Lemaitre): farther galaxies recede faster = expanding universe (basis: redshift); Edwin Hubble

→ CMB: 1960s discovery, Penzias & Wilson, 1965; oldest light, ~380,000 yrs after Big Bang; at ~3000 K electrons joined protons (recombination) so photons flew free = "baby photo of the universe"

→ Hotter body = shorter wavelength (more energy → faster motion → higher frequency)

→ Universe age 13.8 bn yrs

2 · DARK ENERGY & DARK MATTER

Together ≈ 95% of universe

→ Dark matter ≈ 27%: no light emitted, absorbed or reflected; inferred from gravity (galaxy rotation, gravitational lensing = light bending near massive object); invisible scaffold holding galaxies

→ Dark energy ≈ 68%: accelerating expansion; repulsive force pushing space apart (not matter or gravity)

3 · GALAXIES

Galaxy types

→ Spiral (Milky Way)

→ Elliptical (old, cool stars, little gas; minimal star formation)

→ Lenticular (bulge + disk, no arms)

→ Irregular (chaotic, gas-rich)

Active galaxies ~10%

→ Centre >100× brighter than all stars combined; supermassive black hole + accretion disk; radiation infrared → X-rays; jets near speed of light (AGN)

→ Seyfert (most common, lowest energy)

→ Quasar (brightest)

→ Blazar (jet points at Earth)

→ Milky Way is NOT active

4 · STARS

Lifecycle of stars

Red giant: H exhausted, expands, cools → *Planetary nebula: outer layers shed* → *White dwarf: core cools (black dwarf unproven)* → *Supernova: explosion after iron buildup* → *Neutron star or black hole by mass*

"Small stars retire quietly, big stars quit with a bang" ⭐

Mass	Path	End
🌐 < 8 × Sun	Main sequence → Red giant → Planetary nebula	White dwarf
🌐 > 8 × Sun	Red supergiant → Supernova (iron core)	Neutron star (8–20/25×) or Black hole (>25×)

Chandrasekhar limit

→ core < 1.44 × Sun = white dwarf

→ 1.44–3 = neutron star

→ >3 = black hole. Source quotes both 8–20 and 8–25× for neutron stars: use 25 as the cut-off to black hole.

Populations (by metallicity)

→ Pop III: hypothetical, very first stars, hundreds of millions of yrs after Big Bang, no heavy metals
→ Pop II: oldest observed, 5–10 bya, halo/globular clusters
→ Pop I: younger, metal-rich, 4–5 bya, disk and arms, our Sun

Neutron star

→ Collapsed core of massive star (8–25× Sun) after supernova; almost all neutrons; strong gravity + magnetic fields
→ Can accrete mass → collapse into black hole
→ Pulsar = rapidly rotating neutron star, beams from magnetic poles, regular pulses ("lighthouse")

🌐 Black hole (>25× Sun)

→ Nothing, not even light, escapes (Earth's escape velocity printed 11.8 m/s: unit looks a typo, don't memorise)
→ Event horizon: boundary of no return
→ Accretion disk: gas/dust spiralling in, hot from friction (disk → horizon → void)
→ Gravitational waves: ripples in spacetime; LIGO (Laser Interferometer Gravitational-Wave Observatory, Nobel 2017) and Virgo
→ Spaghettification: stretched by tidal forces
→ Hawking radiation: theoretical particle emission → slow evaporation

🎯 5 · SOLAR SYSTEM PYQ 2015 PYQ 2011

Components

→ Sun: central star, H + He, nuclear fusion

Planets

→ Terrestrial: Mercury, Venus, Earth, Mars
→ Gas giants: Jupiter, Saturn
→ Ice giants: Uranus, Neptune
→ Planet Nine/Planet X: hypothesised, explains Kuiper Belt orbits; gas or ice giant, 5–7× Earth mass, ~200× farther than Earth

Dwarf planets (5)

→ Pluto (biggest), Ceres (biggest in asteroid belt), Eris, Haumea, Makemake (newest)
→ Criteria: orbits Sun · round (spherical) · not a moon · has not cleared its orbit

Body	Moons
Mercury, Venus	0 (Venus has quasi-satellite Zoozve)
Earth / Mars	1 / 2
Jupiter / Saturn	101 / 274 (Jupiter has biggest moon Ganymede)
Uranus / Neptune	28 / 16
Pluto	5

Handout conflict: Saturn's moons printed 274 in the moons list but 74 in the inner/outer planets table: use 274.

Region/body	Key facts
Quasi-moon	Asteroid orbiting Sun but staying close to a planet; orbit oblong, less stable, may drift away
Kuiper Belt	Beyond Neptune, 30–50 AU (1 AU = 149.6 million km); icy bodies, dwarf planets; rock, water ice, ammonia, methane; source of comets
Oort Cloud	Spherical shell of icy objects; starts 2,000–5,000 AU, outer edge 10,000–100,000 AU; long-period comets

Asteroids	Rocky; belt between Mars and Jupiter; Vesta largest ~530 km (329 miles); smallest under 10 m (33 feet); total mass < Earth's Moon; Trojans at L4/L5 Lagrange points
Comets	Icy; gas + dust → glowing coma and tail near Sun; from Kuiper Belt or Oort Cloud
Meteoroid → meteor → meteorite	Space particle (rocky/metallic) → burns in air ("shooting star") → survives, hits ground. Of 50,000+ meteorites 99.8% from asteroids; 0.2% split between Mars and Moon

🌐 Edge of the Sun's reach

Heliosphere (solar-wind bubble) → *Termination shock: wind supersonic → subsonic* → *Heliosheath: turbulent, field irregular* → *Heliopause: solar wind = interstellar pressure* → *Bow shock: existence contested by Voyager 1 and 2*

Goldilocks zone (2015, ans c): distance from a star where water stays liquid; search for Earth-like planets (Earth; Kepler-186f). Asteroid vs comet (2011, ans B = 1 and 3): asteroids rock/metal, between Mars and Jupiter (Jupiter–Mars), orbit like planets, no tail; comets ice + dust, tail. "Comets between Venus and Mercury" is wrong. Both formed ~4.5 bn yrs ago. ✅

6 · PLANETS

Planet facts

→ Venus and Uranus rotate East → West
→ Uranus axis parallel to its orbit
→ Jupiter: Great Red Spot 16,000 km, Little Red Spot 3,500 km (storms)

- **Inner (rocky)** → dense, small, few/no moons, weak field, short orbits: Mercury 88 d, Earth 365, Mars 687
- **Outer (gaseous)** → big, many moons, rings, strong fields: Jupiter 12 y, Saturn 29, Uranus 84, Neptune 165
- **Atmospheres**
  → Venus CO_2, super-hot (greenhouse)
  → Mars thin CO_2
  → Jupiter/Saturn H + He
  → Uranus/Neptune methane + ammonia

Handout alert: Planet Nine only hypothesised (5–7 × Earth mass).

🎯 7 · THE SUN PYQ 2022

🌐 Sun

→ Composition: 71% H, 27.1% He, 0.97% O, 0.4% C, traces
→ No "surface": gas gets denser toward the core
→

🌐 Sun = a samosa (inside → out)

Core: fusion, >15 million °C → *Radiative zone: energy by radiation* → *Convective zone: convection cells* → *Photosphere: turns opaque (golden crust)* → *Chromosphere: outer ~10,000 °C* → *Corona: hot plasma, millions of km*

Space-weather term	What it is	Remember
Solar wind	Charged particles streaming from corona	plasma too hot for gravity to hold
Solar flare	Explosion from twisted magnetic fields; X-rays + UV	classes A, B, C, M, X (X biggest); may or may not come with CME
CME (coronal mass ejection)	Plasma + magnetic field blobs from corona	15–18 h to reach Earth
Geomagnetic storm	Earth in path of a CME	Starlink: 40 satellites lost (2022) · Quebec blackout 1989 (GICs damage transformers) · GPS errors (aviation, defence) · polar flights rerouted (radiation) · ISS shielding · radar/missile-warning disruption
Aurora	Solar-wind particles excite O and N → photons	Borealis (N), Australis (S); at poles (magnetic poles near)
Van Allen belts	Charged particles trapped by Earth's field, doughnut-shaped; shield Earth, risk to spacecraft	inner: protons, 1,000–6,000 km · outer: electrons, 13,000–60,000 km
Sunspots	Darker, cooler areas	11-year (and 400-year) cycle; now in Maxima (Solar Maxima/Minima)

PYQ 2022 solar storm (ans c): effects = GPS/navigation failure, power-grid damage, intense auroras, satellite orbit disturbance, polar short-wave radio disruption. Not tsunamis (need vertical water shift) and not forest fires (flares don't raise Earth's temperature). May 2024 storm: auroras as far south as Leh. ✅

🎯 8 · EVOLUTION OF EARTH PYQ 2018 PYQ 2012

Order of formation

Crust → Atmosphere → Ocean → Life

📝 Atmosphere: 3 stages

→ (1) H and He, blown away by solar wind
→ (2) Degassing: volcanoes give CO_2, N and S oxides, water vapour → clouds → rain → oceans
→ (3) Life raises oxygen (photosynthesis)

4.5 bya — Earth forms

~4.2 bya — LUCA (as printed)

3.8 bya — simple life near deep-sea vents (blue-green algae)

1953 — Miller–Urey: organic from inorganic

Source oddity: LUCA ~4.2 bya is printed but is earlier than the 3.8 bya first life (verify).

PYQ 2018 (ans C = 1 and 3): magnetic field reversed every few hundred thousand yrs ✔; life modified early atmosphere ✔; "54% oxygen, no CO_2 at birth" ✘ (early air: water vapour, CO_2, N_2, no O_2). PYQ 2012 (ans B): elements behind origin of life = Carbon, Hydrogen, Nitrogen. ✅

🎯 9 · EARTH FACTS, MAGNETISM, DAY-NIGHT PYQ 2012 PYQ 2013 PYQ 2019 PYQ 2024

Earth fact	Value
Axial tilt	23.5° from vertical (towards Pole/North Star)
Rotation (sidereal day)	23 h 56 m 4.1 s, West → East (counter-clockwise from above North Pole); slowing (wind/wave friction, Moon receding, core slowdown)
Revolution (sidereal year)	365 d 5 h 48 m, elliptical orbit; aphelion = farthest from Sun, perihelion = nearest
Pole drift	Groundwater pumping (NW India, W North America) shifted pole ~80 cm east (0.00000719°), 1993–2010; ice melting + water moving to low latitudes also shifts axis
Magnetic field	Flips every 2–3 lakh years (chrons: normal/reversed); 183 reversals in 83 million yrs (random); magnetic north drifting Canada → Arctic → Siberia since 1990s

Why magnetic field matters

- → Blocks solar wind, stops atmospheric loss
- → Creates magnetosphere (holds Van Allen belts)
- → Causes auroras; helps compasses; guides migrating animals
- → Protects satellites and grids

Pole star 2012 (ans C): village 5 km east → keep pole star on left. Day length 2013 (ans B): revolution on a tilted axis + latitude; no tilt = no seasons, equal days. 21 June 2019 (ans A): Sun does not set at Arctic Circle; equator overhead only at equinoxes. 2024 (ans d): >12 h daylight at Tropic of Cancer and Arctic Circle (2 and 4). Northern days longest at June solstice, shortest at December. ✅

No-year PYQ, axis shift (ans b): melting polar ice → water moves toward equator (mass redistribution, "true polar wander") explains shift; solar flares/CMEs are true but don't affect tilt or rotation. ✅

10 · MOON

Moon

- → Earth's only natural satellite; 384,400 km; 27% (a bit over one-fourth) of Earth's size; gravity drives tides
- → Perigee = closest, Apogee = farthest
- → Tidally locked (same face always)

🏛️ Origins

- → Fission (George Darwin): rejected
- → Giant Impact (Theia, Mars-sized, 4.5 bya): accepted
- → Binary-exchange capture: proposed

Pear — Perigee = Pear-close · Apogee = Away

🎯 11 · ECLIPSES, LATITUDE, TIME PYQ 2024

Eclipses

- → Lunar: Earth between Sun and Moon; Blood Moon = whole Moon in umbra; Blue Moon = 2nd full moon in a month (no eclipse link)

🌐 Solar: Moon between Sun and Earth

- → Annular: Moon farther → ring
- → Total: Moon closer, diamond-ring effect
- → Partial: penumbra only

- → Not every full/new moon gives eclipse: Moon's orbit tilted 5°

Latitude and longitude

- → Imaginary lines to locate places
- → Latitudes: parallel, east-west
- → Longitudes: north-south, meet at poles; also climatic zones and time zones

→ Time: 4 min per 1° longitude; Earth rotates W → E so east is ahead; International Date Line keeps calendar day consistent

A-A-U — Most time zones: France 12 · Russia & USA 11 ("two elevens, a cricket team each") · A-A-U = Antarctica, Australia, UK 9 · Canada 6 · Denmark 5

No-year PYQ	Answer
Anadyr (Siberia, UTC+12) vs Nome (Alaska, UTC-9), ~800 km apart across Bering Strait	Only Statement I right (ans a): different days at breakfast; but Monday in Anadyr is Sunday in Nome, not Tuesday
UK, Denmark, NZ, Australia, Brazil: more than 4 zones?	Four (ans b): Denmark 5 (mainland UTC+1 CET + Greenland), NZ 5 (mainland NZST/NZDT UTC+12/+13 + territories), Australia 6+, UK up to 9 with territories; Brazil only 4 (UTC-2 to UTC-5; now 3 official)

🎯 12 · INTERIOR OF THE EARTH PYQ 2017

Why study interior

→ Changes over and below surface; volcanism, earthquakes
→ Internal structure of solar-system objects
→ Evolution of atmosphere; deep-sea mineral exploration

How we know

→ Direct: deepest mines (Mponeng, TauTona 3.6 km), deepest drill (Kola, 12 km), volcanic eruptions
→ Indirect: meteorites, gravity anomalies (heavy material = higher gravity), magnetic field shifts, best of all earthquake waves

Layer	Facts
🧍 Crust 30–50 km; 0.5–1.0% of volume	Oceanic (SiMa): denser, thinner, 5–30 km, basaltic · Continental (SiAl): lighter, thicker, 50–70 km (70–100 km under Himalaya), granitic · separated by Conrad discontinuity · outer = sedimentary cover, then crystalline igneous + metamorphic (acidic); lower = basaltic and ultra-basic
Mantle Moho → 2900 km; 83%	Density upper 2.9–3.3, lower 3.3–5.7, avg 4.5 g/cm³ · Lithosphere = crust + uppermost mantle, 10–200 km, brittle · Asthenosphere to 400 km, semi-solid, source of magma · lower mantle solid · upper/lower mantle by Repetti
Core 2900–6400 km; 16%	Heaviest, densest; nickel + iron [NiFe] · outer liquid, inner solid, separated by Lehmann's discontinuity

CoolMenReallyGetLove — Conrad (continental/oceanic crust) · Moho / Mohorovičić (crust/mantle) · Repetti (upper/lower mantle) · Gutenberg (mantle/core, 2,900 km) · Lehmann (outer/inner core)

Discontinuity: place where seismic waves change velocity and/or direction abruptly.
Earth = boiled egg: shell = crust, white = mantle, yolk = core.

Mains 2017 (10m): "How does NASA's Juno Mission help understand the origin and evolution of the Earth?" Handout prints the question only; only Jupiter notes (gas giant, storms) given.

🎯 13 · EARTHQUAKES AND SEISMIC WAVES PYQ 2023

Earthquake

→ Abrupt energy release along a fault (sharp break in crustal rock); friction holds, pressure builds, rock slips → shockwaves
→ Focus (hypocentre) = where energy released
→ Epicentre = surface point nearest focus

Distribution

→ Mostly plate boundaries; two belts: Circum-Pacific, Mid-continental
→ Deep-focus at convergent, shallow-focus at divergent boundaries

Scale	Measures
Richter	Magnitude from surface-wave amplitude, log scale 0–10 (5 → 6 = ×10 amplitude)
Mercalli	Intensity (damage) 1–12
Moment magnitude	Energy (fault area, slip, rock rigidity), log, most widely used now

Wave	Type	Motion	Medium	Order
P (primary)	Body, longitudinal	to-and-fro along travel	solid, liquid, gas (faster in solids)	1st, fastest
S (secondary)	Body, transverse	up-down, right angles	solids only	2nd
L (Love)	Surface, shear	horizontal shaking; fastest surface wave	–	3rd; collapses buildings
R (Rayleigh)	Surface, rolling ellipse	vertical + horizontal	–	slowest; tsunamis

Push-pull – P = Push-pull, Prompt. S = Sideways, Solids only. Love shakes Left-right; Rayleigh Rolls like a surf.

Wave behaviour

→ Recorded by seismographs; body waves (P, S) from focus; they generate surface waves at surface rocks

→ Velocity rises with elasticity; direction changes (reflect/refract) at different densities; cooler rigid areas transmit faster than hotter

Shadow zones

→ P: 103°–143° band round Earth

→ S: everything beyond 103°, much larger, a little over 40% of surface

→ S vanish → liquid outer core; P-shadow clue → solid inner core

PYQ 2023 (ans c): P waves recorded earlier than S; P particles vibrate along direction of travel, S at right angles. Both correct. ✅

🎯 14 · EARTH MOVEMENTS AND WEATHERING PYQ 2024

	Endogenic (radioactivity, tidal/rotational friction, primordial heat)	Exogenic (Sun-driven: temperature + rain)
Slow	Diastrophism (bending, folding, warping, fracturing): Orogenic = mountain-building, horizontal force → vertical effect, long narrow belts; Epeirogenic = continent-building, vertical force → horizontal effect, radial up/down	Weathering · Erosion · Deposition · Mass movement = denudation
Sudden	Earthquakes, volcanism	–

Mass movement: only under gravity. Denudation gives: soil formation, ore concentration (iron, manganese, aluminium, copper), landforms. Orogeny, epeirogeny, earthquakes, plate tectonics → faulting and fracturing. 💡

🎯 Epeirogenic examples

→ Uplift: Colorado Plateau, Brazilian Highlands, Australian Shield, Siberian Platform

→ Subsidence: Siberian Basin, Konkan

→ Sudden: 2004 quake submerged Indira Point; NZ 1885 uplift up to 3 m; Japan 1891 subsidence 6 m

Weathering

👥 Chemical (composition changes)

→ Solution/hydration

→ Carbonation (CO_2 + water → carbonic acid)

→ Oxidation (iron, manganese, sulphur rocks)

→ Reduction (red iron oxide → green/blue-grey)

Physical

→ unloading

→ granular disintegration

→ exfoliation (diurnal temperature, "onion peeling", smooth rounded, dry/high places)
→ block separation (jointed rocks)
→ shattering (angular pieces)
→ frost wedging (water freezes, expands)
→ salt weathering (crystallisation, deserts)

Biological

→ burrowing, roots
→ decaying matter gives humic, carbonic and other acids
→ minerals removed by algae and plants

S-C-O-R — "Some Chefs Often Roast" = Solution/hydration · Carbonation · Oxidation · Reduction

PYQ 2024 (ans a): rain weathers rock because rainwater carries dissolved CO_2 and O_2; both statements explain it. ✅

🎯 15 · DRIFT AND SEA-FLOOR SPREADING PYQ 2014

1858	1912	1930s	1960	1967	1968
Pellegrini	Wegener drift	Holmes convection	Hess spreading	McKenzie & Parker	Morgan

💡 Earlier ideas

→ Antonio Pellegrini 1858: identical fossil plants in N American and European coal → once joined
→ Edward Suess (late 19th c.): big ancient continents split into present ones
→ F. B. Taylor: arcuate (bow-shaped) mountain belts of Asia, Europe from creep of continents toward Equator

Wegener 1912

→ One landmass Pangaea + one ocean Panthalassa
→ Tethys Sea split it: Laurentia (Laurasia) north, Gondwanaland south
→ Forces: tidal (westward) + gravity and polar-fleeing (equator-ward)
→ Break-up in Triassic (251–199.6 mya); ~200 mya all continents one supercontinent

Evidence	Detail
Jigsaw fit	Brazil bulge ↔ Gulf of Guinea; Greenland ↔ Ellesmere/Baffin; India–Madagascar–Africa
Rocks/ fossils	Ghana gold placers from Brazil plateau; Mesosaurus fossils only in South Africa and Iraver formations of Brazil (4,800 km apart); 2-bn-yr-old rocks on both coasts
Climate	Tillite; India's Gondwana sediments have counterparts in six southern landmasses
Mountains	Caledonian/Hercynian (Europe) continue as Appalachians

⚠️ Drawbacks

→ No free float of SiAl on SiMa: lithosphere moves over asthenosphere
→ Ocean floor isn't passive; tides/gravity too weak (convection currents the engine)
→ Only equator/west movement (but Antarctica moved away, Australia east, India NE)
→ No reason for Triassic break-up

Hess 1960: sea-floor spreading

→ War-time research: ocean floor full of relief (ranges, trenches)
→ Mid-oceanic ridges most volcanically active; ocean floors not static
→ Rocks equidistant either side of ridge crest similar in constituents and age
→ Shallow-focus quakes at ridges (deep-focus at Himalayan belt)

→ Oceanic rocks much younger than continental (subduct)

Holmes 1930s: convection

→ Radioactive heat in mantle (100–2900 km) seeks escape → convection currents

→ Rising limbs → divergence; falling limbs → convergence

Paleomagnetism

→ Rock minerals lock in direction + intensity of magnetic field (age and location of formation); basalt on sea floor

→ Ocean floor each side of ridges has magnetic stripes: normal, then reversed

→ New rock at ridge pushes older (reversed-era) rock away; polarity reversals recorded in magma

PYQ 2014 (ans c): continental drift and glacial cycles both influenced evolution (unique primates of Australia, S America; genus Homo in last 3 glacial cycles). No-year (ans c, all three): Brazil–W Africa rock belts, Ghana gold from Brazil plateau, Gondwana sediments in six southern landmasses (Africa, S America, Australia, Antarctica, Madagascar, NZ). ✅

16 · PLATE TECTONICS

Theory

→ Suggested McKenzie and Parker 1967; outlined Morgan 1968

→ Lithosphere broken into plates (7 major, many minor) floating on asthenosphere under convection currents (Holmes 1930s)

→ Oceanic: Simatic, thinner (Pacific); continental: Sialic, thicker (Arabian); both (Indo-Australian)

→ Minor plates from stress where majors converge (Mediterranean: Eurasian + African)

→ Explains earthquakes, volcanoes, fold and block mountains

7 major plates

→ Antarctica + ocean

→ North American

→ South American

→ Pacific

→ India-Australia-NZ

→ Africa + E. Atlantic floor

→ Eurasia + ocean

🗺️ Ring of Fire (Pacific boundary, NZ → Chile)

→ 70% earthquakes, 90% volcanoes (75% of active)

→ 20% earthquakes in Mid-continental belt

→ Slowest ridge Arctic (<2.5 cm/yr); fastest East Pacific Rise (>15 cm/yr, ~3,400 km west of Chile)

DCS – Divorce · Collide · Slide (divergent · convergent · transform)

Boundary	Plates...	Creates	Example
Divergent (constructive)	move apart	new crust, mid-ocean ridges, rift valleys, volcanic islands, shallow quakes	Mid-oceanic ridges, Great African Rift (Lebanon → Mozambique), Chagos
Convergent C–C	continent meets continent	fold mountains, no volcanoes (no subduction)	India–Eurasia → Himalayas
Convergent O–C	dense ocean plate subducts	trench, volcanoes, fold mountains (Cordilleran)	Pacific–S American → Andes
Transform (conservative)	slide past	only deformation; usually ⊥ ridges	San Andreas Fault

Himalayas have no volcanoes; Andes and Alps do.

~225 mya	~200 mya	~140 mya	~65 mya	40–50 mya
Tethys separates India from Asia	moves north (Pangaea breaks)	at 50°S	Deccan Traps	collision (some say 60)

Indian plate

→ island off Australia first; Himalayas still rising. Margins: E Arakan Yoma → Java Trench

→ W Kirthar

→ N Himalaya. **Limits of theory:** doesn't explain hotspots; can't place ancient plates or predict future boundaries.

PYQ 2013 (ans d): electromagnetic radiation, geothermal energy, gravity, plate movement, rotation, revolution: all cause dynamic surface change (radiation → temperature change, gravity → downslope movement). ✅

🎯 17 · VOLCANISM PYQ 2024

Volcanism: molten rock from asthenosphere enters lithosphere, may or may not reach surface. **Volcano:** vent in crust erupting magma, gases, ash. Under Earth magma; after eruption lava. 💡

Acidic lava

→ Light-coloured, low density, high silica

→ Highly viscous, flows slowly, little spread

→ Solidifies in vent → explosive, pyroclastic material

Basic lava

→ Dark, high density, low silica

→ Highly fluid, spreads widely

→ Lava plateaus, gentle volcanic mountains

Vents and gases

→ Fumarole: steam + gas (SO_2, CO_2); sulphur-rich = solfatara, CO_2-rich = mofette

→ Gases: carbon, sulphur, nitrogen oxides, water vapour, H_2S, trace chlorine

→ Geysers only in Iceland, New Zealand, Yellowstone (USA)

Distribution

→ Only 10–20% of volcanic activity is above sea

→ Ring of Fire 75% active

→ Mid-Atlantic ridge: Iceland

→ Great African Rift: Kilimanjaro, Kenya

→ Cameroon (only active in W Africa); Pelée (West Indies); Mediterranean: Vesuvius, Stromboli, Ararat, Elburz

→ None in Australia; none in Himalaya

South Asia / India

→ Barren Island: only active lava volcano (last erupted 2022)

→ Narcondam extinct

→ Baratang: India's only mud volcano

→ Chandragup (Pakistan): mud, no lava/pyroclast, "not a true volcano"

Extrusive (surface)	Intrusive (inside; seen after erosion)
Conical and fissure vents, basalt plateaus · mid-ocean ridges · pyroclastic debris · caldera and crater lakes (Crater Lake, Oregon)	Batholith/pluton (huge; Mt Wicklow) · Sill (flat; Great Whin Sill) · Laccolith (dome; Henry Mts) · Lopolith (saucer; Transvaal) · Phacolith (wavy; Cordon Hill) · Dyke (vertical; Alaska)

Sill = sill — Sill = window-sill, flat · Dyke = wall, upright · Laccolith = Loaf (dome) · Lopolith = Lota (bowl/saucer) · Phacolith = wavy

I HaveSomeVada-Pav,Please — Eruption styles calm → crazy: Icelandic (fissure, basalt) → Hawaiian (central vent, most lava) → Strombolian (short bursts, pasty lava) → Vulcanian (dark cauliflower clouds) → Pelean (most dangerous, glowing clouds) → Plinian (ash up to 50,000 ft, deadly pyroclastic flows)

Types of volcano

→ By frequency: Active (St Helens, Merapi) · Dormant (Vesuvius before AD 79) · Extinct (Narcondam)

→ By shape: cinder cone · composite (strato) · shield

→ Vulcanian/Pelean/Plinian ash = hygroscopic nuclei → rain + ash → lahars (mudflows)

Mantle plume and hotspot

→ Rising heat arm in mantle; intra-plate (not at boundaries)

→ Plate moves over it → island chains (Hawaii); cut off from heat → extinct

→ Azores: hotspot + Mid-Atlantic ridge

→ Réunion hotspot: active >65 million yrs; Indian plate passing over it → Deccan Traps

PYQ 2024 (ans d): pyroclastic debris, ash and dust, nitrogen compounds, sulphur compounds: all four are volcanic products. ✅

18 · OROGENY: MOUNTAINS, ISOSTASY

Orogeny = mountain building. By location: continental, coastal, oceanic. By formation: fold, fault (block), volcanic, residual. 💡

Mountain	Formed by	Notes / examples
Fold	Compression of sedimentary strata; loftiest; young near converging boundaries	Himalaya (India–Eurasia), Alps and Atlas (Eurasia–Africa), Andes (S America–Pacific): quakes + volcanoes · old: Appalachians (USA), Aravallis
Block (fault)	Dip-slip faults (vertical displacement), tension/ compression; horst = raised, graben = lowered	Great African Rift (floor graben; Mitumba and Muchinga horsts), Rhine valley–Vosges
Volcanic	Lava, bombs, cinders, ash, dust, mud in layers around vent	"Mountains of accumulation"; Circum-Pacific: Fuji, Mayon, Merapi (Sumatra), Etna
Residual	Left after denudation of fold/fault/ volcanic mountains	Monadnocks (USA)

Isostasy

→ Equilibrium between lithosphere and asthenosphere (buoyancy vs gravity)

→ Crust subsides/uplifts when load added or removed (erosion, mountain building, snow/ice melt)

→ Causes: tectonic (subsidence, uplift) and climatic

→ Can cause earthquakes; ice melt → rebound → cracks → may reactivate Antarctic volcanoes

16-7-48 — Longest ranges (km): Mid-Atlantic Ridge 16,000 · Andes 7,000 · W. Cordillera 4,830 · Great Dividing Range & Trans-Antarctic 3,500 · Urals & Atlas 2,500 · Appalachians 2,414 · Himalayas 2,400 · Altai 2,000. Say "16, 7, 48, 35-35, 25-25, 24-24, 20".

🎯 19 · ROCKS AND ROCK CYCLE PYQ 2022

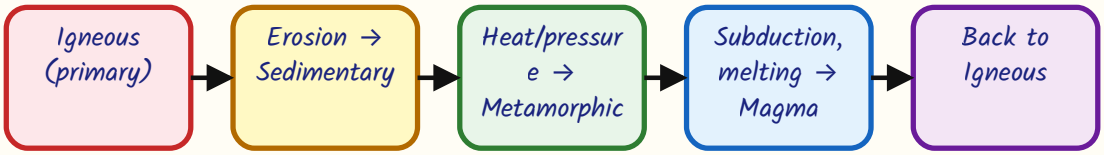
	Igneous (primary)	Sedimentary	Metamorphic
🏛️ Origin	Magma/lava cooling	Denudation → deposits → lithification	PVT (pressure, volume, temperature) change of existing rock; recrystallise
💡 Features	No fossils; plutonic/intrusive (slow, deep, big grains) vs extrusive (fast, surface, small smooth grains); acid (granite) light/less dense vs basic	Layered/strata, fossils, porous; 75% of crust area, 5% of volume; tillite (ice), loess (wind)	Foliation, lineation, banding; may/may not have strata and fossils; heat = thermal, pressure = dynamic types
🎯 Examples	Granite, gabbro, pegmatite, basalt	Sandstone, shale, limestone, coal, chalk, chert, halite	Gneiss, slate, schist, marble, quartzite
Value	Metal ores (iron, nickel, copper, lead, zinc, chromite, manganese, gold, diamond, platinum); building stone; black soil	Petroleum, gas, coal, phosphates, hematite iron ore, limestone; placers (bauxite, manganese, tin); rich soils	–

Sedimentary by mode

→ Mechanical: sandstone, conglomerate, shale, loess
→ Organic: chalk, limestone, coal, geyserite
→ Chemical: chert, limestone, halite, potash

Amygdales: almond bubbles in basalt filled with minerals (copper). Metamorphism: rocks forced down by tectonics or contacted by rising magma. 💡

Rock cycle (continuous; any rock can become any other)



Mineral	Mohs hardness (1 soft – 10 hard) and use
Feldspar	6–7 (glass, ceramics)
Quartz	7 (jewellery, electronics)
Bauxite	1–3 (aluminium)
Cinnabar	2–3 (mercury)
Dolomite	3.5–4
Hematite and Magnetite	5.5–6 (iron)

Mains 2022 (10m) "primary rocks": Intro: igneous from magma/lava, hence primary. Body: non-fossiliferous; plutonic vs volcanic; acid vs basic; examples; economic value (ores, building stone, black soil). Close: all other rocks derive via rock cycle. ✅

20 · EROSION, FLUVIAL CYCLE

Erosion

→ = removal + transportation + deposition (transport separates it from weathering). Agents: flowing water, underground water, wind, glaciers, coastal waves. Five cycles: fluvial
→ glacial
→ marine
→ arid
→ karst.

YMO – River life: Young – V, Mature – meander, Old – delta. Youth: fast, V-valley, waterfalls, rapids, headward erosion · Mature: lateral erosion, meanders, U-profile · Old: deposition, braiding, delta 📌

Fluvial erosion

→ Types: hydraulic action, corrosion (chemical), attrition (load vs load), corrasion/abrasion (load vs rock)

→ Downcutting deepens, lateral widens, headward lengthens → river piracy (Saraswati; Shyok captured Nubra)

→ Rejuvenation: stage change from tectonic (submergence/emergence) or climatic (sea level) activity → terraces

→ Valley: young = deep, narrow, V, vertical downcutting; mature = U, broad base, concave slope; tributary valley hangs above main valley (falls/rapids)

Erosional forms

→ Gorge (narrow, steep; Indus at Nanga Parbat, Brahmaputra at Namche Barwa: antecedent rivers) vs canyon (broader; Grand Canyon, Colorado, Arizona: 483 km long, 2.88 km deep)

→ Waterfall (youth; Jog/Gersoppa on Sharavati 260 m), rapids/cataracts

→ Potholes (swirling boulders drill)

→ Meander: outer = cliff-slope, erosion, concave; inner = slip-off, deposition, point bars

→ Ox-bow lake (cut-off meander; Ganga shifted south leaving ox-bows north)

→ Peneplain (end product) with monadnocks

Depositional forms

→ Alluvial fans/cones

→ Natural levees (breach = flood)

→ Delta: arcuate fan; bird's-foot weak tides; cuspate strong waves

→ Braided channel

→ Estuary: drowned valley; ports, fisheries

Drainage pattern	Description	Example
Dendritic/Pinnate	irregular tree	Indus, Godavari, Mahanadi, Cauvery, Krishna
Trellis	short tributaries at right angles; soft-rock erosion	Seine, Paris basin
Rectangular	main stream and tributaries bend at right angles	Colorado (USA); Mahanadi
Angular	tributaries at acute angles	Himalayan foothills
Parallel	parallel streams, uniform slope	rivers of lesser Himalayas
Radial	from a summit in all directions	Central French Plateau, Kilimanjaro, Amarkantak, Saurashtra
Annular	ring around summit (soft outer stratum)	Black Hills, S. Dakota
Centripetal	all converge into low basin	Ladakh, Tibet, Bagmati (Nepal)

Handout conflict: Mahanadi listed under both dendritic and rectangular. Take dendritic unless the question says rectangular.

🎯 21 · KARST CYCLE PYQ 2025

Groundwater words

→ Porous (pore spaces, sandstone) · Permeable (water passes; clay porous but not permeable)

→ Aquifer stores and transmits water; water table = top of saturation zone

→ Confined (impermeable cover, pressure) vs unconfined

→ Artesian well (rises unpumped); spring; perched water table

👫 Karst

→ Named after a province of Yugoslavia (now Slovenia) on Adriatic coast

→ Thick limestone, dolomite or gypsum; rock dense, highly jointed, thickly bedded

Erosional
→ Sinkholes/swallow holes (3–9 m; dolines, uvalas, polje)
→ Lapies (grooves and ridges)
→ karst window, arch/bridge, caves, blind valley

Depositional (speleothems, in caves)
→ Stalactite hangs from roof
→ Stalagmite rises from floor
→ Column when joined
→ flowstone, cave pearls, draperies

Ceiling–Ground — Stalactite = Ceiling (holds tight to the roof). Stalagmite = Ground (might reach the roof).

PYQ 2025 (ans c): chalk = porous and permeable; clay = porous but not permeable ✅ (USGS: very high porosity, low permeability). Statement III ("clay not porous at all") wrong; II right and explains I.

22 · MARINE CYCLE AND COASTS

Marine erosion
→ Waves aided by winds, currents, tides, storms; depends on wave size, slope, shore height, coast shape, rock, depth
→ Wave pressure compresses air in fissures → rupture; waves use debris as tools (corrasion); solvent action in limestone and chalk

Cats Always Sit Still: C-A-S-S



💡 Erosional features
→ Chasm: narrow, deep, headward erosion along vertical weakness; mouth widens → bay
→ Wave-cut platform: sea cliffs retreat; notch at base (maximum erosion)
→ Sea cave (differential erosion) → sea arch
→ Stack: pillars offshore; further eroded (below low tide) → stumps
→ Blow/spouting hole: air compressed in cave; noise; widen, walls collapse → geo/gloup

💡 Depositional features
→ Beach: temporary debris cover on wave-cut platform
→ Bar: submerged, offshore; encloses lagoon; barrier = overwater counterpart
→ Spit (one end tied to headland) → hook (curved end)
→ Tombolo: bar joining island
→ Marine dunes: by wind, not waves

Coast type	Cause	Examples
Emergence	land up / sea down	bars, spits, lagoons, salt marshes, beaches, cliffs, arches
Submergence	land sinks / sea up	Ria (river valleys drowned, irregular; SW Ireland) · Fjord (glacial U-trough drowned, long steep inlets; Norway) · Dalmatian (ridges parallel to coast; Yugoslavia) · Drowned lowland (low, no indentations; Baltic coast of E. Germany)
Neutral	new material built out; no net relative change	alluvial fan, delta, volcano, coral-reef coasts

Compound	submergence + emergence (either order)	Norway and Sweden
Fault	down-thrown block submerged, steep fault face to sea	Konkan coast (India)

23 · GLACIAL CYCLE

Glacier

→ Moving ice mass, few metres/day; loaded with rock debris
→ Types: continental, ice cap, piedmont, valley

Erosional

→ Cirque/Corrie/Cwm: hollow basin in ridge, steep on 3 sides, open on 1; tarn lake
→ Horn/Pyramidal peak: cut from >2 sides
→ Arête: knife-edge between two cirques
→ Bergschrund: crevasse between moving glacier and stationary ice/rock at upper end
→ Glacial trough: U-shaped valley (uniform horizontal + vertical erosion); flooded = fjord
→ Hanging valley: tributary valley left high, discordant
→ Roche moutonnée ("sheep rock"): smooth side faces ice advance, rough steep side plucked
→ Craig and tail: resistant mass + softer ridge downstream

Depositional

→ Outwash/till plain: stratified debris, clay, sand, gravel
→ Esker: winding ridge of unassorted deposits, like a levee, road-making
→ Kame terraces: broken ridges, humps
→ Drumlin: inverted boat; steep glacier side faces glacier, lee side tapers (shows direction)
→ Kettle holes: depressions in till plain
→ Moraine: ground, lateral, end

24 · ARID CYCLE, LAKES, PLATEAUS

H-R-E — Hamada = rocky (large smoothened rocks) · Reg/Serir = stony gravel and pebbles, easier to cross · Erg/Koum = sandy · Cold deserts (high altitude/latitude). H-R-E: Hard → Rubble → Ergs of sand

Water forms (flash floods + wind)

→ Bolson: intermontane basin
→ Gully (ditch-like, metres–tens of metres); Ravine (narrower than canyon, larger than gully, smaller than valley)
→ Badland: rills erode weak sediments, no vegetation, no agriculture (Chambal ravines)
→ Playa: temporary lake; salt-covered = Salina
→ Pediment: rock-cut surface at mountain foot (erosional)
→ Bajada: moderately sloping depositional plain

💰 Wind processes

→ Deflation: lifting and carrying away dry dust
→ Abrasion: sand-loaded wind grinds rock
→ Attrition: wear of sand particles in transport

Wind erosional

→ Blowouts/deflation hollows (to several km)
→ Mushroom rock (cap on narrow stem)
→ Zeugen = Zzz, lies flat (table-shaped, horizontal layers) · Yardang = Y stands up (vertical layers)
→ Wind window (holes widen) → wind bridge (arch)

→ Inselberg (= monadnock of deserts)
→ Pediplain (= peneplain of deserts)

Wind depositional

→ Ripple marks: saltation
→ Dunes: heaps, few m to 20 m (some hundreds of m high, 5–6 km long); deserts and coasts
→ Barchan: crescent, windward convex, leeward concave and steep
→ Longitudinal: parallel to wind (Sahara, Australian, Libyan, S African, Thar)
→ Transverse: ⊥ wind
→ Parabolic: U-shaped, longer and narrower than barchan, windward concave
→ Star: central peak, 3+ arms, multi-direction wind
→ Loess (ex-situ): fine desert dust carried far (Loess Plateau near Gobi, China)

Lakes

→ Larger, deeper than ponds; most have natural outflow; endorheic = no outflow (lose water by evaporation/seepage)
→ Tectonic: Titicaca, Caspian
→ Rift valley: Tanganyika, Malawi, Dead Sea
→ Glacial: tarns; Scandinavia, Canada
→ Volcanic crater/caldera: Lonar, Krakatoa
→ Erosional: karst, playa, wind-deflated (Great Basin, Utah)
→ Depositional: ox-bow, lagoon (Chilika), moraine-dammed
→ Man-made

Plateaus

→ Flat tablelands, a third of land; mineral-rich (mining); eroded → outliers (old dense rock; iron ore, coal)

- **Tectonic**
  - → Volcanism: Deccan
  - → Crustal shortening: Tibet (intermontane)
  - → Thermal expansion: Ethiopian Highlands
- → Erosional: Allegheny (Appalachians)

TRAPS

"Himalayas have volcanoes" ✘ · "P waves can't pass liquids" ✘ (that's S) · "orogeny = continent-building" ✘ (that's epeirogeny) · "solar flares cause tsunamis/forest fires" ✘ · "Blue Moon = eclipse" ✘ · "Sun overhead on equator on 21 June" ✘ (equinoxes).

Connects to: Himalaya and Deccan story → Indian Geography; tilt and 21 June → Climatology; ridges, trenches, coral reefs → Oceanography; Indus/Brahmaputra gorges → Indian drainage. ✅

Recall drill: cover and answer

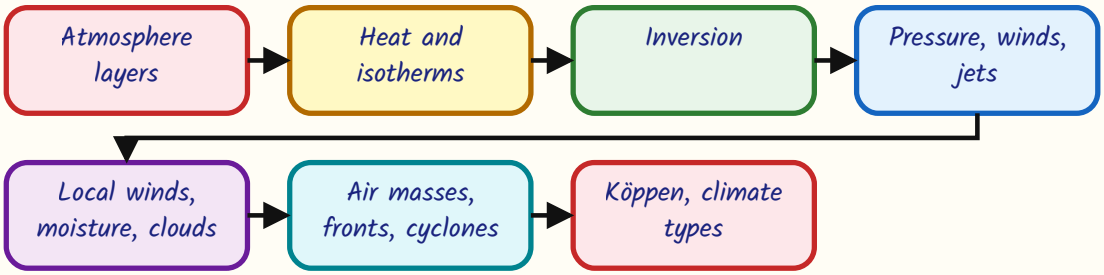
CMB found by whom, when?	Penzias & Wilson, 1965; ~380,000 yrs after Big Bang	White dwarf / neutron star / black hole mass?	<1.44 · 1.44–3 · >3 (Chandrasekhar)
Moons: Jupiter, Saturn, Uranus, Neptune?	101, 274, 28, 16	Earth tilt; sidereal day?	23.5°; 23 h 56 m 4.1 s
Earth volume: crust / mantle / core?	0.5–1% / 83% / 16%	Boundaries top to bottom?	Conrad, Moho, Repetti, Gutenberg, Lehmann
P shadow / S shadow?	103–143° / beyond 103°	Drift / spreading / plates?	Wegener 1912 / Hess 1960 / McKenzie-Parker 1967
Only active lava volcano in South Asia?	Barren Island (India)	Fastest / slowest ridge?	East Pacific Rise / Arctic

CLIMATOLOGY

PYQ Prelims 2011 ×5, 2012 ×3, 2013, 2014, 2015 ×3, 2018, 2019 ×2, 2020, 2021 ×2, 2022, 2023 ×3, 2024 ×6, 2025 + 2 with no year · Mains 2022 (troposphere, 15m)

✎ *Atmosphere, heat, pressure, winds, jet streams, clouds, air masses, cyclones and climate types: 32 Prelims PYQs make this the most-asked chapter.*

At a glance: the Climatology route



Hot clusters

→ heat budget and lapse rate (2012, 2023, 2024)
→ isotherm bending (2024, 2025)
→ westerlies, Coriolis, jet streams (2011, 2015, 2020, 2024)
→ clouds and dew (2019, 2022)
→ cyclone genesis (2015)
→ climate-type identification (2011–2024). **Prepare:** Statement I/II cause–effect pairs (2023, 2024, 2025); Mains: troposphere, jet streams and climate change, tropical vs temperate cyclones. No teacher ink: priority = handout PYQ count.

1 · ATMOSPHERE: LAYERS AND GASES

PYQ 2011 PYQ 2012 PYQ 2024 PYQ 2022

Layer (bottom to top)	Height	Key facts
1 Troposphere (kitchen)	8–18 km (avg 13)	All weather, clouds, jet streams; 18 km at equator (strong convection), 8 km at poles; cools ~1 °C per 165 m
2 Stratosphere (AC floor)	to 50 km	Ozone (lower part) so temp rises with height; almost cloud-free = jets fly here
3 Mesosphere (freezer)	50–80 km	Coldest; meteors burn; noctilucent clouds
4 Thermosphere (antenna)	80–400 km	Temp rises again (UV, X-ray); ISS and satellites; Kármán line 100 km; ionosphere bounces radio
5 Exosphere (terrace)	> 400 km	Particles up to 1,000 °C yet feels cold (too thin to pass heat)

Gases

→ N_2 78% · O_2 21% (≈99%) · Argon 0.93% · CO_2 ~0.04%
→ Water vapour 0–5%: >90% below 5 km; falls equator to poles
→ CO_2 and water vapour only up to 90 km; ozone layer 20–25 km
→ By composition: homosphere (<80 km, uniform) · heterosphere (>80 km, layered by molecular mass)
→ Aerosols/salt = hygroscopic nuclei; microplastics seed ice 5–10 °C warmer than usual
→ Origin of gases: volcanic eruptions, hot springs, chemical breakdown of solids, biosphere

Source conflicts: lapse rate quoted as 1 °C/165 m and 6.5 °C/1,000 m (about equal). Ionosphere printed 75–1,000 km (PYQ note) vs thermosphere 80–400 km. Water vapour 0–5% (layers) vs "around 1% of air" (PYQ note).

PYQs: 2012 (C) cooler aloft because air is heated from the ground up + thinner air; NOT more moisture. 2012 (D) CO_2 absorbs infrared: transparent to incoming, opaque to outgoing radiation. 2023 (C) water vapour absorbs infrared (also: equator insolation 10× poles ✘; IR two-thirds of insolation ✘; IR not visible spectrum ✘). 2024 (a) troposphere thicker at equator because strong convection lifts heat. 2024 (d) atmosphere heated more by terrestrial (long-wave) than incoming radiation; only Statement II right. Undated (d, II and III): without atmosphere nights freeze but days get extremely hot, so "below freezing everywhere" ✘. Mains 2022 (15m) troposphere: clouds, air masses, jet streams sit here; terrestrial radiation is prime heat source. ✅

Ionosphere 2011 (D): charged particles reflect radio waves; NOT ozone; radio wavelength 1 mm–100 km, not "very long". Ionised by solar and cosmic radiation; D, E, F regions; electron density highest in F (day and night; day by solar radiation, night by cosmic rays); D vanishes at night, E weakens; VLF waves bounce in an Earth–ionosphere waveguide; Earth radius 6,370 km. ✅

🎯 2 · HEAT, ALBEDO, ISOTHERMS PYQ 2013 PYQ 2023 PYQ 2024 PYQ 2025

Temperature controls

→ Latitude (slant rays) · duration of daylight · sunspot cycle
→ Transparency (cloudy day cooler, cloudy night warmer)
→ Altitude · mountain orientation · vegetation (transpiration cools)
→ Distance from sea · ocean currents · local winds/urban

4 ways heat moves

→ Conduction: through solid, molecule to molecule
→ Convection: fluid movement (liquid/gas)
→ Advection: horizontal movement of air/water
→ Radiation: electromagnetic waves, no medium

Con-Con-Ad-Rad — Conduction = Contact (spoon in hot chai) · Convection = Currents (boiling milk) · Advection = Aside, sideways (Loo carries heat in) · Radiation = Rays, no medium (Sun to Earth)

💰 Heat budget

→ Earth gets insolation (short-wave) and returns terrestrial radiation (long-wave) = constant temperature
→ Hotter body = shorter wavelength: Sun short-wave, Earth long-wave

Albedo (reflectivity)

→ High: snow/ice = white shirt (reflects, cools)
→ Low: ocean, forest, city = black shirt (absorbs, warms)

→ Energy: 0–40° N/S surplus; beyond 40° deficit; winds and currents carry heat poleward, mostly across 30°–50° (hence storms there)

🛡️ Isotherms = lines of equal temperature (vertical distribution: see layers)

→ January, NH: bend poleward over oceans (warmer), equatorward over land (cools fast)
→ July: reverse (south over oceans, north over land)
→ SH: shift less evident (little land) and opposite in season: NH January = SH July
→ Cause: water has high specific heat; land heats and cools fast

Handout conflict: isotherm text first says "July: isotherms shift towards equator on land", then summarises "July: north over land, south over oceans". Trust the summary and the January rule.

PYQs: 2013 (A) interiors have bigger annual range due to land-water thermal difference only (not altitude, winds, rain; interiors are dry). Maritime influence = moderating; continental = extreme. 2023 (D) land-sea contrast greater in winter; water has higher specific heat. 2024 (a) January: isotherms north over ocean, south over continent; statement 2 (Gulf Stream, N Atlantic Drift as "cold currents") ✘. 2025 (a) both right and II explains I. ✅

3 · TEMPERATURE INVERSION

Inversion: temperature rises with height

→ Needs clear, calm, long nights, no mixing, valleys/basins
→ Common at fronts, in winter, in mountains and valleys

Good
→ Frost protection for plants

Bad
→ Pollution dome
→ Fog: poor visibility, transport
→ Pests and diseases thrive

Pollution dome

→ Pollutants trapped under warm air; cold ground in cold weather cools air; no vertical mixing
→ Effects: poor air quality; respiratory, cardiovascular problems
→ Mitigation: cut emissions, air-quality monitoring/forecasting, public awareness
→ Beijing: China launched US$100-bn drive in 2013 (vehicle rules, relocate industry, clean energy): PM2.5 down 35% by 2017

🎯 4 · PRESSURE, PRESSURE BELTS, FORCES PYQ 2024

Air pressure

→ Weight of air column per unit area; maximum at sea level
→ Heat = expand, rise = low; cool = sink = high
→ Isobars = lines of equal pressure
→ 7 belts = 3 thermal + 4 dynamic; not uniform (land-water)
→ Equatorial low (10° N-S) = ITCZ = Doldrums (calm, trades converge)

Lazy Hens Lay Happily — Low equator (ITCZ/Doldrums 10° N-S) → High ~30° → Low ~60° → High poles

Belt shift (Earth's 23.5° tilt)

→ 5-10° north (up to 25° over India) when Sun over Tropic of Cancer 21 June
→ 5-10° south when over Tropic of Capricorn 22 December
→ Balanced at equinoxes 21 March, 23 September
→ At 30°-40° = Mediterranean pattern: winter westerlies = rain; summer dry trades offshore

3 forces on wind

→ Pressure gradient: speed + direction (high to low; close isobars = fast)
→ Coriolis: direction only; deflects right in N, left in S; zero at equator, max at poles; grows with wind speed
→ Friction: both; only a few thousand metres up; less over sea so faster
→ Winds are named for where they come from (polar easterlies come from east)

Types of winds

→ Planetary/permanent: all forces, all year (trades, westerlies, polar easterlies)
→ Geostrophic: tropopause level, no friction, parallel to isobars (jet streams)
→ Seasonal: monsoon
→ Diurnal: sea/land breeze, mountain-valley breeze

→ Local: specific place, all year or one season

Planetary wind	From to	Notes
Trade winds	subtropical high to equator, easterly	helped trade routes; push ocean currents
Westerlies	subtropical high to sub-polar low, W to E	strongest in S. Hemisphere (little land): Roaring 40s, Furious 50s, Shrieking 60s
Polar easterlies	polar high to mid-latitudes	cold, weaker

Hadley Hot — Hadley (equator–30°, biggest, trades) · Ferrel Fill-in (30°–60°, middle, runs opposite) · Polar Puny (60°–pole, smallest, polar easterlies). Tricellular meridional circulation = 3 cells per hemisphere.

PYQs: 2011 (A) southern westerlies stronger because less land (not higher Coriolis). ✅ 2011 (C) jets fly in lower stratosphere: no clouds/vapour and no vertical wind shear. 2015 (B) westerlies blow 30°–60° (not "30°N–60°S"); moist air for NW India's winter rain is westerlies: they start in horse latitudes, steer extratropical cyclones; Mediterranean disturbance dragged by the jet stream; J&K and NW India get the most. 2024 (c) Coriolis rises with wind velocity and is max at poles, absent at equator.

🎯 5 · GEOSTROPHIC WINDS, JET STREAMS PYQ 2020 PYQ 2024

Geostrophic wind: blows parallel to isobars; pressure gradient force balanced by Coriolis (right in N, left in S). 💡

Jet streams: narrow fast bands, 8–15 km

- **Permanent**
  - → Polar: 50°–60° N/S, boundary cold polar/warmer subtropical air, strongest in winter
  - → Subtropical: 20°–30° N/S, weaker, less consistent
- **Temporary**
  - → Tropical Easterly
  - → Somali
- → Both hemispheres (2020: "NH only" ✘) · formed by temperature gradients of air masses
- → Wavy jets = Rossby waves
- → Steer storms; carry western disturbances (Mediterranean extra-tropical cyclones, by subtropical westerly jet); control South Asian monsoon; mix troposphere–stratosphere (ozone-depleting substances)

Climate change and jets

- → Arctic warms faster = smaller temperature contrast = weaker, more meandering jets
- → Long heatwaves, coldwaves, droughts or floods

Heat dome

- → Stagnant high pressure; air sinks, compresses, warms
- → Mid/high latitudes; continental climates
- → Heatwaves, heatstroke; agriculture, energy hit
- → Climate change: more frequent, intense, longer

Polar vortex (= DANA, Depresion Aislada en Niveles Altos, isolated depression/cut-off low)

- → Cold air held by polar jet; both poles, N pole matters most
- → Arctic Amplification + meandering jets = cold air spills to low latitudes

Stratospheric cooling

- → CO_2 radiates heat to space in the stratosphere (traps it in troposphere)

→ Ozone loss (CFCs, poles) = less UV absorption = less heating

→ GHGs affected stratosphere by ~1885, five decades before troposphere warming was detected

→ Impact: slows ozone recovery, strengthens polar vortex, disrupts jets and weather, extreme events, affects aviation and satellite orbits, key climate-change indicator ("warm troposphere + cooling stratosphere"); more PSCs

🎯 6 · LOCAL AND MONSOONAL WINDS PYQ 2014

Wind	Where	Hook
Chinook	E. Rockies (USA, Canada)	warm dry, "snow eater"
Foehn	Alps (leeward)	warm dry descending
Sirocco (Ghibli, Chili, Khamsin)	Sahara to Mediterranean	hot, dry; rainy in S. Europe
Harmattan	W. Africa (Sahara to Guinea)	dry, dusty NE trade; "Doctor" wind
Loo	N. India, Pakistan, May–June	furious, hot, heatstroke
Bora	Adriatic (Italy, Slovenia, Croatia)	extremely cold dry NE, katabatic
Mistral	Rhône valley (France)	cold NW
Santa Ana	S. California	hot dry; worsens wildfires
Khamsin	Egypt	dusty spring wind, up to 50 days
Purga	Siberia, Russian steppes	icy NE blizzard wind
Pamperos	Pampas (Argentina, Uruguay, Brazil)	violent cold SW

Brrr — Hot/warm: Chinook, Foehn, Sirocco, Harmattan, Loo, Santa Ana · Cold: Bora, Mistral, Purga, Pamperos. Mistral and Bora = "Brrr" winds.

Source note: Khamsin appears twice: as a Sirocco name (Sahara to Mediterranean) and as an Egyptian dusty spring wind.

Monsoon winds: seasonal, tropics, reverse with the Sun's apparent position (detail in Indian Geography). PYQ 2014 (C): seasonal reversal = monsoon climate; "mausim" (Arabic) = season; ~6 months NE, 6 months SW; South Asia, Africa, Australia, Pacific coast of Central America. ✅

🎯 7 · HUMIDITY, CLOUDS, PRECIPITATION PYQ 2019 PYQ 2022 PYQ 2024

Humidity (condensation = vapour to liquid water)

→ Absolute (g/m³) · relative (% of capacity) · specific (g per kg)

→ Dew point = saturation temperature

→ Dew: surface cools below dew point; none on cloudy nights (clouds send radiation back)

→ Frost = deposition (vapour to ice) · Fog = surface cloud · Smog = smoke + fog (sulphurous, photochemical) · Haze = dry particles

→ Atmospheric rivers: narrow moisture corridors (more water than Amazon), pushed by westerlies to mid-latitudes = heavy rain, floods

Clouds	Height	Note
High (Cirrus)	> 6,000 m	wispy ice "chicken feathers"; thin ones trap heat
Mid (Alto-)	2,000–6,000 m	
Low	< 2,000 m	Stratus, stratocumulus, nimbostratus; thick ones reflect
Cumulonimbus	vertical tower	thunder, hail, tornado

PYQs: 2022 (D) high thin clouds trap heat, low thick clouds reflect/cool: both statements ✘. 2019 (D) cirrus thinning + sulphate aerosols in stratosphere = geo-engineering to reduce global warming (cirrus trap long-wave; aerosols reflect sunlight). 2019 (B) no dew on cloudy night: clouds reflect back Earth's radiation. 2024 (a) water vapour decreases with altitude; not max at poles. ✅

→ 15–25 km, polar winter, below −78 °C

Polar stratospheric clouds (PSCs)

→ "Nacreous / mother-of-pearl"; bright at twilight

→ Surfaces for chlorine and bromine reactions = ozone destruction, Antarctic ozone hole

Cloud seeding (weather modification)

→ Agents: silver iodide, potassium iodide, salt (NaCl), dry ice = hygroscopic nuclei

→ Delivery: aircraft or ground generators (rockets, flares)

→ Uses: rain, snowpack, fog dispersal at airports, hail suppression, pollution control

→ Limits: needs moisture; variable, debated; silver iodide build-up; costly

Precipitation	How	Where
Convectional	heated air rises = cumulonimbus; short, intense, thunder	tropics 0–30°
Orographic	air forced up mountains; rain shadow on lee side	mountains
Frontal	warm over cooler: warm front long steady; cold front short intense	mid-latitudes 30–60°
Cyclonic	convection/frontal on a large scale around a low	cyclones

Thunderstorm

→ Stages: cumulus, mature, dissipating

→ Collisions of ice/water: lighter + at top, heavier − at bottom = voltage = lightning (excites N_2, O_2), thunder

→ Tall objects hit most: lightning conductors

Tornado

→ Violently rotating column from thunderstorm to ground

→ Over 300 km/h (long text: 300 miles/h)

→ Tornado Alley (Texas to Oklahoma, Kansas, Nebraska): warm moist Gulf air meets cold dry Canadian + warm dry desert air = supercells

→ Also Europe, Asia, Africa, Australia

Waterspout

→ Air + water droplets over water; less intense

→ Fair-weather (calm, little damage) vs tornadic (dangerous)

🎯 8 · AIR MASSES AND FRONTS PYQ 2020

Air mass: homogeneous temp + humidity, no vertical movement; surface to lower stratosphere; thousands of km; source = high-pressure, calm regions; jets carry them

🧑‍🤝‍🧑 Types (Tropical warm, Polar cold, continental dry, maritime moist)

→ mT: Amazon, SE Asia rain; heavy rain and thunderstorms on coasts, cyclones

→ cT: Sahara, central Australia; heatwaves, arid

→ mP: W Europe (UK, Ireland), Japan Sea coast (from Okhotsk) snow; cool, damp

→ cP: Siberia, Canadian Prairies; cold dry winters

→ cA: N Canada, Greenland, Scandinavia; frigid, polar vortex

→ mE (equatorial): Indonesia, Malaysia, W Africa (Gulf of Guinea); heavy rain

🗺️ Fronts: 3-D boundary of two contrasting air masses; mid-latitudes, rare in tropics/poles

→ Frontogenesis = formation (face-off); Frontolysis = dissipation

→ Temperate cyclones owe genesis to frontogenesis; bigger contrast = thinner front; cold fronts move faster

Occluded: cold front overtakes warm

→ Warm occlusion: air ahead of warm front cooler; clouds cumulus, stratocumulus, nimbostratus; moderate-heavy rain, thunder

→ Cold occlusion: air behind cold front colder; stratocumulus, nimbostratus, altostratus; rain/snow, cooler

→ Stationary = stalemate (light rain for days)

Cold front = Crash

→ (cumulonimbus, thunder, hail)

→ Warm front = Wet blanket (stratus/cirrostratus, steady drizzle, fog)

→ Occluded = cold overtakes warm.

🎯 9 · CYCLONES PYQ 2015 PYQ 2020

🗺️ Cyclone: low-pressure centre; winds counterclockwise in N, clockwise in S

→ Types: tropical, extratropical, subtropical

→ Names: Cyclone (Indian Ocean) · Hurricane (Atlantic) · Typhoon (W Pacific, S China Sea) · Willy-willy (W Australia)

	Tropical	Temperate (extra-tropical)
🏛️ Origin	Thermal	Dynamic (frontal)
Latitude	10–30°	35–65° (more in N. Hemisphere)
Forms over	warm seas only	land and sea
👥 Size	100–500 km	300–2,000 km
Rain	heavy, a few hours	slow, many days/weeks
Wind	100–250 km/h; wind, surge, rain damage	30–150 km/h; flood damage
Life	not more than a week	2–3 weeks
Fronts	absent	created by fronts
India	both coasts, east coast hot-spot	Western Disturbances

Tropical cyclone needs

→ Sea-surface ≥ 26.5 °C, moisture, low wind shear, pre-existing disturbance, Coriolis

→ Depression ≤ 62 km/h → cyclone ≥ 64 knots (74 mph)

→ Structure: eye (calm, clear), eyewall (worst), rainbands

→ Impacts: wind, rain, storm surge, tornadoes

Source conflict: cyclone sea temperature given as ≥ 26.5 °C in the notes and "> 27 °C" in the PYQ note.

Climate change

Tropical

→ Warmer seas = more intense, more Cat 4–5 (Arabian Sea, N Atlantic), fewer overall (N Atlantic slightly more active)

→ Poleward shift; slower, prolonged rain; rapid intensification; sea-level rise worsens surge

Temperate

→ Tracks move poleward (less cold air at low latitudes)

→ Gradient/jet changes alter intensity; more moisture = heavier rain, floods, landslides

PYQs: 2015 (B) no cyclones in S Atlantic / SE Pacific: ITCZ seldom occurs (also strong wind shear, cooler SST, no pre-existing vorticity). 2020 (C) only some cyclones develop an eye; eye is NOT 10 °C cooler (Erin's eye ~21 °F warmer; above 7.5 miles same as outside); jets in both hemispheres. ✅

🎯 10 · KÖPPEN AND CLIMATIC REGIONS PYQ 2011 PYQ 2024

AllBigCitiesDeserveExtraHalwa — A tropical humid · B dry · C warm temperate · D cold snow-forest · E polar · H highland. Second letters: f no dry season · w winter dry · s dry summer · m monsoon.

Handout conflict: Köppen group table labels Df = humid continental, Dw = subarctic; regional table labels Taiga = Df, Manchurian = Dw. Remember names and places, not D-codes. Rainfall numbers in the regional table (e.g. 150–1000 cm for Af) look scrambled: do not memorise.

Climate (code)	Where	Hooks
Equatorial (Af)	10° N–S: Amazon, Congo, Indonesia, PNG	24–27 °C, tiny range; 4 pm convectional rain; evergreen selvas, mahogany, rosewood, ebony; leached soil; slash-and-burn
Savannah (Aw)	10–30°, interiors: Sudan, Venezuela, Brazil, N Australia	hot rainy summer, dry winter; elephant grass, baobab; Masai; "Big Game country"
Tropical monsoon (Am)	10–30°, E margins: India, Vietnam	3 seasons; 200–250 cm; teak, sal; paddy
Hot desert (BWh)	W margins, subtropical high, cold currents: Sahara, Kalahari, Mojave, Atacama	xerophytes (waxy/thorny leaves, deep roots, succulents), date palms at oases; Bedouin, Tuareg, Bushmen
Cold desert (BWk)	interiors: Gobi, Patagonia	wide range, cold winters; bunchgrasses
📝 Steppe (BS)	45–60°, mid-continental: Prairies, Steppes, Downs	Chernozem; wheat/maize, ranching
Mediterranean (Cs)	30–45°, W margins: California, Chile, SW S Africa, SW Australia	hot dry summer, winter rain; citrus, olives, vineyards; cork oak, eucalyptus
👫 Marine West Coast (Cfb, British type)	45–60°, W margins: NW Europe, SW S America, NZ	mild, rain all year, small range; deciduous forest; dairy
China type (Cw/Cfa; Gulf, Natal)	30–45°, E margins: SE China, Japan	warm moist summer, cool dry winter; typhoons; most productive mid-latitude farming
Taiga/Siberian (Df)	50/60–70°: N Canada, Eurasia	−30 to −40 °C winters; conifers (softwood, pure stands); fur, lumbering
Manchurian/ Laurentian (Dw)	E margins, N Hemisphere only: Manchuria, Japan, NE America	cold dry winters, wet warm summers
Polar (ET, EF)	Greenland, N Canada, Alaska, Antarctica	summer < 10 °C; mosses, lichens; Eskimos, Lapps, Yakuts

Climate PYQ	Answer / rule
👫 2011 rainforest regrows slowly	Soil poor in nutrients: leached, nutrients stored in plants, rapid decay (thin humus)
2011 African–Eurasian desert belt	Subtropical high-pressure cells (not warm currents); rain shadow (Gobi), cold currents (Namib, Atacama) also cause deserts; hot deserts 20°–30°, W coasts
2012 savannah	Definite dry and wet season; 800–1,600 mm rain; less than monsoon
2015 clear morning, thunder, short rain	Equatorial: 26–28 °C, annual range as low as 3 °C, ≥ 2,000 mm, ITCZ two rainfall peaks
2018 desert leaf traits	Hard waxy, tiny, thorns: all three (succulents; cacti spines)
👫 2021 keeps savannah forest in check	Fire, grazing herbivores, seasonal rainfall (2, 3, 4); acacia; maize, millets, sorghum, cassava; groundnuts, cotton, E Africa coffee
👫 2021 fast litter decay, bare soil, epiphytes	Tropical rain forest (canopy 75 m+, 3 layers)
👫 2023 rainforest soil rich?	Statement I ✘ (leached); II ✔

🆕 2024 low annual/daily range, rain all year, 50–250 cm	Marine West Coast
🧑‍🤝‍🧑 No year dust in subtropical/temperate	I ✔ (deserts: Sahara, Arabia, Central Asia, China; scarce vegetation, high wind); II ✘ (strong dry winds); equatorial humid/ forested, polar ice-covered

11 · TRAPS AND CONNECTIONS

Traps

→ jet streams only in NH ✘
→ land–sea contrast greater in summer ✘ (winter)
→ westerlies 30°N–60°S ✘
→ ozone reflects radio ✘
→ more moisture aloft = colder ✘
→ Gulf Stream is cold ✘
→ eye 10 °C cooler ✘
→ rainforest soil rich ✘.

Connects to

→ pressure belts and 23.5° tilt / 21 June → Geomorphology
→ Western disturbances, monsoon, Loo → Indian climate
→ ocean currents shape isotherms and deserts → Oceanography
→ climate regions → Economic Geography.

Recall drill: cover and answer

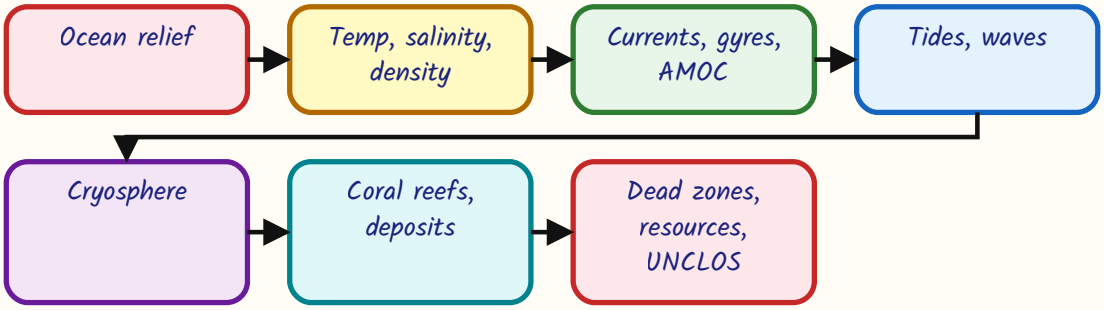
Troposphere height; lapse rate?	8–18 km (avg 13); 1 °C per 165 m (≈6.5 °C/km)	Ozone layer height; coldest layer; jet-flight layer?	20–25 km; Mesosphere; lower stratosphere
Kármán line; energy surplus/deficit boundary?	100 km; 40° N/S	Coriolis: zero, max, direction?	Equator; poles; right in N, left in S
Roaring / Furious / Shrieking?	40s / 50s / 60s (S. Hemisphere westerlies: less land)	Polar vs subtropical jet?	50°–60° (strong in winter) / 20°–30°; both hemispheres
Cold front vs warm front clouds?	Cold: cumulonimbus; Warm: stratus, cirrostratus	Cyclone sea temperature; why none in S Atlantic?	≥ 26.5 °C; ITCZ seldom occurs
Chinook: where, nickname?	E. Rockies; snow-eater	PSC altitude and temperature?	15–25 km; below −78 °C
Cloud seeding agents?	Silver iodide, potassium iodide, salt, dry ice	Main cause of desert belts?	Subtropical high-pressure cells (plus rain shadow, cold currents)

OCEANOGRAPHY

PYQ Prelims 2013 ×2, 2015 ×2, 2021 ×2 · Mains 2017 (oceanic salinity, 15m)

✎ *Ocean floor, temperature-salinity-density, currents, tides, waves, corals, deposits, resources and UNCLOS: 6 Prelims PYQs, heavy on current-affairs themes.*

At a glance: the Oceanography route



Pattern: the exam asks why: counter-current flows east, tides occur, fishing grounds sit where warm and cold currents meet. Current-affairs heavy: AMOC weakening, coral bleaching, dead zones, marine heatwaves, Southern Ocean carbon sink (good Mains material). Priority printed as MEDIUM-HIGH (6 Prelims, 1 Mains). No teacher ink. ⭐

1 · OCEANS AND RELIEF

Basics

- → Hydrosphere ~71% of Earth's surface
- → 5 oceans: Pacific, Atlantic, Indian, Arctic, Southern
- → Average depth 3,800 m vs lithosphere average height 840 m
- → Relief from tectonic, volcanic, erosional, depositional processes

Part (house)	Facts
Continental shelf (front porch)	gentle 1–2°; to ~200 m (up to 500 m); continental crust + sediment (sand, silt, clay); oil and gas, fisheries; North Sea, Gulf of Mexico; Great Barrier Reef (>2,300 km); width few km to hundreds of km
Continental slope (steep stairs)	3–6° (up to 25°); ~200–3,000 m; submarine canyons (Monterey, Hudson); Blake Plateau; deep-sea ecosystems (pressure, dark, cold)
Continental rise (landing)	gentle sediment apron at slope base; deep-sea fans, turbidity currents; Amazon rise
Abyssal/deep-sea plain (floor)	>2,000 m (2,000–6,000+); flat, fine sediment (clay, silt, sand); polymetallic nodules: Clarion–Clipperton Zone (Pacific, Hawaii–Mexico)
Trench (cellar)	>6,000 m; convergent boundaries, subduction, quakes; Mariana

Source overlap: slope runs to ~3,000 m but abyssal plain starts >2,000 m, and the notes also print "deep ocean 2–4 °C": treat depths as approximate.

💡 Other features

- → Seamounts: ≥ 1,000 m underwater volcanoes (Davidson)
- → Guyots (table mounts): flat-topped seamounts, wave-eroded then submerged (Emperor Seamounts)
- → Hydrothermal vents: mineral-rich water; black smokers; life may have begun here

Term	Handout example
Strait (narrow waterway joining two waters)	Palk (India–Sri Lanka)
Isthmus (land strip joining landmasses)	Kra (Malay Peninsula–Asia)
Archipelago	Japanese (W Pacific)

Lake	Baikal (Russia), Wular (India)
Bay	Bengal
Cape	Comorin = Kanyakumari (southernmost tip of Peninsula); Good Hope (Africa)
Lagoon (behind barrier island/reef)	Chilika (brackish, Odisha)
Island	Lakshadweep (Arabian Sea)
Gulf (constricted entrance vs bay)	Kutch (Gujarat)
Peninsula (3 sides water)	Indian (Arabian Sea, Bay of Bengal, Indian Ocean)

🎯 2 · TEMPERATURE, SALINITY, DENSITY PYQ 2017

	Temperature	Salinity	Density
Key numbers	26.7 °C equatorial mean; 12.7 °C at 45° N/S; poles freezing; deep 2–4 °C; mixed layer to 200–400 m; thermocline = rapid fall	deep ocean ≈ 34.7 ppt (avg ~35); Great Salt Lake 220 · Dead Sea 240 · Lake Van 330	cold > warm; high salinity > low; deep > surface (pressure)
Pattern	warmest at equator; Indian Ocean warmest; N Hemisphere warmer (more land)	highest at ~20°–30° (high evaporation, little rain); lower at equator and poles	temperature matters more than salinity
Factors	latitude; currents (Gulf Stream, Kuroshio warm); atmospheric circulation; continentality; depth; season	evaporation/precipitation; rivers; sea-ice (freezing leaves salt); currents (Gulf Stream saline, Labrador cold fresher); winds	temperature, salinity, pressure

Sun + No Rain = Salty – Subtropics (hot, dry) saltiest; equator and poles diluted. Enclosed seas keep getting saltier: 220 – 240 – 330 = Great Salt Lake → Dead Sea → Lake Van. Continentality: sea ringed by land is warmer.

Mains 2017 (15m) salinity: Intro: dissolved salts, avg ~35 ppt, deep 34.7. Variation: latitude (subtropical high), evaporation vs rain, rivers, sea ice, currents, inland seas (220/240/330). Effects: density → thermohaline circulation, stratification (halocline), nutrients, habitats and productivity. Close: melting-ice freshwater now alters salinity and circulation. ✅

Ocean acidification

→ pH 8.2 (pre-industrial) → 8.11 (1985) → 8.04 (2024); ~30% more acidic in 200 yrs

→ More in Arctic than tropics (cold water absorbs more CO_2)

→ Causes: CO_2 dissolving, volcanoes, agricultural runoff

→ Effects: coral bleaching, weaker shells/exoskeletons, phytoplankton changes

🎯 3 · FRESHWATER PYQ 2013 PYQ 2021

Freshwater share

→ Only ~3% of Earth's water is fresh (of freshwater, other ~70% after ice: groundwater ~30%, rivers/lakes ~1%); 2.5% unavailable (ice, atmosphere, soil, polluted, too deep); 0.5% available

→ Ice caps/glaciers ~68–69% · groundwater ~30% (largest active resource; equals a lake 180 m deep over the Earth) · rivers/lakes ~1%

Handout conflict: surface water (lakes, rivers, swamps) given as 0.3% in 2013 note and ~1% in 2021 note. 2013 (C): after ice, most freshwater is groundwater. 2021 (B): ice caps/glaciers hold more than groundwater; rivers/lakes not more.

🎯 4 · OCEAN CURRENTS PYQ 2015 PYQ 2021

🆕 Currents: continuous flow of large water volumes in a definite direction, all year; move heat tropics to poles

→ Surface: top ~400 m, only ~10% of ocean water; wind-driven

→ Deep (thermohaline): density-driven; down-welling at poles, up-welling in tropics

Circumpolar

→ Continuous loop round poles, isolate polar waters

→ Antarctic Circumpolar Current, West Wind Drift

→ Factors: latitude, wind, Coriolis, gravity, temperature, salinity, continental margins

Equatorial system

Trades push water E to W (+ Earth's spin) → *Coriolis: N and S Equatorial Currents* → *Water piles up in the west* → *Gravity: flows back east* → *Equatorial Counter Current (3-10° N)*

2015 (B): counter-current flows east due to convergence/pile-up of the two equatorial currents, not Earth's rotation directly. NECC 100–150 m deep, in Atlantic, Indian, Pacific. Counter Equatorial Current (CEC). Not the Equatorial Undercurrent (EUC) (eastward along equator, ~200 m deep in W Pacific, 100 m in E Pacific). ✅

West = Warm & Wild – Western boundary currents (Gulf Stream, Kuroshio, Brazil, East Australian, Agulhas): warm, narrow, fast, strong. East = cold & Easygoing: Eastern boundary (Canary, Benguela, California, Peru, West Australian): cold, broad, slow, nutrient upwelling.

Gyres (wind + Coriolis + coastlines): N Hemisphere clockwise, S anticlockwise

→ N Atlantic (clockwise): Gulf Stream, N Atlantic Current, Canary, N Equatorial; Sargasso Sea in middle = only sea without a coastline (Sargassum weed)

→ S Atlantic (anti): Brazil, Benguela, S Equatorial

→ N Pacific (clockwise): Kuroshio, N Pacific, California

→ S Pacific (anti): East Australian, Peru, S Equatorial

→ Indian (anti): Agulhas, West Australian, S Equatorial

2021 (C), both right: tropics: west of ocean warmer (trade winds carry warm water west); temperate: east warmer (westerlies carry warm equatorial water to continents' west coasts). ✅

Water masses: big homogeneous bodies with own temp, salinity, density from a source region; keep properties far

→ Move by gravity, density, wind stress, Earth's rotation (3-D); backbone of global circulation

→ Set habitat: warm clear = coral; cold nutrient-rich = kelp, high phytoplankton and productivity; deep = specialised life

→ Warm salty surface water north (Gulf Stream) → cools, sinks in N Atlantic = NADW → flows south at depth

→ Regulates Atlantic circulation of heat, climate, weather, rain (also written AOC)

AMOC (Atlantic Meridional Overturning Circulation)

→ Weaker than any time in past millennium (warming; melting-ice and rain freshwater disrupt sinking)

→ If weak: cooler W Europe, more extreme events (heat, cold, storms), stronger US east-coast hurricanes, less CO_2 uptake, sea-level rise, fisheries shifts

Southern Ocean carbon anomaly

→ Absorbs ~40% of CO_2 taken up by oceans (stronger sink than models)

→ Glacier-melt + rain freshwater = light "lid", blocks mixing, traps carbon-rich water 100–200 m down

→ "Fleeting reprieve": lid weakens = CO_2 release, faster warming

NEW Ekman current

→ Wind-driven; water ~45° right (N) / left (S) of wind

→ Layers turn = Ekman spiral

→ Gives coastal upwelling (nutrients, fisheries), heat redistribution, El Niño/La Niña links

NEW Climate change on currents

→ Weaker thermal/salinity gradients = thermohaline slowdown

→ Less upwelling nutrients; less oxygen in downwelling zones; shifted winds; ice melting from below; biodiversity decline

Marine heatwaves: prolonged unusually warm sea

→ Episodes doubled since 1982; warming rate nearly doubled since 2005 (Copernicus)

→ Causes: climate change, high pressure, shifting currents, El Niño, feedback loops

→ Effects: coral bleaching, ecosystem disruption, current changes

🎯 5 · TIDES AND WAVES PYQ 2015

Tide: rise (high/flood) and fall (low/ebb) of seawater

→ Cause: gravity of Moon (main) and Sun + centrifugal force of Earth = bulges near and far side

Timing

→ Most places semi-diurnal: 2 highs, 2 lows a day; same tide every 12 h 26 min

→ Moon takes 24 h 52 min to return over a tide centre

→ Diurnal: Gulf of Mexico, Gulf of Thailand

→ Height depends on: Sun-Moon-Earth distances, funnel-shaped bays/estuaries, local winds, pressure (low pressure raises sea)

→ Use: nutrient circulation, temperature moderation, flushing pollution

2015 (D): tides = Sun's gravity + Moon's gravity + Earth's centrifugal force (all three). ✅

SPrung Straight – Spring tide = syzygy: new and full moon, Sun-Moon-Earth aligned; highest highs, lowest lows, ~20% higher; twice a month; nothing to do with the season.

Nearly right-angle – Neap tide = quadrature: quarter moons, forces oppose; lower highs, higher lows.

Tidal bore

→ Rare: incoming tide makes wave travelling upstream in narrow estuary/shallow river; absent at neaps

→ India: Hooghly; world's biggest: Qiantang (China)

Climate change on tides

→ Higher sea level = larger areas tidal

→ May amplify tidal ranges; threatens mangroves

→ Wind gives energy; gravity pulls crests down; falling water lifts troughs

📜 Waves: energy moves, not water; particles move in small circles

Tsunami: very long seismic sea waves

→ Causes: earthquakes at subduction zones (size ∝ vertical shift), volcanoes, landslides, meteorite impact, nuclear tests

→ 500–1,000 km/h in deep water, < 1 m high (invisible); slows and grows near coast; many waves

→ Warning: sea-level change, odd ocean behaviour

Rogue (freak) waves

→ Unusually large, spontaneous, even in calm seas

→ Constructive interference: wave systems in phase (rare, random)

→ Focusing by currents (Agulhas Current opposing waves) and topography (ridges, shelves)

Seiche: standing wave in lake/bay/reservoir

→ Causes: wind, pressure change, quakes/landslides

→ Resonance with natural rhythm grows waves

→ Basin size, shape, depth decide size and duration

6 · CRYOSPHERE

Cryosphere = Earth's ice

→ 69% of freshwater; albedo; stores methane/CO_2; feeds rivers; affects circulation; GLOFs in Himalaya, Alps, Andes

Melting effects

→ Thermohaline change; winds/jet streams

→ Freshwater (short-term up, long-term down)

→ Lower albedo; GHG release

→ Heavy elements and pathogens released

→ Isostatic rebound and volcanic activity

Arctic

→ Weaker polar vortex = cold spells in N America/Europe; more temperate storms

→ New shipping routes (thick ice drifts into Northwest Passage); resource access

→ Hundreds of Greenland lakes burst since 2008; trapped mercury leaks (reaches Arctic by air and ocean circulation)

Antarctica

→ Jet-stream weakening; winter heatwaves (mid-July −25 to −30 °C instead of −50 to −60 °C); wildlife and research

→ Oceans: acidification; SST rise 0.06 °C/decade (1980s) → 0.27 °C/decade now; circulation and nutrient changes

7 · CORAL REEFS

Corals: marine invertebrates, colonies of polyps; secrete calcium carbonate; "rainforests of the sea"

→ Symbiotic algae zooxanthellae give energy by photosynthesis

Kinds

→ Hard (stony, reef-builders): Acropora, Porites
→ Soft: octocorals, sea fans
→ Cold-water: deep, polar/temperate
→ Black corals: 30–300+ m, Gulf of Mannar

Warm, Clear, Shallow, Salt-stable, Light-snack, Mild-waves — 20–29 °C · clear sunlit water · <50 m deep · salinity 32–40 ppt · low nutrients (oligotrophic; runoff feeds algae that choke them) · moderate waves

Fringing = Friend — Fringing = Friend, hugs shore (boat channels) · Barrier = Bodyguard, stands off, lagoon between · Atoll = Wedding ring around lagoon (sinking volcanic island/seamount)

Bleaching: corals expel zooxanthellae (white; recover or die); more disease

→ Causes: heat, acidification, pollution, UV, tectonic exposure/drowning, overfishing (fish keep reefs clean), disease, tourism
→ Great Barrier Reef: 4th mass bleaching ongoing since early 2023
→ India: bleaching in Gulf of Mannar and Andamans; Lakshadweep, March 2024
→ Climate: acidification (harder to build skeletons); stronger storms (damage, smothering sediment)

Source conflict: 4th mass bleaching start printed as January 2023 once and February 2023 once.

8 · OCEANIC DEPOSITS

Land, Life, Liquid, Cosmos — Terrigenous (land: rivers, glaciers, wind; margins) · Biogenous (organisms): calcareous ooze $CaCO_3$ (warmer, shallower), siliceous ooze (diatoms, radiolarians; colder, deeper) · Hydrogenous (chemical): manganese nodules, phosphorite · Cosmogenous (meteorite dust)

9 · DEAD ZONES PYQ 2013

Dead zones (hypoxic): oxygen too low for most life

→ Cause: nutrient pollution (N, P: farm runoff, sewage, industry, even wind-blown dust: Madagascar bloom)
→ Chain: algal bloom → dies, sinks → bacteria use up oxygen
→ Effects: marine life suffocates, fisheries/tourism lose, toxins and bacteria contaminate seafood

Examples

→ N Gulf of Mexico (Mississippi runoff, annual)
→ Baltic Sea (chronic, nutrient pollution)
→ Black Sea (natural, poor circulation)

→ Fix: cut nutrient pollution (farming rules, wastewater treatment), monitoring, public awareness

🎯 10 · OCEAN RESOURCES AND FISHING GROUNDS PYQ 2013

Resources

Biotic
→ Fish: tuna, salmon (Pacific); cod (N Atlantic); sardines (Mediterranean, Peru)
→ Shellfish; seaweed (Japan nori); marine pharma; pearls

Abiotic
→ Offshore oil/gas: North Sea, Gulf of Mexico, Persian Gulf
→ Tidal, OTEC, wave, offshore wind
→ Dissolved salts (NaCl); manganese nodules (source of "dark oxygen"); phosphorite (Morocco, Peru, USA)

Services
→ Absorb ~90% of excess heat, 25–30% of human CO_2 (blue carbon: mangroves, seagrass)
→ Phytoplankton make 50–60% of Earth's oxygen

✅ 2013 (C) fishing grounds: best where warm and cold oceanic currents meet (plankton mixing): Newfoundland banks, Sea of Japan. Helpers: coastal upwelling, indented coasts, shelf food from erosion + sunlight, nearby forests for boats, rugged land pushing people to fishing, temperate water below 20 °C.

📜 11 · LAW OF THE SEA: UNCLOS 1982

📌 **12 - 24 - 200** — Territorial sea 12 nm (22.2 km) · Contiguous zone 24 nm (44.4 km; customs, fiscal, immigration, sanitary laws) · EEZ 200 nm (370.4 km; sovereign rights over resources) · High seas beyond (navigation, overflight, fishing, research). A dozen → a day → a double century.

💡 UNCLOS (United Nations Convention on the Law of the Sea), adopted 1982: main legal framework on maritime boundaries, navigation, resources, protection of marine environment.

12 · TRAPS AND CONNECTIONS

Traps
→ spring tide = in spring ✘
→ counter-current due to Earth's rotation ✘
→ salinity highest at equator ✘
→ density driven more by salinity ✘ (temperature)
→ tsunamis tall in deep ocean ✘
→ rivers/lakes hold most freshwater after ice ✘ (groundwater)
→ atolls form on continents ✘ (sinking volcanic islands).

Connects to
→ trenches, guyots, hotspots → Geomorphology
→ Coriolis, trade winds, westerlies, cyclones, El Niño → Climatology/ Indian climate
→ resources, ports, fisheries → Economic Geography
→ Palk Strait, Chilika, Gulf of Kutch, Cape Comorin → Indian Geography.

Recall drill: cover and answer

Average ocean depth; shelf and slope gradient?	3,800 m; 1–2° / 3–6°	Trench depth; example?	> 6,000 m; Mariana
Mean temperature equator vs 45°?	26.7 °C / 12.7 °C	Saltiest listed water bodies?	Lake Van 330 > Dead Sea 240 > Great Salt Lake 220
Ocean pH now?	8.04 (2024); pre-industrial 8.2	Why does the counter-current flow east?	Convergence/pile-up of the two equatorial currents
Only sea with no coastline?	Sargasso Sea (N Atlantic gyre)	Tide repeats after; spring tide when?	12 h 26 min; new and full moon (syzygy)
Tidal bore in India; world's biggest?	Hooghly; Qiantang (China)	Tsunami speed in deep ocean?	500–1,000 km/h, under 1 m high
Coral temperature / depth / salinity?	20–29 °C / <50 m / 32–40 ppt	UNCLOS zones?	12 · 24 · 200 nm

INDIAN GEOGRAPHY

PYQ Prelims 40 PYQs: physiography 15 (2012, 2014 ×2, 2015, 2016, 2017 ×2, 2018 ×2, 2020, 2022, 2023 + 3 undated) · drainage 12 (2011, 2013, 2014, 2015, 2016 ×2, 2017, 2021 ×2, 2022, 2023 + 1 undated) · climate 3 (2011, 2017 + 1 undated) · forests and soils 10 (2011, 2013, 2015 ×3, 2016, 2021, 2023 ×2 + 1 undated) · Mains none quoted

✎ *Location, six physiographic units, rivers, climate and monsoon, forests and soils: the largest PYQ bank of the six chapters (40 Prelims), all map-and-list questions.*

At a glance: the Indian Geography route



Pattern: "which pairs / rivers / peaks / states?": tributaries of Godavari and Brahmaputra, hills of which range, which state has what, which forest has which tree. Place every fact on a map. Mains hooks (none quoted): Western Disturbances, cloudbursts, urban flooding, ELLA, river-linking, mangroves, social forestry. No teacher ink: priority = PYQ count. Image-only pages (peaks, glaciers, passes, waterfalls, lakes) are kept below as tables.

🎯 1 · LOCATION OF INDIA PYQ 2015 PYQ 2018 PYQ 2017

Fact	Number
Area	32,87,263 km² · 7th largest · 2.4% of world area
Extent	N–S (Indira Col, Kashmir → Kanyakumari) 3,214 km · E–W (Rann of Kutch → Arunachal) 2,933 km
Latitude / longitude	8°4′N (6°45′ for islands) to 37°6′N · 68°7′E to 97°25′E · wholly Northern Hemisphere
🧑‍🤝‍🧑 Southernmost point	Indira Point (Pygmalion Point), 6°45′N
Time	1° longitude = 4 min · E–W gap ≈ 2 hrs (~30 × 4) · Chai Bagan time = 1 hr ahead of IST (British, NE India, to raise productivity)
Tropic of Cancer	splits India in two; area north ≈ 2× south · 8 states: Gujarat, Rajasthan, MP, CG, Jharkhand, West Bengal, Tripura, Mizoram
Borders	land 15,106.7 km through 17 states · coastline 11,098 km, 13 states/UTs · longest = Bangladesh · shortest = Afghanistan (Wakhan)
🧑‍🤝‍🧑 No border / no coast	Telangana, MP, CG, Jharkhand, Delhi, Chandigarh, Haryana; all others = frontline states (border management)

U C My Old Aunt — Standard Meridian 82°30′E passes through 5 states: Uttar Pradesh · Chhattisgarh · Madhya Pradesh · Odisha · Andhra Pradesh (handout mnemonic).

Neighbours (clockwise) → Pakistan · Afghanistan · China · Nepal · Bhutan · Bangladesh · Myanmar · Sri Lanka · Maldives · Indonesia

Neighbour	Bordering Indian states/UTs
Pakistan	J&K, Ladakh, Punjab, Rajasthan, Gujarat
Afghanistan	Ladakh
China	Ladakh, HP, Uttarakhand, Sikkim, Arunachal
Nepal	UP, Uttarakhand, Bihar, Sikkim, West Bengal
Bhutan	Assam, Sikkim, Arunachal, West Bengal
Bangladesh	West Bengal, Assam, Meghalaya, Mizoram, Tripura
Myanmar	Arunachal, Nagaland, Manipur, Mizoram

Source doubt: coastline is **11,098 km** here, but the cyclone note says "over **7,500 km**" (as printed). Use 11,098 km for coastline.

🎯 2 · PHYSIOGRAPHY: SIX UNITS PYQ 2012 PYQ 2014 PYQ 2023

My Parents Do Play Cricket Indoors — handout's six units top to bottom: Mountains · Plains · Desert · Plateau · Coasts · Islands (Northern and North-eastern Mountains · Northern Plains · Indian Desert · Peninsular Plateau · Coastal Plains · Islands).

Northern and NE mountains

→ **Origin:** Indian + Eurasian plates converge (Tethys closes); Tethys sediments fold

→ Multiple upliftments = **parallel ranges**; steep south, gentle north (**hogback**); ridge–valley

→ **2,500 km** long; width **500 km** (Kashmir) → **200 km** (Arunachal); **13** states/UTs

→ Main Himalaya: **Nanga Parbat** (Indus gorge) to **Namche Barwa** (Brahmaputra gorge) = two **syntaxial (knee-joint) bends**; then **Purvanchals**

→ **Transhumance:** seasonal herding to **bugyals**: **Bakarwals, Gaddis, Bhotiyas, Drokpas**

Belt	Also called	Notes
Trans-Himalaya	Karakoram, Ladakh, Zaskar	highest, cold dry · most glaciers in **Karakoram** · **K2** (Godwin-Austen) · **Ladakh** = highest Indian plateau · Zaskar south of Indus · Kailash entirely outside India
Great Himalaya	Himadri, inner	avg **6,100 m**; tallest peaks
Middle Himalaya	Himachal, Lesser	**3,500–4,500 m** · Pir Panjal, Dhauladhar (southernmost HP range), Kumaon, Mussoorie, Mahabharat · **Karewas** (clay–silt–sand–gravel of Pleistocene lakes, between Pir Panjal and Zaskar) → saffron (zafran), walnuts, almonds · Bugyals
Outer Himalaya	Shiwaliks	lowest; nearly unbroken, gap of **80–90 km** at Teesta valley

Foothills (top to bottom)

→ **Bhabar** (stones, rivers vanish) → **Terai** (swamp, rivers reappear) → plains

Duns (W)
→ valleys between Shiwaliks and Himachals (HP, Uttarakhand); longitudinal
→ Dehradun, Patli Dun, Kota Dun

Duars (NE)
→ floodplain foothills north of Brahmaputra valley (Assam, Arunachal)
→ alluvial, fertile, dense vegetation

Chos / Choes / Raos
→ seasonal Shiwalik streams, NW Punjab foothills; sandy beds
→ less fertile than Duars; need irrigation

Purvanchals

→ crescent, convex to west; perpendicular to Himalaya; rough, forested = hard to develop

→ Mishmi, Patkai Bum, Naga, Kohima, Manipur hills (Loktak), Barail, Mizo (Lushai)

→ Barail: separates Manipur and Naga hills; divides Brahmaputra and Barak basins

Regional Himalaya	Limits	Peaks / valleys (as printed)
Punjab / Kashmir / Himachal (west)	Indus → Sutlej	Nanga Parbat 8,126 m · K2 8,611 m · Dhaulagiri I 8,167 m · Kashmir, Kullu, Kangra valleys
Kumaon (central)	Sutlej → Kali	Nanda Devi 7,816 m (highest entirely in India) · Nanda Kot 7,818 m · Trishul 7,120 m · Valley of Flowers (UNESCO) · Ganga source
Nepal (eastern)	Kali → Teesta	Everest 8,848 m · Kanchenjunga 8,586 m (3rd highest) · Lhotse 8,516 m · Annapurna I 8,091 m
Assam (easternmost)	Teesta → Arunachal / Purvanchals	Namche Barwa 7,756 m · Manas NP (UNESCO)

Source conflicts: Nanda Devi 7,816 m vs Nanda Kot 7,818 m (so Nanda Devi is "highest entirely in India" only by that label; the 2022 PYQ note says Nanda Devi 7,817 m). Kamet is printed as "highest of Uttarakhand" though Nanda Devi (Uttarakhand) is higher. Learn ~7,816 and rely on the label "highest entirely in India".

Region	Peak: what the handout says
Trans / Himadri	Nanga Parbat (Ladakh Himadri, >8,000 m, syntaxial bend) · K2/Godwin-Austen (Karakoram, 2nd highest in world) · Broad Peak, Gasherbrum I and II (Gasherbrum group, all >8,000 m) · Nun (J&K), Kun (Ladakh): Himadri
HP	Reo Purgyil: highest of HP, Zaskar Range
Uttarakhand	Satopanth (lake at foot) · Kamet: highest of Uttarakhand and of Zaskar · Trishul: 3 peaks like a trident · Nanda Devi: Greater Himalaya, UNESCO World Heritage (with its NP)
Shiwalik peaks	Karoh: highest of Haryana · Amsot: highest of UP · Someshwar: highest of Bihar
East Himalaya	Kanchenjunga: highest Himalayan peak in India, Sikkim, biosphere reserve + NP, India's first UNESCO mixed heritage site · Sandakphu: highest of West Bengal (Darjeeling, tea) · Tiger Hill (Himachal range) · Kangto: highest of Arunachal
Purvanchal / NE	Dapha Bum: highest of Purvanchals (Mishmi) · Saramati: Nagaland · Mt Esii/Iso: Manipur · Phawngpui (Blue Mountain): Mizoram · Betlingchhip (Thaidawr Tlang): Tripura · Nokrek: Garo, NP + biosphere reserve · Shillong Peak: Khasi, highest of Meghalaya · Tumjang/Sielkal/Thingtuibung: Barail, highest of Assam
Peninsula (north → south)	Parasnath: highest of Jharkhand, Jain pilgrimage, Chotanagpur · Guru Shikhar: highest of Aravallis and Rajasthan · Gorakhnath: highest of Gujarat, Girnar hills · Dhupgarh: highest of Satpuras and MP · Sadbhawna Shikhar (Goodwill Peak): highest of Vindhyas · Kalsubai: highest of Maharashtra (W. Ghats) · Mahabaleshwar: origin region of Krishna · Sonsogor: highest of Goa (Maharashtra–Goa–Karnataka trijunction)
Western Ghats south	Mullayanagiri: highest of Karnataka · Kudremukh (minerals, tiger reserve, NP) · Brahmagiri: source of Kaveri · Doddabetta: highest of Tamil Nadu and Nilgiris · Anaimudi: highest of Western Ghats and Kerala · Agasthyamalai: TN–Kerala border
Others	Saddle Peak: highest of Andaman and Nicobar · Mt Harriet (now Mt Manipur) · Bailadila: Chhattisgarh, Deccan · Gaurlata: highest of Chhattisgarh, Central Highlands · Doligutta: highest of Telangana · Mahendragiri: Eastern Ghats · peak No. 45: highest of Odisha, Eastern Ghats (name blank in handout) · Arma Konda / Jindhagada: highest of Eastern Ghats and Andhra Pradesh

Peak pegs — Trishul = trident (3 peaks) · Kamet = the "Kamet-ition winner" of Uttarakhand · Doddabetta = "Dodo-big-bat" of the Nilgiris · Anaimudi = "Anna" is tallest in the Ghats' family (Kerala) · Kalsubai = Maharashtra's "kal-subah" (tomorrow morning) climb.

Tarim → Yarkant, Skarmi

Glaciers (colour = basin on handout map)

Indus system
→ Hispar, Chogo, Basha, Lungma, Rupal (Nanga Parbat)
→ Hunza, Batura, Khurdopin, Biafo, Baltoro → Shigar
→ Siachen, Rimo → Nubra/Shyok; Chong Kumdan, Drang Drung → Shyok/Stod
→ Shafat · Sonapani → Kutle Nala · Barashigri → Chandra · Suru (Nanga Parbat side)

Ganga basin
→ Gangotri → Ganga · Chorabari → Mandakini · Milam → Gori Ganga · Pindari → Pindari · Kamet · Dhauliganga
→ Yamunotri and Bandarpunch → Yamuna

Brahmaputra basin
→ Zemu → Teesta

Pass	State/UT	Hook
Aghil	Ladakh	Karakoram; historic trade; military
Karakoram	Ladakh	ancient silk route
Khardung La	Ladakh	connects Leh and Siachen
Mig La	Ladakh	printed as "world's highest motorable pass"
Chang La	Ladakh	connects Pangong Tso to rest of Ladakh
Umling La	Ladakh	printed as 2nd highest motorable road
Zoji La	Ladakh–J&K	connects Ladakh and Kashmir valley; snow-blocked
Burzil La	J&K (now PoK)	trade
Banihal	J&K	Jawahar Tunnel (NH44)
Shinku La	HP	NPD route (Nimmu–Padum–Darcha), all-weather to Ladakh
Baralacha La	HP	Leh–Manali highway
Rohtang	HP	Atal Tunnel under it; Leh connectivity
Shipki La	HP	India–China border; Sutlej enters India
Mana	Uttarakhand	highest motorable pass of Uttarakhand
Niti	Uttarakhand	northernmost pass of Uttarakhand
Lipulekh	Uttarakhand	Nepal claims it, handout says India's; Mansarovar Yatra
Nathu La	Sikkim	Indo–China trade; Mansarovar Yatra
Jelep La	Sikkim	connects India and Bhutan
Bum La	Arunachal	Dalai Lama entered India 1959
Sela	Arunachal	Sela Tunnel; connects Tawang
Bomdila	Arunachal	Tawang connectivity; used in 1962 war
Yongyap	Arunachal	near entry of Dihang; British Stilwell Road
Diphu	Arunachal	tri-junction India–China–Myanmar
Chaukan, Pangsau	Arunachal	Indo–Myanmar; Stilwell Road passes Pangsau
Haldighati	Rajasthan	Aravallis; Maharana Pratap vs Mughals
Asirgarh / Burhanpur	MP	Satpuras; Asirgarh fort; key to the Deccan (north–south link)
Thalghat	Maharashtra	Mumbai–Nashik highway (Western Ghats)
Bhorghat	Maharashtra	Mumbai–Pune highway
Chorla Ghat	Goa	near Goa–Maharashtra–Karnataka junction
Palghat (Palakkad)	Kerala–TN	widest gap in Western Ghats; monsoon gateway to TN; between Nilgiris (north) and Annamalai (south)
Shenkotta	Kerala–TN	Western Ghats gap

NORTHERN PLAINS

Northern plains

→ Great depression south of Himalaya filled by Indus, Ganga, Brahmaputra alluvium; 3,200 km W to E

→ Flat, slight slope north to south; one of the most densely populated regions; rivers give irrigation, transport, hydropower; Delhi, Varanasi, Patna

→ Climate: semi-arid (west) → humid subtropical (east)

→ Problems: soil erosion, water pollution, groundwater depletion

📜 Four sections

→ Rajasthan plain: bagar, saline lakes

→ Punjab–Haryana: Bari and Bist doab, canals

→ Ganga plain: Yamuna eastward; Rohilkhand, Awadh, Mithila, Magadha, Bengal; delta

→ Brahmaputra plain: chars / point bars, floods, tea, Sundarbans delta

Term	Meaning
Bhabar	alluvial cones at foothills; rivers vanish
Terai	swampy; rivers re-emerge
Bhangar	older coarse alluvium away from channel; kankar nodules; less fertile
Khadar	newer alluvium near rivers; renewed yearly; very fertile
Reh / Kallar	barren saline patches (UP, Haryana); spread by over-irrigation
Bhur	elevated land along the Ganga (Ganga–Yamuna doab)
Bagar	semi-arid plain from Aravalli foothills, Rajasthan; Rohi = fertile patches in it
Doab	between two rivers: Bari, Bist, Rechna, Chaj
Betlands	Khadar plains of Punjab

Old Bhangar, New Khadar — Bhang-ar = bhang, old and coarse, kankar. Khad = manure: renewed every year, super fertile.

INDIAN DESERT: THAR

Thar (Great Indian Desert; local Marusthali)

→ Cause: sub-tropical high pressure + rain shadow of Aravallis; extension of Rajasthan plains

→ ~2,00,000 km²; 85% India, 15% Pakistan; Rajasthan, Gujarat, Punjab, Haryana

→ Borders: Indus plain (west) · Punjab plain (north, NE) · Aravalli (SE) · Rann of Kutch (south); natural India–Pakistan border

→ Features: dunes (some >150 m), sandy plains (Indira Gandhi Canal), low barren hills bhakars, seasonal "rivulets", saline lakes Sambhar, Didwana

→ Source of salt, limestone, gypsum

🎯 PENINSULAR PLATEAU PYQ 2023

Peninsular plateau

→ Oldest and largest unit; irregular triangle; broken from Gondwanaland; rich in metallic minerals; includes Deccan Traps

→ Meghalaya plateau cut off by Rajmahal–Garo (Malda) Gap / Ganga Suture Zone during northward drift

→ Two halves: Central Highlands and Deccan Plateau, split by the Narmada rift

Part	Where / what
Mewar / Marwar	NW, Rajasthan, near Aravallis; scrub
Malwa	Aravallis (NE) to Bundelkhand (E) to Vindhyas (S); Chambal badlands; black soil, cotton

Bundelkhand	between Vindhyas and Ganga plain (MP, UP); sal, teak, kheria
Baghelkhand	north Deccan, MP/CG; forests; coal
Chotanagpur (Eastern Highlands)	Hazaribagh, Ranchi, Rajmahal Hills; coal, mica; sal
Meghalaya (NE tip)	Garo, Khasi, Jaintia, Mikir, Rengma; Mawsynram wettest on Earth
Deccan	largest, central; avg 600 m; Deccan Traps; Chhattisgarh plateau and plains (Dandakaranya, Bastar; Mahanadi rice bowl)

Range	Facts (as printed)
Aravalli	relict of world's oldest mountains · NE edge = Delhi Ridge · highest Guru Shikhar · Gujarat, Rajasthan, Haryana, Delhi
Vindhyas	Jobat (Gujarat) → Sasaram (Bihar); east as Bharner and Kaimur · Gujarat, MP, UP, CG, Bihar
Satpuras	higher than Vindhyas · Rajpipla, Mahadev, Maikala · Dhupgarh highest · Gujarat, Maharashtra, MP, CG
Western Ghats (Sahyadris)	Satmala, Harishchandra, Balaghat, Ajanta, Baba Budan, Nilgiri, Annamalai, Cardamom, Palani · climatic AND drainage divide · 6 states: Gujarat, Maharashtra, Goa, Karnataka, Kerala, TN
Eastern Ghats	Garjhat, Mahendragiri, Eramalla, Nallamalla, Seshachalam, Velikonda, Palkonda, Javadi, Shevroy, Kalrayan, Biligiri Rangan · highly dissected by rivers · highest Jindhagada (Araku) · Odisha, Telangana, AP, Karnataka, TN
Meeting point	Eastern and Western Ghats meet at the Nilgiris

Thal-Bhor-Chorla-Pal-Shenkotta — Western Ghats gaps north → south: Thalghat · Bhorghat · Chorla · Palghat · Shenkotta. Ghats compare: Western = Wall (continuous, high, wet windward, divide); Eastern = Entry-gate pieces (broken by rivers, low, drier, scrub).

COASTAL PLAINS PYQ 2022

Coast	Facts
Western plain	mixed shoreline (emergence, submergence, stable)
Kachchh	emergence (tectonic + depositional) → Rann of Kachchh salt marsh; Banas and Luni inland drainage; Banni grassland (one of India's largest)
Kathiawar (Saurashtra)	Deccan Trap black soil; Gir: only Asiatic lions; Narmada, Tapi form estuaries
Konkan (Daman–Goa)	submergence; wider shelf; sea cliffs, marine erosion
Karnataka plain (Goa–Mangalore)	submergence
Malabar (Mangalore–Kanyakumari)	stable (deposition); kayals = lakes, lagoons, backwaters
Eastern plain	Subarnarekha → Kanyakumari; emergence by deposition (long rivers into Bay of Bengal); Northern Circars (Mahanadi–Krishna) + Coromandel (south of Krishna)
Utkal / Andhra / TN	Chilika (biggest lake, Utkal) · Kolleru (once lagoon, now inland) + Krishna–Godavari delta · plain ends at Pulicat → Kanyakumari

Eastern and western coasts meet at Kanyakumari. ⭐

ISLANDS PYQ 2019 PYQ 2018

Islands	Facts
Andaman and Nicobar	500+ islands, 6°–13°N, 92°–93°E, Bay of Bengal · extension of Arakan Yoma (plate collision; a few volcanic) · Saddle Peak highest · 10° Channel separates Andaman and Nicobar · 6° Channel: Great Nicobar–Sumatra · Barren Island = only active lava volcano in South Asia (Indian plate under Burmese); Narcondam extinct · Baratang = only active mud volcano · Havelock (Swaraj Dweep) largest
Lakshadweep	36 coral islands and reefs, ~400 km off Kerala, Arabian Sea · coral atolls on submerged volcanic Lakshadweep–Chagos Ridge; low, sandy, no rivers or permanent freshwater · Andrott largest · Minicoy (Maliku) southernmost, split by 9° Channel

10 - 9 - 6 — Channels = own number: 10° (Andaman | Nicobar) · 9° (Lakshadweep | Minicoy) · 6° (Great Nicobar | Sumatra): count down from the north.

🎯 PYQ BANK: PHYSIOGRAPHY

Year	Asked	Answer / trap
2012	Young fold mountains: deep gorges, U-turn courses, parallel ranges, steep gradients	D, all four (folding of crust; highest ranges; steep slopes = recent; old fold mountains are lower)
2014	Pairs: Cardamom–Coromandel? Kaimur–Konkan? Mahadeo–Central India? Mikir–NE?	C: Mahadeo and Mikir only. Kaimur = eastern Vindhya (~483 km, Katangi near Jabalpur → Sasaram, Rohtas; Rewa, Mirzapur). Cardamom (Yela Mala) = south W. Ghats, SE Kerala + SW TN, UNESCO
2014	Ten Degree Channel separates	A Andaman and Nicobar
2015	Easternmost and westernmost states	D Arunachal and Gujarat
2016	Place–region: Khajuraho–Bundelkhand ✔, Tirupati–Rayalaseema ✔; Bodhgaya–Baghelkhand ✘, Shirdi–Vidarbha ✘, Nashik–Malwa ✘	C (2 and 5). Malwa = W MP + SE Rajasthan; Baghelkhand = NE MP + small W UP; Vidarbha = E Maharashtra (Nagpur, Amravati divisions)
2017	Kohima → Kottayam: minimum states	B = 7 (Nagaland, Assam, WB, Odisha, AP, TN, Kerala)
2017	Himalaya in 5 states only? W. Ghats in 5 only? Pulicat in 2 only?	B: only 3 right. Himalaya >5 states (J&K, HP, Uttarakhand, Sikkim, Arunachal + hill parts of Assam, WB); W. Ghats 6 (30–50 km inland); Pulicat = AP (40%) + TN (60%), 2nd largest brackish lagoon after Chilika, Sriharikota barrier island (Satish Dhawan Space Centre)
2018	Barren Island: active ✔; ~140 km east of Great Nicobar ✘; last erupted 1991, inactive since ✘	D (1 only). Erupting again since 1991, last as recent as 2022
2018	City on longitude nearest Delhi	A Bengaluru (both 77°E)
2020	Siachen Glacier is…	D north of Nubra Valley (eastern Karakoram, ~70 km / 44 mi)
2022	Namcha Barwa–Garhwal ✘ · Nanda Devi–Kumaon ✔ · Nokrek–Sikkim ✘	B (2 only). Namcha Barwa in SE Tibet inside Great Bend of Yarlung Tsangpo; Nokrek biosphere on Tura Range, Meghalaya Plateau
–	Repeated falls in sea level → marshland	D Rann of Kutch (shoreline of emergence)
2023	Amarkantak = Vindhya + Sahyadri? Biligiri = easternmost Satpura? Seshachalam = southernmost W. Ghats?	D none: Amarkantak = Vindhya + Satpura; Biligiri Rangan in SE Karnataka; Seshachalam = Eastern Ghats
–	Peninsular Block: W-coast submergence ✔; residual hills (Velikonda, Mahendragiri) ✔; deep V-valleys ✘	B: ancient stable craton; Eastern Ghats denuded; peninsular rivers mature
–	UP has most neighbouring states · Rajasthan longest int'l border · Sikkim only one-neighbour state	A as printed (see conflict)

Source conflict: last PYQ prints answer A (statement 1 not correct), but its explanation says West Bengal has the longest international border and Meghalaya also has only one neighbouring state, which would make statements 2 and 3 wrong. Check atlas / class notes.

🎯 3 · DRAINAGE

PYQ 2011 · PYQ 2013 · PYQ 2014 · PYQ 2015 · PYQ 2016 · PYQ 2017 · PYQ 2021 · PYQ 2022 · PYQ 2023

Factors governing drainage

- → Topography (slope, relief)
- → Nature of rocks (basalt low permeability; granite impermeable)
- → Amount of water (rain + snow)
- → Periodicity (perennial / seasonal)
- → Geological time
- → Anthropogenic: deforestation, urbanisation, dams

⏳ Terms

- → Catchment (drainage basin / watershed): whole land area draining to one point
- → Drainage divide: elevated landform separating catchments
- → Basin size: major >20,000 km² · medium 2,000–20,000 · minor <2,000
- → Inland drainage: does not reach sea (Kachchh, shoreline of emergence)

Type	Meaning	Example
Consequent	flows down original slope	Himalayan streams
Subsequent	tributary on softer rock	Appalachian
Antecedent	river older than the uplift; cuts gorges	Indus, Brahmaputra (through Himalaya); Ghaghara
Superimposed	pattern from older landscape laid onto new rocks	Subarnarekha
Concordant / Discordant	follows structure (stable regions) / does not (uplift, erosion past)	

Big numbers

→ 77% drains to Bay of Bengal · 23% to Arabian Sea
→ Divide: Delhi Ridge – Aravallis – Vindhyas – Satpuras – Western Ghats
→ Himalayan rivers 40–60 mn yrs; peninsular 200 mn yrs
→ Basin areas: Indus 11.78, Ganga 8.61, Brahmaputra 5.8 lakh km²; Godavari 3.12 lakh km² = largest peninsular basin

	HIMALAYAN rivers = teenagers	PENINSULAR rivers = grandparents
Age, valley	young, V-valleys; shift channels (avulsion, e.g. Kosi); origin: Shiwalik river divided into Indus and Brahmaputra	mature (~200 mn yrs), broad shallow graded valleys
Flow	perennial (snow + rain); meanders; big deltas; flooding	seasonal (except Kaveri); straight courses; small basins; Narmada and Tapi make estuaries, others deltas

HIMALAYAN RIVERS: INDUS PYQ 2021

Indus

→ Origin Bokhar Chu, Kailash; 3,180 km, rises W Tibet; in India only Ladakh
→ Riparian: Ladakh, J&K, HP, Punjab; gorge near Nanga Parbat
→ Panjnad = I–J–K–R–B–S (left tributaries, as printed)
→ Right bank: Shyok (Rimo glacier, Siachen tongue; joins at Keris, PoK), Gilgit · Others: Zanskar, Suru

River	Origin / hook
Jhelum	Verinag; meanders; Wular Lake; joins Chenab in Pakistan
Chenab	near Baralacha La (Lahaul–Spiti); joins Sutlej in Pakistan
Ravi	west of Rohtang, Kullu hills; joins Chenab in Pakistan
Beas	Beas Kund near Rohtang; gorges at Kati and Largi (Dhauladhar); entirely in India; meets Sutlej in Punjab
Sutlej	Rakshasthal near Mansarovar; enters India at Shipki La; joins Indus in Pakistan as Panjnad

GANGA SYSTEM

Ganga

→ Origin: Gangotri (Bhagirathi) + Satopanth (Alaknanda); Uttarakhand, UP, Bihar, Jharkhand, WB; Padma in Bangladesh
→ NW-1: Haldia → Prayagraj

Panch Prayag (up → down)

→ Vishnu · Nand · Karna · Rudra · Dev

Feeders

→ Dhauli (Ganga) · Nandakini · Pindar · Mandakini · Bhagirathi

Vasudev Ne Karna ko Rasta Dikhaya – Panch Prayag: Vishnuprayag · Nandprayag · Karnaprayag · Rudraprayag · Devprayag. Feeders: Dhul Nandakini ke Pair Mein Bikhri = Dhauli(Ganga) · Nandakini · Pindar · Mandakini · Bhagirathi.

Rama Ghoomte hue Ghaghara Ganda Kar Mat – LEFT-bank tributaries: Ramganga · Gomti · Ghaghara · Gandak · Kosi · Mahananda (+ Burhi Gandak between Gandak and Kosi).

Left bank	Facts
Ramganga	Chamoli; meets Ganga at Kannauj
Gomti	Pilibhit; meets Ganga at Ghazipur
Ghaghara	Tibet; antecedent; joins in Bihar; tribs Sarda, Saryu, Rapti, Choti Gandak
Gandak	Tibet–Nepal border; Kali Gandak, Mayangadi, Trishuli; joins near Patna
Burhi Gandak	West Champaran; Kanwar Tal (India's biggest oxbow; Sikrahna)
Kosi	Sapt Kosi (7 tribs: Arun, Tumar, Sun Kosi); Sorrow of Bihar (shifting course); Nepal; joins near Kursela
Mahananda	near Tiger Hill; "last major right-bank tributary" to join Ganga in India (handout wording)
🛡️ Right bank	Yamuna (Yamunotri; joins at Prayagraj; Chambal, Sind, Ken, Betwa) · Tons/Tamsa · Son (Amarkantak, radial drainage with Narmada, Johila; joins near Patna) · Phalgu (Gayaji)
Distributary	Hooghly (Giria → Sagar Island, estuary) · Damodar (Chotanagpur → Hooghly near Kolkata) · Mayurakshi, Ajay

Source conflict: Mahananda sits in the left-bank list but is described as the "last major right bank tributary". Treat it as printed (left list).

🎯 BRAHMAPUTRA AND BARAK PYQ 2011 PYQ 2014 PYQ 2016 PYQ 2017

Brahmaputra ("son of Brahma"; Luit, Dilao)

→ Source Chemayungdung glacier (also Angsi), SW Tibet, Kailash; near sources of Indus and Sutlej; Mariam La separates it from Manasarovar; Indus-Tsangpo Structure Zone between Great Himalaya and Kailas Range

→ U-turn at Namcha Barwa → enters as Dihang (Siang); Lohit and Dibang join; Sadiya → Dhubri

→ Name chain: Tsangpo – Dihang – Brahmaputra – Jamuna – Padma – Meghna

→ Majuli island where Subansiri joins; Assam floods

Tributaries

→ Right: Subansiri · Kameng · Manas · Beki · Raidak · Jaldhaka · Teesta (+ Sankosh)

→ Left: Lhasa · Nyang · Parlung Zangbo · Lohit · Dibang · Burhi Dihing · Dhansiri · Kolong

SKM-ST – Brahmaputra right-bank tributaries: Subansiri · Kameng · Manas · Sankosh · Teesta.

Teesta (2017)

→ 315 km; Pahunri (Teesta Kangse) glacier, above 7,068 m; Sikkim, West Bengal, Bangladesh, Bay of Bengal

→ Fed by Thangu, Yumthang, Donkha ranges; Rangpo joins; border of Sikkim–WB to Teesta Bazaar

→ Main tributary Rangeet (originates in Sikkim)

→ Source is not the same as Brahmaputra (originates in Sikkim)

Barak

→ 900 km; Manipur hills; enters plains near Lakhipur; Manipur, Nagaland, Mizoram, Assam

→ Splits into Surma + Kushiyara → Meghna (after meeting Padma, Ganga + Brahmaputra flow)

→ Separated from Brahmaputra by Barail Hills; inland waterway; rich flora and fauna

🎯 PENINSULAR DRAINAGE

PYQ 2013 PYQ 2015 PYQ 2021 PYQ 2022 PYQ 2023

Peninsular rivers

→ By direction: East-flowing Mahanadi, Godavari, Krishna, Kaveri · West-flowing Narmada, Tapi · NE-flowing Chambal, Ken

→ ~200 mn yrs, mature, broad shallow graded valleys

Seonath Man Hasdo On the Joke I Told — Mahanadi: left Seonath · Mand · Hasdo · right Ong · Jonk · Tel · and Ib (left). ("I" = Ib, "Told" = Tel.)

River	Facts
Mahanadi	Sihawa, Raipur (CG); CG + Odisha (53% / 47% as printed); Hirakud Dam; gorge near Tikkarpara; delta
Godavari	Longest peninsular; Vridha / Dakshin Ganga; 2nd longest river of India after Ganga; Trimbakeshwar (Nashik); Maharashtra, Telangana, CG, AP, Yanam · right: Manjara, Pravara, Manair (16.1% of basin) · left: Purna, Pranhita (Wainganga + Penganga), Indravati, Sabari (59.7%); below Rajahmundry splits into Gautami + Vashishtha; KG delta
Krishna	Mahabaleshwar; Maharashtra, Karnataka, Telangana, AP · right: Koyna, Panchganga, Dudhganga, Ghataprabha, Malaprabha, Tungabhadra · left: Bhima, Dindi, Halia, Musi
Kaveri	Brahmagiri (Kodagu); perennial (SW monsoon upper + NE monsoon lower); Kerala 3%, Karnataka 41%, TN 56% (as printed); delta; splits at Tiruchirapalli into Kollidam + Kaveri, rejoin at Srirangam island · right: Lakshmana Tirtha, Kabini, Suvarnavathi, Bhavani, Noyyal, Amaravati · left: Harangi, Hemavati, Shimsha, Arkavati
🧑‍🤝‍🧑 Others (east)	Brahmani (S. Koel + Sankh near Rourkela; rises near Nagri, Ranchi, ~600 m; Eastern Ghats start south of it) · Subarnarekha (Ranchi plateau, Nagri ~600 m; Odisha–WB boundary; estuary) · Pennеru (Nandi Durg, Chikballapur; Gandikota canyon, Kadapa, "Grand Canyon of India", Erramala hills; estuary) · Ponniyar, Vaigai (short) · Vamsadhara (Kalahandi, Eastern Ghats; between Rushikulya and Godavari; Kalingapatnam) · Nagavali (Lakhbahal, Kalahandi, 1,300 m; Langulya)
Narmada	Amarkantak (Maikala, ~1,080 m); rift valley between Vindhya and Satpura, few tributaries (short, trellis); Marble Rocks Gorge; estuary ~21 km wide below Bharuch, Gulf of Khambhat; Hiran, Kolar, Hatni, Choti Tawa; MP, Maharashtra, Gujarat; meander near Bharuch
Tapi	"Narmada's twin"; Multai (MP); Purna, Girna, Panjara; Satpura north, Mahadev east, Ajanta/Satmala south
Luni	west of Ajmer; Sagarmati → Luni after Sarsuti joins; ephemeral; best inland-drainage example in the world → Rann of Kachchh; Rajasthan's biggest system
Sabarmati	Aravallis → Gulf of Khambhat; gorge at Dharoi
Mahi	Vindhyas (MP) → Gulf of Khambhat; only river to cross the Tropic of Cancer twice
🧑‍🤝‍🧑 Short west rivers	parallel drainage off W. Ghats, no deltas: Gujarat (Auranga, Par, Daman Ganga) · Goa (Mandovi, Zuari, Rachol) · Maharashtra (Ulhas, Kalindi) · Karnataka (Kalinadi, Sharavati – Jog Falls, Tadri, Netravathi) · Kerala (Bharathapuzha, Periyar, Pamba)
Inland	Ghaggar-Hakra (Ghaggar in India, Hakra in Pakistan after Ottu barrage; ends in Thar) · Luni, West Banas (Rajasthan → Gujarat → Rann of Kachchh)

River-linking project

→ Idea: Sir Arthur Cotton (19th c.); revived 1960s, momentum 1980s (scarcity, floods)

→ NWDA set up 1982 to study feasibility; Supreme Court intervened several times

→ Himalayan component 14 links, peninsular 16 links; some states propose intra-state links; under debate

→ Ken-Betwa pilot: cabinet approval 2014, hurdles (environment, inter-state)

→ Pattiseema lift scheme (Godavari → Krishna; Polavaram right canal to Krishna at Vijayawada), inaugurated 2016

State	Waterfall (height m) – river
Karnataka	Kunchikal 455 (Varahi) · Barkana 259 (Sita) · Jog 253 (Sharavati) · Magod 198 (Bedti) · Shivanasamudra 98 (Kaveri) · Hebbe 168 (Thunga) · Shimsha 46

Odisha	Barehipani 399 (Budhabalanga) · Khandadhar 244 (Korapani Nala) · Koilighughar 61 (Ahiraj)
Kerala	Meenmutty 300 (Kallar) · Soochipara 200 (Chaliyar) · Athirapally 25 (Chalakuddy → Periyar)
Chhattisgarh	Amritdhara 90 (Hasdeo) · Teerathgarh 91 (Munga) · Chitrakote 29 (Indravati)
MP	Chachai 130 (Bihad → Tons) · Dhuandhar 30 and Kapildhara 30 (Narmada) · Sonemuda 15 (Son)
Jharkhand	Lodh 143 (Budha) · Hundru 99 (Subarnarekha) · Panchghagh 45 (Banai)
Others	Nohsngithiang (Seven Sisters) 315, Meghalaya · Vantawng 229, Mizoram · Kempty 40 (Yamuna, Uttarakhand) · Dudhsagar 320 (Mandovi, Goa) · Thalaiyar (Rat Tail) 297 (Manjalar/Vaigai, TN) · Maharashtra: Thoseghar, Someshwar 10 (Godavari)

Tallest first — Kunchikal 455 (Karnataka) > Barehipani 399 (Odisha) > Dudhsagar 320 > Nohsngithiang 315 > Meenmutty 300 > Thalaiyar 297 > Jog 253.

Lake	State/UT	Hook
Wular	J&K	freshwater, part of Jhelum
Dal	J&K	Srinagar
Pangong Tso	Ladakh	high-altitude saline; Indo-China border
Tso Moriri	Ladakh	high-altitude saline
Renuka	HP	largest natural lake of HP
Roopkund	Uttarakhand	near Trishul; skeletons at bottom
Nainital / Bhimtal	Uttarakhand	natural freshwater / artificial FW
Gobind Ballabh Pant Sagar	UP	Rihand dam; largest artificial lake of India
Sambhar	Rajasthan	playa lake
Pushkar, Udaipur lakes	Rajasthan	artificial; Pushkar religious
Kanwar	Bihar	largest oxbow lake of Asia; Burhi Gandak
Lhonak, Cholamu	Sikkim	high-altitude glacial
Gurudongmar	Sikkim	highest lake of India
Deepor Beel	Assam	Brahmaputra basin
Loktak	Manipur	between west and east Manipur hills; phumdis; Keibul Lamjao NP, sangai deer

Named on map, no notes: Parshuram Kund, Nal Sarovar, Bhojtal, Lonar, Salim Ali, Husain Sagar, Kolleru, Pulicat, Stanley, Vembanad, Ashtamudi, Chilika.

🎯 PYQ BANK: DRAINAGE

Year	Asked	Answer / rule
2011	Why Brahmaputra makes a U-turn	B syntaxial bending of young Himalaya. Tethys ~70 mn yrs ago, receded from early Eocene; maximum push from Aravalli (W) and Shillong Plateau (E); Arakan Yoma bends near Namcha Barwa, Sulaiman near Nanga Parbat (hairpin)
2013	Why Narmada flows west	A linear rift valley only (not "slope west")
2014	Through Arunachal: Barak, Lohit, Subansiri	B Lohit + Subansiri; Barak = Manipur, Nagaland, Mizoram, Assam
2015	Godavari tributaries: Vamsadhara, Indravati, Pranahita, Pennar	D Indravati + Pranahita
2016	Brahmaputra tributaries: Dibang, Kameng, Lohit	D all three
2016	Recent river linking	B Godavari–Krishna (Pattiseema)
2017	Teesta: same source as Brahmaputra? Rangeet a tributary? Flows into Bay on the border?	B only 2 right
2021	Which joins the Indus directly: Chenab, Jhelum, Ravi, Sutlej	D Sutlej (others join the Sutlej system first)
2021	Rise from Eastern Ghats: Brahmani, Nagavali, Subarnarekha, Vamsadhara	B Nagavali + Vamsadhara (Brahmani, Subarnarekha rise on Ranchi plateau)
2022	Gandikota canyon	C Pennar
2023	Jhelum through Wular ✔; Krishna feeds Kolleru ✘; Gandak meander → Kanwar ✘	A only one. Kolleru lies between Krishna and Godavari; Kanwar = Burhi Gandak (Sikrahna)
–	River: antecedent, 3 countries, Tibetan origin, irrigation, no distributaries	C Sutlej (Brahmaputra and Indus have distributaries; Teesta rises in Sikkim)

🎯 4 · INDIAN CLIMATE PYQ 2011 PYQ 2017

Monsoon climate

→ ~75% rain in monsoon months; tropical south to very cold north, desert west to heavy rain east

Controls

→ Latitude ~6°N–35°N (tropical)

→ Himalayas: block Central Asian cold; shape monsoon

→ Pressure and winds: summer low pulls SW winds; winter high → dry NE winds; sub-tropical westerly jet → Western Disturbances

→ Ocean currents (Indian Ocean Equatorial Current: warm, humid coasts)

→ Continentality (inland extremes)

→ Relief: W. Ghats barrier, rain shadows

WiSAR — Four seasons: Winter (mid-Nov–Feb) · Summer / pre-monsoon · Advancing monsoon (Jun–Sep) · Retreating monsoon (Oct–Nov).

Winter (mid-Nov–Feb)

→ Fog (can turn to smog), inversion, pollution domes, frost, cold waves in north

→ Offshore NE trade winds

→ WD rain = "Mawath / Mahawat" (Mewat) vital for Rabi

→ Fewest tropical cyclones (ITCZ shifts, high pressure on land and ocean)

Western Disturbances

→ Extratropical storms from the Mediterranean; steered east by the sub-tropical westerly jet; lose moisture on the Himalaya

→ Winter and pre-monsoon rain in Punjab, Haryana, UP, Delhi; Himalayan snow (feeds Ganga / Indus perennial flow); moderate winter cold; hail, thunderstorms

→ Climate change: possibly more intense (warmer, moister air), shifted timing/path, now seen in summers, interacting with cyclones and monsoon → extreme rain

Summer

→ Sun north → ITCZ shifts (~25°N)
→ Loo, dust storms (Aandhi; WD can bring showers); heat waves
→ Pre-monsoon showers (warm continental + maritime air meet): 1% of annual rain; lots of cyclonic activity

Pre-monsoon showers: what they help — Mango showers (Karnataka/Kerala) · Coffee / Cherry-blossom showers (Karnataka, Kerala) · Norwester / Kaal Baisakhi (WB, Bihar, Jharkhand, Odisha: rice, tea, jute) · Bardoli Chheerha / Bordoisila = tea showers (Assam).

Cyclones in India

→ Coast >7,500 km (as printed); season April–Dec, peaks May–June and Oct–Nov
→ Bay : Arabian Sea ≈ 4 : 1 (bigger warm-water area, lower wind shear); 5–6 per year, 2–3 severe
→ Climate change: intensity up, not necessarily frequency; Arabian Sea activity rising; 2022 study: Bay numbers may fall, intensity rise

Naming

→ WMO/ESCAP Panel on Tropical Cyclones, since September 2004
→ 13 countries × 13 names: Bangladesh, India, Iran, Pakistan, Oman, Maldives, Myanmar, Sri Lanka, Thailand, UAE, Yemen, Qatar, Saudi Arabia
→ RSMC New Delhi assigns; names retired after use; a South China Sea cyclone entering the Bay keeps its name

IMD colour warning	Lead time
Pre-cyclone watch (no colour)	72 h
Cyclone Alert: Yellow	≥ 48 h
Cyclone Warning: Orange	≥ 24 h (landfall point forecast)
🧍‍♂️ Post-landfall outlook: Red	≥ 12 h

Mnemonic (handout): 72 – 48 – 24 – 12, halving each step: Watch → Yellow → Orange → Red. ⭐

Urban heat island

→ Urban areas much hotter than rural surroundings
→ Causes: concrete/asphalt, less vegetation, vehicle/industrial/AC heat, anthropogenic energy, urban geometry ("heat canyons")

🌐 Advancing monsoon (Jun–Sep; "mausim"; seasonal wind reversal; 75% rain)

Drivers

→ Land–sea differential heating (primary)
→ ITCZ shifts north
→ Sub-tropical westerly jet (weaker/northward = stronger monsoon)
→ Tibetan Plateau heating → Tropical Easterly Jet (TEJ)
→ Somali jet (moisture across Arabian Sea)
→ ENSO · IOD · MJO
→ Arctic sea ice (less ice → more rain in central and NE India, less in west and south peninsula; mechanism unknown)

→ Monsoon trough = low over India attracting SW winds
→ Burst = sudden onset with thunder, lightning, downpour; ~1 June, Kerala

Two branches

→ Arabian Sea (stronger): Malabar, Konkan, Kathiawar

→ Bay of Bengal (blocked by Arakan): Utkal and North-Eastern

→ Meet around Agra–Ferozepur

→ Tamil Nadu dry: rain shadow of W. Ghats; branches parallel to its coast

→ Break: usually August: weak Tibetan Low + winds parallel to west coast; flooding common

Cloudbursts

→ ≥ 10 cm rain in an hour over a small area; flash floods, landslides

→ Orographic lift + rapid condensation + cold-air downdraft; more in mountains, also plains

→ Himalaya, Western Ghats; monsoon; hard to predict (dense radar, high-res models); more with warming (warmer air holds more moisture)

Urban flooding

→ Impervious surfaces (concrete, asphalt) + overwhelmed drains

→ Mumbai 2005 (944 mm in a day) · Chennai 2015 (drainage, water-body encroachment) · Bengaluru (unplanned growth) · Hyderabad (lake encroachment) · Mumbai mangrove loss · heavy rain in Mumbai/Kolkata linked to climate

El Nasty / La Nice — El Niño = El Nasty (warm east Pacific → weak SW monsoon, drought in India/Australia; NE monsoon stronger). La Niña = La Nice (cold east Pacific → good monsoon, Australian floods; NE monsoon weaker).

Phenomenon	What it is	India link
El Niño	trades weaken; warm water sloshes east; Humboldt (Peruvian) current replaced, less upwelling	weak SW monsoon; Australia drought/fire; Peru floods; California storms
La Niña	trades strengthen; cold east Pacific; more upwelling	better monsoon; floods in Australia, E. Africa, SE Asia
ELLA years	transition between El Niño and La Niña (IMD–IITM Pune study)	severe, prolonged summer heat; more heatwaves in NW and central India; early identification helps forecasting
El Niño Modoki	"similar but different" (Japanese); warmth in central Pacific (dateline El Niño), cooler east and west	different global pattern
PDO	North Pacific; each phase 20–30 yrs (ENSO = 2–7 yrs); positive = warm east	modulates ENSO links to monsoon; amplifies or weakens El Niño/La Niña
IOD	west–east SST gap in the Indian Ocean	Positive IOD (warm west near Arabian Sea, cool east) → stronger monsoon, even in El Niño years; negative → weaker, more break days
Atlantic Niño / Niña (Atlantic Ocean Dipole / Zonal Mode)	Niño: east equatorial Atlantic warms, west cools; Niña = reverse	not as wide-ranging as Pacific
Arctic Ocean Dipole	Arctic sea-level pressure pattern (Arctic Dipole Anomaly)	–
MJO	eastward cloud-rain pulse, 30–60 days; active (wet) vs suppressed (dry)	active phase enhances monsoon; modulates cyclone formation; shorter cycle = better monsoon impact

Retreating monsoon (Oct–Nov)

→ ITCZ shifts; pressure rises over north India; "October heat" (humid, clear)

→ NE monsoon rain: coastal Andhra + most of Tamil Nadu (NE trades over Bay); El Niño: NE stronger; La Niña: NE weaker, SW stronger

→ Season of most severe cyclones in Bay (hot sea, trough moves south, wind change)

Rainfall map

→ >200 cm: windward W. Ghats (S. Gujarat → Kerala), NE (Assam, Meghalaya, Arunachal; funnelling: Cherrapunji, Mawsynram), Andaman and Nicobar

→ 100–200: Himalayan foothills, east India, east TN

→ 50–100: parts of Deccan, NW India

→ <50: west Rajasthan, W. Ghats rain-shadow, Ladakh

Year	Asked	Answer
2011	La Niña vs El Niño: (1) La Niña = cold equatorial Indian Ocean? (2) La Niña no effect on monsoon?	D neither: La Niña = cooling of tropical west coast of S. America (Peru/Chile cold current returns); helps monsoon; Australia floods
2017	IOD: (1) W Indian Ocean vs Eastern Pacific? (2) can influence El Niño's effect	B only 2: IOD = W (near Arabian Sea) vs E (near Bay of Bengal) Indian Ocean
–	Andaman and Nicobar climate: humid tropical coastal ✔; SW + NE monsoon ✔; max rain Dec–May ✘	B (1 and 2); max rain May–Sept (SW monsoon); ocean winds onshore both ways

🎯 5 · FORESTS PYQ 2015 PYQ 2016 PYQ 2023 PYQ 2011

Forest	Climate	Where	Trees / traits
Wet evergreen	>200 cm; 25–27 °C	W. Ghats (south of Mumbai), NE, A&N	no undergrowth (dense canopy); jackfruit, rosewood, mahogany, aini, ebony, agarwood
Semi-evergreen	175–225; 24–27 °C	W. coast, Assam, lower E. Ghats, Odisha, Andamans	wet evergreen + moist deciduous mix; evergreen due to undergrowth; kadam, cashew, cedar, chestnut, thorny bamboo
Dry evergreen	~100 cm all year	TN coast	short; tamarind, neem, toddy palm
🧍‍♂️ Moist deciduous	100–200; ~27 °C	around W. Ghats, NE, Shiwalik (Terai-Bhabar), E. MP, CG, Chotanagpur, Odisha	shrubby undergrowth; teak, sal, mahua, tendu, bamboo (also shisham, sandalwood, khair, kusum, arjun, mulberry); leaf-fall 6–8 weeks; humidity 60–75%
Dry deciduous	75–100	rainier peninsula, plains of UP, Bihar, MP, Rajasthan	transition to thorn; teak, tendu, red sanders, sal, khair
Thorn	<50; 25–30 °C	Rajasthan, SW Punjab, W. Haryana, Kachchh, Saurashtra, some UP, lee of W. Ghats (MH, Karnataka, Telangana, AP, TN)	low scattered; acacia, babul, neem, khair, cacti, wild date
Littoral and swamp (mangroves)	brackish water	Sundarbans (Sundri), A&N, Mahanadi–Godavari–Krishna deltas	viviparity, pneumatophores; sundri, keora, agar, rhizophora; ~7% of world mangroves; fresh + brackish

Rainfall ladder (cm) – >200 wet evergreen → 175–225 semi-evergreen → 100–200 moist deciduous → 75–100 dry deciduous → ~100 spread through year dry evergreen (TN coast) → <50 thorn. Wetter = evergreener, taller; drier = shorter, thornier.

🛡️ Mangroves protect coast

→ Erosion, storm surge, tsunami (2004 PYQ); trap sediment (protect coral, seagrass)

→ Fisheries nurseries; rot-resistant timber, fodder, medicine; tourism

→ Handout alert: capacity to adapt to rising seas is being compromised; roots salt-tolerant (not salt-sensitive), do not store freshwater

Mountain forests

Himalaya (north)

→ base moist deciduous → 1,000–2,000 m wet temperate with pine (British replaced oak with pine for railway sleepers)

→ up to ~3,000 m blue pine, spruce, grassland

→ 3,000–4,000 m silver fir, juniper, rhododendron + bugyals → tree line → alpine tundra → snow line; northern slopes little vegetation

Western Ghats (south) → evergreen base → temperate evergreen "sholas" (Nilgiris, Anamalai, Palani) both slopes; no alpine tundra (height limit)

🎯 Andaman and Nicobar forests (2015 PYQ)

→ ~2,200 plant varieties, 200 endemic, 1,300 not on mainland; forest cover 86.2%

→ South Andaman: epiphytes (ferns, orchids) · Middle: moist deciduous · North: wet evergreen + woody climbers

→ North Nicobar: no evergreen; central/south Nicobars mostly evergreen; grasslands only in Nicobars

→ Mangroves 1,136 km² (Andaman) + 35 km² (Nicobar)

Forest status	Figure (handout)
🧑‍🤝‍🧑 Forest + tree cover	8,27,357 km² = 25.17% (target 33% for stability)
🧑‍🤝‍🧑 Forest / tree cover	7,15,343 km² = 21.76% / 1,12,014 km² = 3.41%
Change vs 2021	+1,445 km² (forest +156, tree +1,289); lower than +1,540 in 2021
Mangrove	4,992 km² (−7.43 km²) · bamboo area +5,227 km²
🗺️ States	>33% forest: 19 states/UTs · >75%: 8 (Mizoram, Lakshadweep, A&N, Arunachal, Nagaland, Meghalaya, Tripura, Manipur)
🧑‍🤝‍🧑 Highest % cover	Lakshadweep 91.33 · Mizoram 85.34 · A&N 81.62
🧑‍🤝‍🧑 Largest area	forest + tree: MP, Arunachal, Maharashtra · forest: MP, Arunachal, Chhattisgarh · bamboo: MP, Arunachal, Maharashtra
Max increase	total: Chhattisgarh, UP, Odisha · forest: Mizoram, Gujarat, Odisha · tree: Gujarat, Bihar, Kerala
Megacities	Delhi has the highest forest cover among the six
Worries	decline in Andhra Pradesh, Maharashtra · Western Ghats ESA −58.22 km² in 10 years · NE −327.30 km²
🆕 New in 2023 report	agroforestry chapter · "forest characteristics" chapter back after 10 yrs · bamboo in tree cover · first estimate of 5–10 cm DBH small trees · 751 districts (636 in 2021)
FAO	India +2,66,000 ha/yr 2010–2020, 3rd (after China, Australia) · GFRA 2025: 9th in total forest area, 3rd in annual gain · Asia the only continent with forest gain since 1990

Social forestry

→ Increase tree cover + rural livelihoods through local people; term first used in India in 1976 by the National Commission on Agriculture

→ Types: community, urban (Miyawaki = dense native planting, Japanese), extension, agroforestry (trees with crops), farm forestry

→ Benefits: jobs, fuel/fodder, NTFPs, carbon, erosion control, habitat, less pressure on natural forests

🎯 6 · SOILS PYQ 2013 PYQ 2021 PYQ 2023

Soil formation factors → Relief · parent rock · climate · biota · time · humans

A Red Black Arid Laterite Saline Peaty Forest — ICAR's 8 soils: Alluvial · Red & yellow · Black · Arid desert · Laterite · Saline · Peaty/marshy · Forest & mountain. Pairs: Black = cotton + cracks (self-ploughing); Red = iron; Laterite = tea–coffee–cashew (acidic, leached); Alluvial = the breadbasket.

Soil	Where	Rich / poor	Crops and hooks
Alluvial	N. plains, coastal plains; >40% of India, most common	rich: potash, lime, phosphoric acid; poor: nitrogen, humus	sugarcane, paddy, wheat, pulses; Bhangar / Khadar

Black (Regur)	Maharashtra, Saurashtra, MP, CG; Godavari–Krishna; 3rd most common	rich: Mg, potash, iron, lime, alumina; poor: phosphoric, N, organic	cotton, sugarcane, pulses; cracks when dry; high water retention; volcanic (Deccan / Rajmahal trap lava, fissure volcanic rock)
Red and yellow	low-rain E and S Deccan; W. Ghat piedmont; yellow in parts of Odisha/CG	rich: iron, potash; poor: N, phosphorus, humus	crystalline igneous rocks; yellow = hydrated iron; wheat, cotton, pulses, tobacco, oilseeds, potato
Laterite	wetter margins; south, W. Ghats of Maharashtra, Odisha, WB, NE; also Karnataka, TN, MP, Bihar, Assam, Meghalaya	iron oxide + potash high (notes); poor: organic, N, phosphate, calcium	acidic; tea, coffee, cashew (tapioca: PYQ); rusty red
Arid	Rajasthan / Thar	little humus, N; phosphorus adequate; kankar	millets, fodder, guar; Indira Gandhi Canal
Saline (Usara)	arid/semi-arid Rajasthan, Punjab, UP, Haryana, Kachchh, Maharashtra	salts of Na, K, Mg	some coconut, salt-resistant fodder (White Revolution)
Peaty / marshy	heavy rain: Odisha, Sundarbans, Bihar, UP, Kerala	humus/organic 40–50%	rice if drained
🧍‍♂️ Forest and mountain	Himalaya, W and E Ghats	varies	valley sides loamy–silty (rice, wheat); upper slopes coarse (horticulture)

Source conflict: laterite is "rich in iron oxide and potash" in the soil table but "poor in nitrogen, potash, lime, organic matter; rich in iron and aluminium" in the 2013 explanation. Exam rule: "laterite rich in N?" = no; laterite is red, cashew + tapioca ✔.

Soil degradation

→ Erosion (deforestation, bad tillage, overgrazing)
→ Chemical (fertiliser/pesticide, salinisation, irrigation acidity)
→ Physical (mining, construction, machinery)
→ UNCCD: affects 15 million km² (larger than Antarctica), growing ~1 million km²/yr; excessive degradation → desertification

India: a sub-continent

→ Himalaya = natural border
→ Own plate: India = Gondwanaland; rest of Asia on Eurasian plate (Laurasia)
→ Almost all major climate and forest types (desert to tundra)
→ Distinct history and culture

🎯 PYQ BANK: FORESTS AND SOILS

Year	Asked	Answer / rule
2023	Rainforest soil nutrient-rich because of fast decomposition	D: I false, II true; soil leached and poor, trees take nutrients as fast as produced
2011	Mangroves as safety hedge (2004 tsunami)	D: extensive roots hold against storms and tides
2013	Laterite: red ✔; rich in N and potash ✘; Rajasthan/UP ✘; tapioca + cashew ✔	C (1 and 4)
2015	Region with mangrove + evergreen + deciduous	D Andaman and Nicobar
2015	Tropical wet evergreen in: Arunachal, HP, Mizoram	C Arunachal + Mizoram (also Assam, Meghalaya, Nagaland, Tripura, WB, A&N, W. Ghats slopes in TN, Karnataka, Kerala, Maharashtra)
2015	Teak dominant in	A tropical moist deciduous
2016	Red Sanders: found in a part of S. India ✔; important in tropical rainforest ✘	A (1 only): Pterocarpus santalinus, endemic to southern Eastern Ghats; wood not aromatic, not Santalum sandalwood; dry deciduous
2021	Black cotton soil forms from weathering of	B fissure volcanic rock
2023	Jackfruit, Mahua, Teak: how many deciduous?	B two (jackfruit evergreen)
–	Mangrove rationale for climate resilience	D bio-shield + livelihood; roots salt-tolerant, not salt-sensitive; do not store freshwater

Traps

→ "highest of..." lists are state-specific (Karoh Haryana, Amsot UP, Someshwar Bihar are Shiwalik; Kangto Arunachal; Shillong Peak Meghalaya)
→ Narmada flows west because of the rift, not slope (2013)
→ Nokrek = Meghalaya, not Sikkim; Namcha Barwa ≠ Garhwal (2022)
→ Amarkantak = Vindhya + Satpura (2023)
→ IOD is Indian Ocean W-E, not Pacific (2017)
→ La Niña helps the monsoon (2011)
→ Rainforest soil is poor (2023)
→ Red Sanders is not aromatic sandalwood. Connects to: Human Geography, Geomorphology, Climatology.

Recall drill: cover and answer

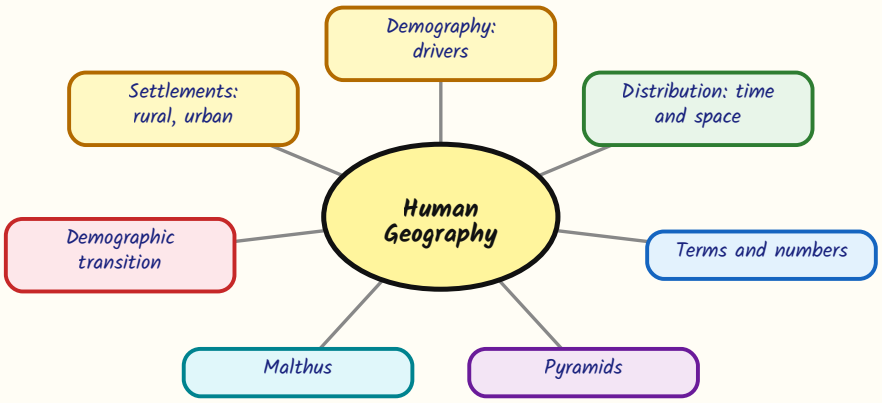
Standard meridian and 5 states?	82°30'E: UP, CG, MP, Odisha, AP (U C My Old Aunt)	Area, rank, coastline, land border?	32,87,263 km², 7th; 11,098 km; 15,106.7 km, 17 states
Channels 10, 9, 6?	Andaman|Nicobar · Lakshadweep|Minicoy · Great Nicobar|Sumatra	Highest: India Himalayan peak; entirely in India; Maharashtra; Kerala; TN?	Kanchenjunga; Nanda Devi; Kalsubai; Anaimudi; Doddabetta
Panch Prayag in order?	Vishnu, Nand, Karna, Rudra, Dev	Left-bank Ganga tributaries?	Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda
Why does Narmada flow west? Brahmaputra U-turn?	Linear rift valley; syntaxial bend at Namcha Barwa	Drainage split; river that crosses Tropic of Cancer twice?	77% Bay : 23% Arabian Sea; Mahi
Cyclone alert lead times; peak months?	72 – 48 – 24 – 12 h; May-June, Oct-Nov	IOD positive; El Niño; ELLA?	good monsoon; weak SW, stronger NE; heatwave years
Soils: alluvial share, black, laterite crops?	>40%; cotton, volcanic; tea, coffee, cashew	Forest + tree cover; target?	21.76% + 3.41% = 25.17%; 33%

HUMAN GEOGRAPHY

PYQ Prelims 2013 (tribe-state pairs), 2014 (Changpa) + 2 undated (TFR definition; low birth rate / ageing countries)

✎ *Population terms and numbers, pyramids, Malthus, demographic transition, tribes, rural and urban settlements: short, definition-heavy, 4 Prelims PYQs.*

At a glance: the Human Geography set



Pattern: definitions (TFR, sex ratio, dependency) and where-is-which-tribe; theory: Malthus, Demographic Transition. Prepare: five DTT stages as a birth/death table; state-wise tribes. Mains hooks (none quoted): demographic dividend, ageing, urbanisation, gentrification, climate-driven settlement shifts. No teacher ink: priority = PYQ count. ⭐

1 · DEMOGRAPHY

Demography = study of population size, structure, distribution, dynamics (birth, death, migration, age, fertility). 💡

Drivers of growth

- → Birth rate: live births per 1,000 people per period
- → Death rate: deaths per 1,000; low = growth
- → Migration: immigration (in) raises, emigration (out) lowers
- → Fertility rate: live births per 1,000 women of childbearing age; TFR = average children per woman
- → Age structure: youthful = rapid growth; ageing = slow/decline
- → Economic and social: health care, education, jobs, cultural norms

Temporal distribution

- → World: UN, peaks 10.3 billion in mid-2080s
- → 2025 shares: Asia 62.18% · Africa 19.36% · Europe 9.3% (-0.09% vs 2024) · N. America 7.71% · S. America 5.47% · Oceania 0.58%
- → India: most populous; 1.46 billion (UN); peaks early 2060s at 1.7 billion; latest SRS: TFR 1.9

Handout alert: "1.31 billion according to the 2011 census" is printed next to UN's 1.46 billion. Check the census figure before quoting. 📌

Spatial distribution: drivers

- → Geography/climate: Indo-Gangetic plain, temperate zones
- → Resources: river valleys (Nile, Ganga)
- → Economic opportunity: Mumbai, New York, Tokyo
- → Political/social stability: Canada, Australia
- → History: coasts, trade routes
- → Patterns: urban > rural; Asia most populous; coast > inland

Term	Meaning / India number
TFR	average children a woman has in her reproductive years at present birth rates (PYQ: "average live births a woman would have by end of child-bearing age") · 2.1 = replacement rate
IMR	infant (<1 yr) deaths per 1,000 live births · India 25.5 (2021)
Sex ratio	males per 1,000 females (world); India counts females per 1,000 males · 2011: 943 · NFHS-5: 1020
👫 Child sex ratio	juvenile sex ratio, age 0–6 · 2011: 914 · NFHS-5: 920
Life expectancy	average years if current mortality continues · 70.19 yrs (2022)
Wasting / Stunting / Underweight	low weight-for-height / low height-for-age / low weight-for-age
Dependency ratio	dependants (0–14 and >65) ÷ working-age (15–64)
Demographic dividend	working-age ≫ dependants · 56% in 2022–23 → peak 68.9% in 2030
Second dividend	lower fertility + longer life → older people with few dependants and big savings fund investment, boosting growth

Big Mummy's Up-to-Juggle Mania — Replacement 2.1 ("two parents → two kids + a little extra"). India's TFR: NFHS-5 2.0 → SRS 1.9 → projected 1.29 by 2050. Only 5 states have TFR > 2.1: Bihar · Meghalaya · UP · Jharkhand · Manipur.

Stunt = Short — Stunting = short for age; Wasting = thin for height (Waste = Weight-for-height); Underweight = light for age (Under = Weight-for-age).

2 · POPULATION PYRAMIDS

Pyramid = graph of age and sex distribution; shows growth pattern, development, future challenges. 💡

Shape	Features	Example
Expansive 🔺	broad base, narrow top; rapid growth; high birth rates and dependency; low life expectancy	Nigeria; India = expansive with a stabilising base
Constricted 🏺	narrow base (low birth rate), bulging middle (working age), narrow top; growth slowing; top wider than base = ageing, long life	Japan
Stationary ▮	even across ages, narrow only at the very old; birth and death rates low; stable or very slow growth	USA

3 · MALTHUS

Thomas Robert Malthus (English cleric)

→ "An Essay on the Principle of Population", 1798

→ Population grows geometrically (2–4–8–16); food arithmetically (1–2–3–4) → population outruns food

Checks

→ Preventive: moral restraint, delayed marriage, celibacy, contraception (under our control)

→ Positive: famine, disease, war (after the catastrophe, not in our control)

⚠ Criticisms

→ Technology (Industrial and Green Revolution)

→ Ignores social/economic complexity; "equated sociological and biological needs"

→ Modern reading: environmental sustainability and resource depletion, not just food

Multiply vs Add — Babies multiply (×2), bread adds (+1). Preventive = "Pre" = before birth; Positive checks = "Pos-t-mortem", death does the checking.

4 · DEMOGRAPHIC TRANSITION THEORY

Stage	Name	Birth	Death	Growth	Example
1	High stationary (pre-industrial)	High	High	slow, fluctuating	tribal societies, pre-18th c.
2	Early expanding (transitional)	High	Falling	rapid	many African countries
3	Late expanding (industrial)	Falling	Low	slowing	India, Brazil
4	Low stationary (post-industrial)	Low	Low	stable/declining	USA, much of Europe
5	Decline (proposed)	Very low	Low	declining	Japan, South Korea, Germany, Italy

Death Drops first, Birth Boards later — Stage 1: both high. Stage 2: death falls → boom. Stage 3: births fall → boom slows. Stage 4: both low. Stage 5: births below deaths.

Handout alert: the assumption line says "fertility and mortality reduce over time. Former before latter", but the stage table shows mortality falls first (Stage 2), fertility next (Stage 3). Exam rule: follow the stage table.

Till 1921 — stagnant

1921–51 — increasing

1951–81 — explosion

1981 on — late expanding (growth with stability)

India's transition: 1921–81 = early expanding stage; 1981 onwards = late expanding.
Why DTT matters: development ↔ lower birth/death; ageing and pensions; urbanisation; women's education and rights lower fertility; environmental pressure vs labour shortages.

5 · TRIBES AND PYQ

PYQ 2013 PYQ 2014

PYQ	Asked	Answer
2013	Limboo–Sikkim ✔ · Karbi–Himachal ✘ · Dongaria–Odisha ✔ · Bonda–Tamil Nadu ✘	A (1 and 3). Limbu = Kirati, Limbuwan (E Nepal, N Sikkim, W Bhutan). Dongria Kondh of Niyamgiri hills, Odisha; god Niyam Raja; Niyam Dongar ~4,000 m
2014	Changpa: live mainly in Uttarakhand ✘; rear pashmina goats ✔; Scheduled Tribe ✔	B (2 and 3). Semi-nomadic Tibetan people, Changtang, Ladakh; yaks, horses, sheep; pashmina/cashmere; smaller number in W Tibet, partly relocated for Changtang Nature Reserve; half a million nomads in Changtang (1989)
Undated	Total fertility rate is defined as	D average live births a woman would have by end of child-bearing age (not birth rate, not births per couple, not birth rate minus death rate)
Undated	Low birth rate / ageing / declining: Italy, Japan, Nigeria, South Korea, South Africa	A = Italy, Japan, South Korea (1, 2, 4)

Changpa = Chang-Pa-shmina — Ladakh's high plateau herders (Changtang). Dongria = "Dongar" (hill) in Niyamgiri hills. Karbi ≠ Himachal; Bonda ≠ Tamil Nadu.

6 · SETTLEMENTS

Rural: factors

- → Physical: relief, climate, water, soil, resources
- → Economic: farm productivity (Punjab), jobs, roads/rail
- → Social-cultural: kinship, ritual, land ownership (zamindari)
- → Political: policies, security; Green Revolution reshaped Punjab/Haryana
- → Environmental: floods, cyclones (coastal Odisha)
- → Climate change: coastal villages shift inland; drought/flood displacement (parts of Brazil, Burkina Faso)

Huddled vs scattered — Good relief + good climate → huddled together; bad → scattered. Compact (north Indian plains, wet points of Rajasthan, Bundelkhand, Nagaland) · Semi-compact (Indus basin, Gujarat plains, parts of Rajasthan) · Hamleted (socially separate units: panna, para, palli, nagla, dhani; Ganga plains, lower Himalayan valleys, Chhattisgarh plains (tribal), parts of south) · Dispersed (Uttarakhand, HP, Ladakh, Meghalaya).

Rural patterns (shape)

→ Linear: most common; roads, forest paths, rivers; T-shaped, star or checkerboard at crossings
→ Oval/circular: around temple, oasis, pond
→ Triangular: deltas, doabs
→ Rectangular/grid: plains
→ Step: hills

Urban term	Meaning	Handout example
Town	bigger than village, smaller than city; local administrative, commercial functions	–
City	large, dense, diverse economy, advanced infrastructure	Mumbai, Lucknow, Guwahati
Metropolis	major economic, political, cultural centre	New York, Delhi
Megacity	>10 million	Tokyo, Greater Mumbai, Greater Chennai
Conurbation	several towns/cities + suburbs merged	Delhi NCR
Megalopolis	several big cities grown together; labour-market separation	BosWash (Boston–Washington)
Suburb	residential, outskirts, less dense, depends on city	Bandra; Gurugram earlier
Urban sprawl	uncontrolled, low-density, car-based spread into rural land	–
Slum	densely populated, substandard housing, squalor, no basic services	Dharavi
CBD	commercial and business centre ("downtown"); high land values	Chandni Chowk, Connaught Place
Edge city	business, shopping, entertainment outside the CBD in former suburb/rural land	Gurgaon (for Delhi)
Gentrification	redevelopment → higher property values → lower-income residents displaced	Dharavi redevelopment
Satellite city	planned city to relieve a major city	–

Climb the ladder — Town → City → Metropolis → Megacity (>10 million) → Conurbation → Megalopolis. CBD = the city's wallet (downtown); Edge city = the mall that moved to the edge; Gentrification = the café that raised the rent.

🎯 MISSED PYQS (INDIAN GEOGRAPHY)

Asked	Answer / explanation
Andaman and Nicobar climate: humid tropical coastal ✔; SW + NE monsoon ✔; max precipitation Dec–May ✘	B (1 and 2): humid tropical maritime; both SW and NE winds onshore; max rain May–Sept (SW monsoon). (Heat = LP = rain; NH has heat Mar–Sept)
🧑‍🤝‍🧑 Peninsular Block: W-coast submergence by tectonics ✔; residual Veliconda, Mahendragiri hills ✔; deep V-shaped valleys ✘	B: ancient stable craton; Eastern Ghats denuded; peninsular rivers mature on old hard rock
Mangroves for climate resilience: store freshwater/paddy ✘; salt-sensitive roots ✘; bio-shield + livelihood ✔	D (3 only): roots are salt-tolerant, do not store salt water
🏛️ River: antecedent, 3 countries, Tibetan origin, irrigation, no distributaries	C Sutlej (Brahmaputra and Indus have distributaries; Teesta originates in Sikkim)
🗺️ States: UP most neighbours · Rajasthan longest int'l border · Sikkim only one-neighbour state: which not correct?	A as printed; explanation: West Bengal has the longest int'l border; Meghalaya also shares border with only 1 state

Source conflict

→ the last PYQ's printed answer (A, statement 1 wrong) disagrees with its own explanation (statements 2 and 3 wrong). Check atlas / class notes. Traps: TFR ≠ birth rate

→ Changpa = Ladakh, not Uttarakhand

→ sex ratio in India = females per 1,000 males. Connects to: Indian Geography (Changpa/Ladakh, Dongria/Niyamgiri, Thar/Himalaya settlements), Economic Geography.

Recall drill: cover and answer

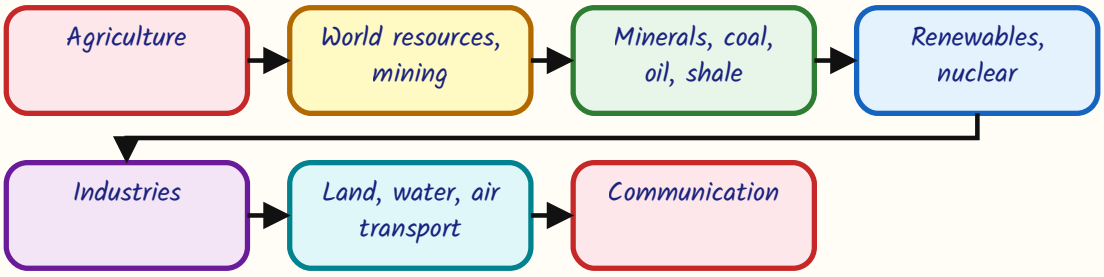
TFR replacement; India NFHS-5 and SRS?	2.1; 2.0 and 1.9	States with TFR above 2.1?	Bihar, Meghalaya, UP, Jharkhand, Manipur (Big Mummy's Up-to-Juggle Mania)
Sex ratio 2011; child sex ratio 2011?	943; 914 (NFHS-5: 1020 / 920)	Working age; dependants; dividend peak?	15–64; 0–14 and >65; 68.9% in 2030
Pyramids: Expansive, Constricted, Stationary?	Nigeria (India stabilising) · Japan · USA	Malthus: growth, work, year?	population geometric, food arithmetic; Essay on the Principle of Population, 1798
DTT: which falls first; India stage?	Death rate (Stage 2); late expanding (Stage 3) since 1981	Changpa; Dongria Kondh?	Changtang, Ladakh, pashmina; Niyamgiri, Odisha
Rural types and urban ladder?	Compact, semi-compact, hamleted, dispersed; town, city, metropolis, megacity (>10 m), conurbation, megalopolis	Low birth rate PYQ answer?	Italy, Japan, South Korea

ECONOMIC GEOGRAPHY

PYQ Prelims 31 PYQs: agriculture 19 (2011 x4, 2012 x2, 2013, 2014 x2, 2015, 2017, 2018, 2019, 2020, 2021 x2, 2022, 2024, 2025) · minerals and energy 9 (2012, 2013 x2, 2016, 2018, 2022, 2023 x2 + 1 undated on rare earths) · transport 3 (2014, 2015, 2023) · industries 0 · Mains none quoted

✎ *Which crop where, which mineral where, which energy source and how much, then industries, transport and communication: the heaviest handout (54 pages) and a VERY HIGH PYQ bank.*

At a glance: the Economic Geography route



Pattern: crop-to-region, mineral-to-state, "statement" questions on fertiliser, salinisation, permaculture, conservation agriculture, plus current affairs (cocoa, turmeric, REE, NCMM). No teacher ink: priority = PYQ count. Image-only pages (crop map, waterways) kept as tables.

1 · AGRICULTURE: SEASONS PYQ 2013

Arable land: India has the largest arable area in the world, ≈ 1.597 million km² (394.6 million acres).

	Kharif	Rabi	Zaid
Time	Jun–Oct; sow with SW monsoon, harvest Sep–Oct	Oct/Nov–Mar/Apr; sow Oct–Dec, harvest Mar–Apr	Mar–Jun; harvest Jun–Jul

Weather	hot, humid, heavy rain	cool dry growth, warm at harvest; Western Disturbance rain helps	warm, dry; irrigation needed
Crops	paddy, maize, jowar, bajra, pulses (moong, urad, arhar), cotton, groundnut, soybean, sugarcane	wheat, barley, oats, gram, mustard, peas, linseed	watermelon, muskmelon, cucumber, vegetables, fodder, some moong/urad

Kar-Rain · Rest-of-winter · Zara-garmi — Kharif = Kar-Rain (monsoon crops) · Rabi = Rest-of-winter (WDs help) · Zaid = Zara-garmi (summer gap).

Pairing pegs

- → Cotton + black soil (Maharashtra, Gujarat, Telangana)
- → Jute + West Bengal (wet, 120–200 cm)
- → Tea = acidic slopes (Assam, WB, TN, Kerala)
- → Coffee = Karnataka (shade; Arabica and Robusta)
- → Rubber = Kerala
- → Coconut = Kalpavriksha (Kerala)
- → Bajra = toughest (tolerates 42 °C), Rajasthan.

Trap (2013 PYQ): wheat is Rabi, not Kharif. Kharif = Arabic for autumn; generally Jun–Sep (state-wise differs); sown ~June, harvested Sep/Oct. Kharif also: rice, maize, bajra, ragi, soybean, groundnut, cotton.

CROP TABLE (temp °C · rain cm)

Crop	Temp · rain	Soil	Top Indian states	Top world	Remarks
Rice	20–32 · 150–300	deep clayey, alluvial	UP, Telangana, WB	India, China, Bangladesh, Indonesia, Vietnam	staple; Aus, Aman, Boro in WB; high humidity
Wheat	10–25 · 50–100	loamy, well-drained	UP, MP, Punjab	China, EU, India, Russia, USA	winter crop; bright sun at ripening
Jowar	20–27 · 50–75	light sandy–loamy	Maharashtra, Karnataka, MP	India, Niger, China, Mali, Nigeria	Kharif and Rabi; fodder + biofuel
Bajra	20–27 · 50–75	sandy loam	Rajasthan, UP, Gujarat	India, Niger, China, Nigeria	drought-resistant, up to 42 °C
Millets (jowar, bajra, ragi)	20–27 · 50–75	wide range	Rajasthan, Karnataka, MP	India, Niger, China, Mali, Nigeria	antioxidants, diabetes/heart, protein; low cost; marginal farmers
Maize	15–27 · 50–80	well-drained loam	MP, Karnataka, Bihar	USA, China, Brazil, EU, Argentina	food, alcohol, feed, biofuel
Cotton	21–30 · 50–100	black (regur)	Maharashtra, Gujarat, Telangana	China, India, Brazil, USA, Australia	cash crop; ≥ 210 frost-free days
Jute	24–35 · 120–200	alluvial	WB, Bihar, Assam	India, Bangladesh, China	retting; plastic alternative
Sugarcane	20–27 · 100–150	loamy	UP, Maharashtra, Karnataka	Brazil, India, China, Thailand, Pakistan	very water-intensive; ethanol + sugar
Pulses	15–27 · low rain	well-drained	MP, Maharashtra, UP	India, Canada, Myanmar, Brazil, Ethiopia	N-fixing; intercropping; India 766 kg/ha vs world 1,015
Oilseeds	15–30 · 30–75	loamy–sandy	Rajasthan, MP, Maharashtra	USA, China, Brazil, India	soybean, mustard, groundnut

Crop	Temp · rain	Soil	Top Indian states	Top world	Remarks
🎯 Palm oil PYQ 2021	25–28 · >200	tropical well-drained	AP, Telangana, Kerala	Indonesia, Malaysia, Thailand, Colombia, Nigeria	deforestation concern; push in NE
🎯 Tea PYQ 2022	16–32 · 150–300	acidic, slopes	Assam, WB, TN, Kerala (+ Tripura, HP)	China, India, Kenya, Sri Lanka, Vietnam	plantation; labour-intensive
Coffee	15–28 · 150–200	acidic, fertile	Karnataka, Kerala, TN	Brazil, Vietnam, Colombia, Ethiopia, Indonesia	shade-grown; Arabica and Robusta
🎯 Cocoa PYQ 2024	18–32 · 100–250	laterite	AP, Kerala, Karnataka, TN	Ivory Coast, Ghana, Indonesia, Nigeria, Ecuador	shade (under coconut/ areca); ~70% of world beans from W. Africa (Ivory Coast, Ghana, Nigeria, Cameroon)
Rubber	25–35 · 200–450	laterite, acidic	Kerala, TN	Thailand, Indonesia, Vietnam, India, China	latex
Cashew	20–30 · moderate	sandy loam	Maharashtra, AP, Odisha, Karnataka, Kerala	Vietnam, India, Ivory Coast, Brazil	export cash crop
🎯 Turmeric PYQ 2025	20–30 · >150	sandy loam	Maharashtra, Telangana, TN, Karnataka	India, China, Myanmar, Bangladesh	>30 varieties, >20 states
Mango	24–30 · 75–250	deep loamy	UP, AP, Telangana, Karnataka, Bihar	India, China, Thailand, Indonesia, Mexico	India largest; Alphonso, Dasheri, Langra
Coconut	20–32 · 100–200	sandy, coastal alluvial, laterite	Kerala, TN, Karnataka, AP	Indonesia, Philippines, India, Sri Lanka, Brazil	Kalpavriksha; humid coast
Red gram (tur)	20–30 · 60–100	loam–black	Maharashtra, Karnataka, UP, MP	India, Myanmar, Malawi, Tanzania, Kenya	Kharif; deep-rooted; N-fixing
Mustard	10–25 · 25–75	alluvial–loamy	Rajasthan, UP, Haryana, MP	Canada, China, India, Germany, France	Rabi; frost-sensitive; sarson, rai

📰 *In the news · 2025 (verify dates)* **National Turmeric Board:** HQ Nizamabad (Telangana), inaugurated June 2025.

Sugarcane moved south: tropical climate, higher sucrose, longer crushing season (Oct–May south vs Nov–Feb north; no extreme summer heat), cooperatives more successful in the south. ⭐

🎯 TYPES OF FARMING PYQ 2011 PYQ 2012

Type	Rainfall	Growing season	Focus
Dry farming	<75 cm	<75 days	moisture conservation, drought-resistant crops
Dryland	75–115 cm	75–120 days	soil moisture retention, drought-tolerant crops, adaptive strategies
Wetland	>115 cm	>120 days	rainwater harvesting, moisture conservation, crop selection
Irrigated	–	–	protective (saves crops from drying) vs productive (raises yield)

75–115–120 ladder. Dry/dryland farming gives ~80% maize, 90% bajra, ~95% pulses, 75% oilseeds. Groundnut, sesamum, pearl millet = predominantly rainfed (2012 PYQ, D): sesamum on marginal dry sloping land; ~80% groundnut area rainfed; main rainfed cereals sorghum, bajra, ragi. ⭐

Micro-irrigation PYQ 2011

→ Drip, sprinklers, foggers; improves water-use efficiency

→ Pros: saves water, stops soil erosion, no levelling, any soil, uniform watering, fertigation and chemigation, labour saving, low pressure (drip), higher yield, bigger fruit

→ Cons: high investment, pipe clogging, spacing needed, technical installation

→ 2011 (C): reduces nutrient loss ✔; checks falling groundwater table ✔; "only means of irrigation in dryland farming" ✘

→ Drip: water drop by drop at root zone; emitters inline, online, microtubes, pressure-compensated; for wide-spaced crops · Micro-sprinkler: sandy/loamy soils, horticulture, small grasses, nurseries and lawns

SRI (System of Rice Intensification)

→ Madagascar, 1980s; raises yield while using fewer resources

→ AWD (Alternate Wetting and Drying) instead of continuous flooding in nurseries

→ Organic manures; yields +30–50%; less methane

AGRICULTURE REVOLUTIONS

Revolution	When · who	Win	Cost
Green	late 1960s–early 1980s · M.S. Swaminathan	HYV seeds + fertiliser + irrigation; wheat/rice jump; food self-sufficiency	regional (Punjab, Haryana, W. UP), social (big farmers), pollution, water-table fall
White (Operation Flood)	1970s · Verghese Kurien	world's largest milk producer; cooperatives (Amul)	quality, cold chain, processing
Blue	1980s	fish output; ~17% of agri exports (2021–22)	overfishing, pollution
Yellow	1980s	oilseeds; fewer edible-oil imports	climate and price swings
Evergreen	Swaminathan's proposal	sustainable mix of traditional + modern (ecology + technology)	scale, policy support

Colour = crop

→ Green grains
→ White milk
→ Blue fish
→ Yellow oilseeds
→ Golden fruit/honey/horticulture
→ Golden fibre jute
→ Silver eggs
→ Silver fibre cotton
→ Pink onion/pharma/prawn
→ Red tomato/meat
→ Round potato
→ Brown coffee/leather/cocoa
→ Grey fertiliser
→ Protein lentils.

🌐 AGRICULTURAL REGIONS OF THE WORLD

Type	Key points
Nomadic herding	4th-world communities; camels–Bedouins · cattle–Masai · horses–Kazakhs · reindeer–Chukchis
Livestock ranching	subsistence: African savannah; commercial: Australia (sheep, cattle), Brazil (beef)
📚 Shifting agriculture	mostly 4th-world communities; sedentary = rotation of crops, not fields
Intensive subsistence	small holdings, multiple seasons; with paddy = wetter margins, without paddy = drier; primitive (manual, no surplus) vs non-primitive (tech, surplus)
🎯 Mixed subsistence / mixed farming PYQ 2012	crops + livestock together (2012, C); not "mixed cultivation"; Europe (Ireland–Russia), N. America east of 98° meridian, Argentine pampas, SE Australia, S. Africa, NZ; dense, urbanised, industrial societies
Extensive commercial grain	breadbasket of Anglo-America, Wheat Triangle of Eurasia; temperate grasslands
Commercial dairy	wetter temperate margins; India = tropical yet leading milk producer
Plantation	tropics; years of yield from same stand: coffee (Brazil), cocoa (Ghana), rubber (Indonesia, Malaysia), spices (India), tea (China); non-traditional (2–3 yrs): cotton, cane
Mediterranean	gardens and vineyards
📚 Horticulture / truck	Bulgaria (roses), Netherlands (tulips); now India (marigold, rose, lily, brinjal)
Cooperative	pool resources except land; Belgium, Netherlands, Denmark; IFFCO, Amul
Collective	USSR Kolkhoz; state owns land; free-riding problem

⏳ AGRICULTURAL TERMS PYQ 2017 PYQ 2020

Term	Meaning
Organic	no synthetic pesticides/fertilisers
📚 Permaculture	mimics natural ecosystems; mulching; multicropping; agroforestry example
Agroforestry	trees and shrubs in farmland
Regenerative	renew soil health and ecosystems
📚 Conservation agriculture	minimum tillage, residue cover, rotations
Hydroponics · Aquaponics · Aeroponics	nutrient water, no soil · fish + hydroponics · air/mist
Vertical farming	stacked layers, often indoors
Precision · Smart	GPS and IoT field management · AI, drones, sensors
Crop rotation · Intercropping	alternate crops for fertility · crops in proximity for pest control, resource use
Cover cropping	protects and enriches soil between crops: mustard, clover, radish
📚 Silviculture · Viticulture · Apiculture · Sericulture	forest plantations · grapes · bees · silkworms
📚 Horticulture · Floriculture	fruits, vegetables, flowers, ornamentals · flowers for the floral industry
Climate-smart	raises productivity and resilience, cuts emissions
Biopharming	GM plants/animals making pharmaceuticals
Agrivoltaic	solar power + farming: land use, rural energy access

Natural farming (Fukuoka method)

→ Minimal human intervention, works with nature; no synthetic chemicals, fertilisers, pesticides, excess tilling

→ Biodiversity, soil health, local natural resources

→ Masanobu Fukuoka (Japan), The One-Straw Revolution, 1975

→ ZBNF (Zero Budget Natural Farming): low-cost, local preparations, no external inputs; weeds, insects, animals not enemies; emulates natural conditions

💰 Land use: 9 categories (Land Revenue records)

→ Forests · Barren and wasteland · Non-agricultural use

→ Permanent pastures (mostly Panchayat/govt) · Misc. tree crops and groves (mostly private)

→ Culturable wasteland (fallow >5 yrs) · Current fallow (≤1 yr) · Fallow other than current (1–5 yrs) · Net sown area

→ Records: Land Revenue Dept (Survey of India); Digital India Land Records Modernization Programme; land degradation: see Soils of India

🎯 AGRICULTURE PYQ BANK

Year	Asked	Answer / rule
2011	Micro-irrigation statements	C (1 and 3)
2011	Lower Gangetic plain, humid-hot: best crop pair	Paddy and jute (C): jute >25 °C, humidity 70–90%; rice 21–37 °C, tolerates max 40–42 °C
2011	State: arid north, cotton centre, cash crops > food crops	Gujarat (B)
2011	Effect of salinisation	Makes some soils impermeable (B): salt breaks aggregates, pores collapse; advanced stage = barren, desertified land
2012	Mixed farming	Crops + livestock (C)
2012	Rainfed: groundnut, sesamum, pearl millet	All three (D)
2013	Kharif: cotton, groundnut, rice, wheat	1, 2, 3 (B); not wheat
2014	Kinnaur–areca? Mewat–mango? Coromandel–soya?	None (D); MP = soybean
2014	Maize: starch, biodiesel oil, alcohol	All three (D)
2015	Excess nitrogen fertiliser	Acidity ↑ and nitrate leaching (C); not more N-fixing microbes
2017	Conservation agriculture	Minimum tillage, residue cover, rotations (D: 2, 4, 5)
2018	Organic matter ↓ water holding? Soil no role in sulphur cycle? Irrigation salinises?	Only 3 (B)
2019	Agri soils emit nitrogen oxides; cattle ammonia; poultry reactive N	All three (D); ~90% N_2O from soils
2020	Permaculture vs chemical farming	1, 2, 4 (B); chemical farming not easy in semi-arid areas
2021	Palm oil: native to SE Asia ✘; lipstick/perfume raw material ✔; biodiesel ✔	B (2 and 3); native to hills of Guinea
2021	Least water-efficient crop	Sugarcane (A): 1,800–2,200 mm/season; sunflower 672.4; pearl millet 350
2022	Tea states: AP, Kerala, HP, Tripura	All four (D); Kerala major, rest minor
NEW 2024	Two largest cocoa producers	Côte d'Ivoire and Ghana (~70% of world cocoa)
NEW 2025	Turmeric 2022-23: India largest producer/ exporter; >30 varieties; Maharashtra, Telangana, Karnataka, TN	All three (D)

Source conflict: 2024 cocoa PYQ prints "Answer: D" but the correct pair is option (c) Côte d'Ivoire and Ghana (explanation: ~70% of world cocoa). Take the pair.

🌐 2 · WORLD RESOURCES, MINING

Resource = availability + utility

→ Anything usable from the environment for needs and wants

→ **Availability:** accessible (technology, economic feasibility, cultural acceptance)

→ **Utility:** some benefit

Classes

→ Origin: natural / human-made · biotic / abiotic

→ Exhaustibility: renewable / non-renewable

→ Ownership: individual / community / national / international (high seas, outer space)

Mining condition: cost of extraction + purification < value of final product. Types: **open-cast** (near surface), **shaft** (deep). 💡

🎯 MINERALS OF INDIA PYQ 2013 PYQ 2018 PYQ 2022 PYQ 2023

State pegs

- → Iron = Odisha–Jharkhand–Chhattisgarh–Karnataka–Goa belt
- → Copper = Rajasthan (Khetri)
- → Bauxite = Odisha
- → Gold = Kolar and Hutti (Karnataka)
- → Diamond = Panna (MP) only
- → Mica = Jharkhand
- → Monazite/thorium = Kerala
- → Uranium = Jaduguda (Jharkhand)
- → Lead-zinc-silver = Rajasthan (Zawar, Rampura Agucha)
- → Chromite = Odisha >96%.

Mineral	Where in India	Hooks
Iron ore	Odisha (Mayurbhanj, Keonjhar, Sundargarh); Jharkhand (Singhbhum); Chhattisgarh (Durg, Dantewada, Bailadila); Karnataka (Ballari, Hospet, Kudremukh); Goa, Maharashtra, Telangana, AP, TN	hematite, magnetite, limonite, siderite; steel
Manganese	Odisha, Karnataka, MP, Maharashtra	steel, batteries
Copper	Rajasthan 52.25% (Jhunjhunu, Khetri, Alwar) · MP 23.28% (Balaghat) · Jharkhand 15.14% (Singhbhum)	chalcopyrite; world leaders Chile, DR Congo, Peru
Bauxite	Odisha (Kalahandi, Koraput), Gujarat, Jharkhand (Lohardaga), Maharashtra, Chhattisgarh (Amarkantak), MP; new finds in Meghalaya	aluminium smelters near power plants (huge electricity)
Gold	Karnataka (Kolar, Hutti), Jharkhand, Chhattisgarh, Rajasthan, AP (Jonnagiri); Bihar preliminary	Kolar to restart 24 yrs after closure
Silver	by-product; Rajasthan ≈ 60% (Zawar, Rampura Agucha)	lead–zinc by-product
Uranium	Jharkhand (UCIL Jaduguda); potential Rajasthan, Chhattisgarh, Maharashtra, HP, Andhra	natural U has 0.72% U-235; U-235 fissile, U-238 not; enrichment raises concentration
Thorium (monazite)	India ≈ 25% of global deposits; Palakkad and Kollam (Kerala), Visakhapatnam, Mahanadi delta	cleaner; U-233; reactors can use up to 99% of fuel
Tin	Chhattisgarh (Dandkaranya)	bronze
Lead / Zinc	Rajasthan largest (HZL mines: Rampura Agucha, Sindesar Khurd, Rajpura Dariba, Zawar, Kayad)	zinc: galvanising; Zn–S batteries as lithium alternative
Nickel	Odisha (Sukinda), Jharkhand, Nagaland; ~0.2% of global reserves; no active mine	stainless steel, batteries
Cobalt	potential: Jharkhand, Odisha, Rajasthan, Nagaland, MP; no domestic production	
Lithium	J&K, Rajasthan, Karnataka (Mandya, Yadgiri); first mine expected in Korba, Chhattisgarh; no operating mine	
🎯 Titanium PYQ 2023	Kerala, TN, AP, Odisha (ilmenite, rutile)	ilmenite $FeO.TiO_2$, rutile TiO_2
Rare earths	monazite, bastnäsite, xenotime; coastal placers in Kerala, TN, Odisha, AP; 13.07 million t monazite (5th largest)	17 elements = 15 lanthanides + scandium + yttrium; China near-monopoly → Minerals Security Partnership
Mica	Jharkhand (Hazaribagh, Koderma), AP (Nellore), Rajasthan	Feb 2025: Major Mineral with feldspar, barytes, quartz
Tantalum, silicon	no production; silicon from quartz in Haryana, Rajasthan, AP	

💰 Tungsten	Karnataka, Rajasthan (Degana)	Madurai mining scrapped after protest
Antimony, germanium, indium, helium	~100% import dependent (antimony complete; germanium 100%; indium complete; helium 100%)	China limited exports (antimony July 2024; gallium July 2023)
Gallium, tellurium, selenium	by-products of bauxite/copper	ALCO plans 10 tpy gallium plant at Damanjodi
Diamond	~90% Panna (MP); AP (Krishna basin), Chhattisgarh	production only in Panna
Limestone	Rajasthan, MP (Satna, Katni), AP, Gujarat	$CaCO_3$ sedimentary rock; world: China, India, USA, Russia, Brazil
Others	Chromite (>96% Odisha) · Pyrite (none since 2003) · Borax (small, Puga valley) · Cryolite (nil)	

2013 PYQ (B): natural gas in Gondwana beds ✘; mica abundant in Kodarma ✔; Dharwars famous for petroleum ✘ (Dharwars: iron, manganese, lead, zinc, gold; Champion reef, Kolar). Natural gas areas: KG basin, Assam, Gulf of Khambhat, Cuddalore (TN), Barmer. Kodarma = ruby mica. 2018 (C): Bhilwara (Rajasthan) has iron ore; MMDR Amendment Act 2015 (from Jan 2015): states auction non-coal mines, first auction Dec 2015; Rajasthan ~57 minerals, sole producer of lead-zinc, selenite, wollastonite; copper 809.09 mt = 54% of resources, 41% of output (2nd leading producer). ✅

Handout alert (2018 PYQ): explanation prints both "states can auction" and "states cannot", and says AP/Jharkhand "do not have gold mines" while the table lists them as gold states. Take answer C and the gold table.

2022 PYQ (b: 1, 2, 4): monazite = rare earths + thorium (prescribed substance, DAE); only govt bodies process/export: IREL (under DAE); in-situ monazite in Odisha, AP, TN, Kerala, WB, Jharkhand (AMD); not the entire coast. REE PYQ (a: 1, 3): NCMM launched 2025 ✔; China first but India not second ✘; REE = 17 elements, not 13. ✅

In the news · 2025 (verify dates) NCMM: National Critical Mineral Mission launched 2025 for critical-mineral self-reliance; REE and China export curbs are the context.

🎯 COAL, OIL, GAS, SHALE PYQ 2012 PYQ 2013 PYQ 2016

Coal

→ Plant remains → coalification; world: China (Shanxi, Inner Mongolia), India, USA, Australia, Indonesia

→ Indian coal: non-coking (non-metallurgical), high ash 40–45%, low sulphur ~0.5–1% (~0.7%), high ash-fusion ~1,500 °C; ~75% of electricity from coal thermal plants

Gondwana (bituminous)

→ Valleys: Damodar, Mahanadi, Godavari, Son

→ Jharkhand-Bengal: Raniganj, Jharia, Bokaro, Giridih, Karanpura

→ Singrauli (MP/UP) · Korba (CG) · Talcher, Rampur (Odisha)

→ Chanda-Wardha, Kamptee, Bander (Maharashtra) · Singareni (Telangana) · Pandur (AP)

Tertiary

→ Anthracite: Kalakot (J&K)

→ Assam (Makum, Jaipur, Nazira) · Arunachal (Namchik-Namphuk) · Meghalaya (Darangiri, Cherrapunji, Mewlong, Langrin)

→ Nagaland: rat-hole mining despite SC ban, Art. 371A

→ Lignite: Neyveli (TN), Gujarat, Puducherry

2012 (B): India imports coal because power plants lack domestic supply AND steel needs coking coal; not "saving reserves". 2013 (B: 1, 2): high ash ✔, low sulphur ✔, low ash-fusion ✘. Also: India 5th largest reserves, 3rd largest producer and importer; sulphur removed by washing. ✅

Oil and gas

→ Crude: Mumbai High, Barmer (Rajasthan), Cambay (Gujarat), Assam-Arakan; KG belt has reserves but no production; new find in Andaman Sea region (being evaluated)

→ Natural gas: Assam (Digboi, Rudrasagar), Mumbai High, Gujarat, Rajasthan, Tripura, KG; also Andaman Sea

🎯 Shale oil and gas PYQ 2016

→ Unconventional, in low-permeability shale; hydraulic fracturing; world: USA, China, Argentina, Canada, Russia

→ India basins: Cambay, Ganga valley, Assam-Arakan, KG, Kaveri

→ 2016 PYQ (D): Cambay, Cauvery, KG all ✔; six basins: Cambay, Assam-Arakan, Gondwana, KG onshore, Cauvery onshore, Indo-Gangetic

→ Problems: water contamination, induced seismicity, land disturbance, methane leaks

🎯 ENERGY: RENEWABLES, NUCLEAR PYQ 2023

Source	India facts	World / notes
All renewables	283.46 GW up to March 2026	
Solar	157 GW by May 2026; target 280 GW by 2030; top: Rajasthan, Gujarat, Maharashtra, TN, Karnataka	China, USA, India, Japan, Germany; thermal + electric; desert panels: warm area (darker than sand), logistics and maintenance costs, sand cover
Wind	56 GW; top: Gujarat, TN, Karnataka, Rajasthan, Maharashtra; no offshore farm	China, USA, India, Germany, Brazil; offshore winds faster (no friction) but cost cancels gain
Bio-energy	10.75 GW; Maharashtra, Karnataka, UP, Punjab, Bihar	USA, Brazil, Indonesia, China, India; waste-to-energy (incineration → steam), biogas (anaerobic digestion: methane + CO_2)
Geothermal	potential 10,600 MW; Manikaran (HP); target 10 GW by 2030; not yet tapped; SCCL 20 kW pilot, Manuguru, Bhadradri Kothagudem (Telangana)	USA, Indonesia, Philippines, Turkiye, New Zealand
Tidal	Kachchh (Gujarat) 50 MW; Durgaduani Creek (WB) 3.75 MW; started, stopped due to cost	S. Korea, France, Canada, China, Russia
OTEC	65 kW at Kavaratti (Lakshadweep) for desalination	only 2 pilot plants: Japan (Okinawa), USA (Makai, Hawaii)
Nuclear	8,180 MW → 22,480 MW by 2031-32; +70% in last 10 years	fusion and fission; only fission used; 2021: USA, China, France, Russia, S. Korea

📝 3 stages of India's nuclear programme

→ 1: PHWR, natural uranium (U-235)

→ 2: Fast Breeder, plutonium (Pu-239), Kalpakkam

→ 3: thorium, U-233, KAMINI, Kalpakkam

U → Pu → Th – Use uranium, Produce plutonium, Then thorium. Stages 2 and 3 are Kalpakkam-linked.

2023 PYQ (C): India depends on coal (~75% power from thermal) despite uranium ✔; "uranium enriched ≥60% needed" ✘, ~5% enrichment is enough. ✅

Source conflict (handout vs mind map): mind map prints older figures, renewables 220.10 GW and solar 105.65 GW (31 Mar 2025, MNRE); handout text says 283.46 GW (Mar 2026) and 157 GW (May 2026). Use the handout; 280 GW by 2030 is the same in both.

🎯 4 · INDUSTRIES (no PYQ)

Sector	What	Examples
Primary	extraction/harvest	agriculture, fishing, forestry, mining
Secondary	processing/manufacture	light (consumer goods), heavy (capital goods: steel, ships), construction
Tertiary	services	trade, tourism, transport, finance, health, education, IT, professional
Quaternary	knowledge	R&D, information services, advanced training
Quinary	top decisions	government, NGOs, culture and arts

P-S-T-Q-Q – Dig → Make → Serve → Think → Decide.

👥 Size

→ (investment / turnover): Micro <₹2.5 cr / <₹10 cr
→ Small ₹2.5–25 cr / ₹10–100 cr
→ Medium ₹25–125 cr / ₹100–500 cr
→ Large above. **Ownership:** public, private, joint, cooperative. **Use:** consumer vs capital goods.

Location factors

→ Raw material · labour · market · transport/infrastructure · energy · capital · government policy
→ **Climate/environment:** circular-economy push, co-locate near recycling centres, "urban mining"
→ Social/cultural · **technology:** reshoring (robots replace cheap labour, plants return to high-income countries)
→ **Global:** nearshoring, friend-shoring · ecological sustainability

Weber's weight rule

→ **Weight-losing** (bulky raw material → light product, e.g. ore) → sit at the raw material
→ **Weight-gaining** (product heavier/bulkier than inputs) → sit near the market
→ **Footloose:** weight does not matter
→ **Isotim** = equal transport-cost line · **Isodapane** = equal total-cost line (transport of inputs + outputs + labour); best site on the lowest isodapane

World industrial regions	Handout lines
North America	NE USA (steel, autos) · West Coast (tech, aerospace) · South USA (oil, aerospace) · Ontario–Quebec
Europe	**Ruhr, Alsace-Lorraine, Benelux** · UK Midlands · Poland–Czechia–Slovakia–Hungary (Western, Northern, Eastern Europe)
Asia	East Asia (China, Japan, S. Korea, Taiwan) · **South Asia** (Maharashtra, Gujarat, TN, Karnataka) · SE Asia (Singapore, Malaysia, Thailand, Indonesia)
Africa	N. Africa · **Gauteng and Western Cape** (Southern) · W. Africa (Nigeria, Ghana)
Australia · S. America	east Australia · Brazil (São Paulo–Rio–Minas Gerais), Argentina, Chile

Indian region	Industries	Key cities
Mumbai–Pune	textiles, autos, chemicals, engineering, pharma, IT	Mumbai, Pune, Thane, Nashik
Kolkata–Hooghly	jute, textiles, chemicals, engineering, iron and steel, paper	Kolkata, Howrah, Durgapur, Haldia
Delhi–NCR	IT, electronics, autos, consumer goods	Delhi, Gurugram, Noida, Faridabad
Ahmedabad–Vadodara	textiles, chemicals, pharma, petrochemicals	Ahmedabad, Vadodara, Surat, Bharuch
Chennai–Bengaluru	autos, electronics, IT, heavy engineering	Chennai, Bengaluru, Hosur, Coimbatore
Kollam–Thiruvananthapuram	cashew, coir, software, biotech	Kollam, Thiruvananthapuram
Chotanagpur	iron and steel, coal, heavy engineering	Jamshedpur, Ranchi, Bokaro, Dhanbad

Visakhapatnam–Guntur	shipbuilding, petrochemicals, fertiliser, steel	Visakhapatnam, Guntur

🎯 5 · ROADS AND RAILWAYS PYQ 2014

Roadways

→ Network >63 lakh km (Jan 2025), among the largest in the world

→ NH = 2% (NHAI, MoRTH) · State highways 4% (state PWDs) · District roads 14% · rural roads (Gram Panchayats, PMGSY) · urban roads (municipalities)

→ NH44 (earlier NH7) Srinagar–Kanyakumari = longest · NH48 (earlier NH8) Delhi–Mumbai

→ Since 1 April 2018 length counts each lane separately

→ NH numbers: even = North–South (rise east → west: NH2 east, NH68 west) · odd = East–West (rise north → south: NH1 north, NH87 south) · sub-highway = prefix (NH101 from NH01); new scheme after 2010

Railways

→ >68,000 km; state-owned, 19 zones (General Manager each), Ministry of Railways

Gauges

→ Broad 1,676 mm: most used, long-distance, freight

→ Metre 1,000 mm: rural; Unigauge launched 1 April 1992

→ Narrow 762 or 610 mm: hills, being phased out

→ Standard 1,435 mm: metro, monorail, tram

DFCCIL dedicated freight corridors

→ Eastern: Ludhiana (Punjab) → Dankuni (near Kolkata), ~1,839 km (coal, steel)

→ Western: Dadri (UP) → JNPT (Mumbai), ~1,506 km

→ Future: East–West (Kolkata–Mumbai), North–South (Delhi–Chennai), East Coast (Kharagpur–Vijayawada), Southern (Chennai–Goa)

→ Also suburban (Mumbai, Delhi), metro (DMRC, BMRCL), proposed Mumbai–Ahmedabad high-speed rail

Metre = 1,000 — Metre = 1,000 mm (name = number). Broad = the biggest, 1,676. Standard 1,435 = city metros.

2014 PYQ (D): NH4 Chennai–Hyderabad? NH6 Mumbai–Kolkata? NH15 Ahmedabad–Jodhpur? All ✘: old NH4 became NH48; NH6 (Jorabat–Selling) is in the NE; NH15 starts at Baihata, Assam. ✅

Pipelines

→ Liquids and gases only; less flexible than rail and road; safe, cost-effective over long distances

→ Naharkatia–Noonmati–Barauni (crude) · Mumbai High–Mumbai–Ankleshwar–Koyali (crude + gas) · Salaya–Koyali–Mathura (crude; to Panipat, Jalandhar)

→ Hazira–Bijapur–Jagdishpur (GAIL): largest underground pipeline in the world; extended to Delhi via Dadri

→ Also carry ores (slurry), water, local milk

Ropeways

→ Suspended cables for difficult terrain; early 20th c. (mining, logging); first major passenger ropeway Auli, 1969; also Darjeeling, Gulmarg, Vaishno Devi

→ Parvatmala (Budget 2022–23, PPP, MoRTH): 8 projects ~60 km initially; Jan 2024: 400 projects, ₹1.25 lakh crore in 5 yrs

→ Varanasi urban ropeway · Gaurikund–Kedarnath (9.7 km, 3,584 m)

🎯 6 · WATER TRANSPORT PYQ 2023

Why cheap: low friction, fuel efficient; no separate infrastructure except terminals. 💡 IWAI established 1986; National Waterway Act 1986; 111 notified inland waterways (13 in operation).

Waterway	Route	Length
NW-1	Ganga–Bhagirathi–Hooghly (Haldia–Allahabad/Prayagraj)	1,620 km
NW-2	Brahmaputra (Dhubri–Sadiya)	891 km
NW-3	West Coast Canal: Champakara–Udyogmandal	205 km
NW-4	Krishna (Muktiyala–Vijayawada)	82 km
NW-73 / NW-100	Narmada / Tapi	226 / 436 km
NW-97	Sunderbans	172 km
NW-10, 68, 83, 85, 91, 111	Amba 45 · Mandovi 41 · Rajpuri creek 31 · Revdanda–Kundalika 31 · Shastri–Jaigad 52 · Zuari 50	km

Ocean transport

→ ~95% of foreign trade by volume, 70% by value

→ 13 major ports + 200+ minor/intermediate; newest major Galathea Bay (Great Nicobar, transshipment)

→ Vadhavan (14th, near Mumbai, Palghar): approved June 2024, ₹76,220 crore, yet to be built

→ Major ports: Port Trusts (Major Port Trusts Act 1963) · minor: State Maritime Boards

Kind Men Join Very Many New Clubs — West coast, north → south: Kandla (Deendayal) · Mumbai · JNPA (Nhava Sheva, leading container port) · Vadhavan · Mormugao (Goa, iron ore) · New Mangalore (only major port of Karnataka) · Cochin (Vallarpadam transshipment).

Very Clever Kids Visit Pretty Kolkata — East coast, south → north: V.O. Chidambaranar (Tuticorin) · Chennai · Kamarajar (Ennore) · Visakhapatnam (natural harbour) · Paradip (bulk) · Kolkata–Haldia (Syama Prasad Mookerjee Port, only riverine major port, Hooghly).

2023 PYQ (B, two pairs): Kamarajar = first major port registered as a company (Companies Act, started 2001) ✔; Mundra (Adani) = largest private port, also overtook JNPT as largest container port ✔; Visakhapatnam largest container port ✘. ✅

Handout alert: the 2023 PYQ explanation says "Kamarajar Port at Tuticorin", but the port list puts Kamarajar at Ennore and V.O. Chidambaranar at Tuticorin. Use the list.

7 · AIR TRANSPORT

18 Feb 1911 — first commercial flight, Allahabad–Naini

1948 — govt takes 49% of Tata-owned Air India

1953 — nationalised: Indian Airlines domestic, Air India international

1990s — liberalised

2021 — Air India sold to Tata Group (handed over Jan 2022)

157 operational airports, 133 with AAI (Airports Authority of India). >90% traffic with two companies (duopoly concern). Private operators: GMR, Adani, GVK (Delhi, Mumbai, Bengaluru). ⭐

8 · COMMUNICATION

1837	mid-19th c.	1851	1854	1881	1927	1936
Indian Post Office Act	Railway Mail Service	telegraph Kolkata–Diamond Harbour	Post Office Act, uniform rates; telegraph links Kolkata, Bombay, Madras, Peshawar	first telephone exchange, Kolkata	Indian Broadcasting Company (Mumbai, Kolkata)	All India Radio

1959	1982	1986	1990s	1995	2013	2015
Doordarshan	colour TV	Speed Post	private telecom and TV	Internet	telegraph ended	Digital India

Notes

→ **Post:** before it, communication went with transport or messengers; first organised system late 18th c. (British); Act 1854 set urban and rural post offices; Speed Post (1986) to compete with private couriers; network among largest after 1947

→ **Radio:** private clubs early 1920s; British took over 1936 as AIR (key in WWII); **Quit India** used radio; multilingual after independence

→ **TV:** private channels broke Doordarshan monopoly (1990s liberalisation)

→ **Telephone:** urban and government offices at first; early 2000s mobile revolution; one of the fastest 5G rollouts

→ **Digital India (2015):** infrastructure, connectivity, e-governance, online education, telemedicine

→ **Satellite:** ISRO communication satellites for TV, telephony, data, internet (see Science and Tech)

2015 PYQ (A: 1, 2, 3): IRS satellites used for crop productivity, groundwater, mineral exploration ✔; telecom, traffic ✘. IRS-1A launched 17 March 1988 (polar sun-synchronous, from Baikonur); data used under NNRMS. ✅

Traps

→ wheat is Rabi

→ palm oil native to Guinea hills, not SE Asia

→ excess N = acidity + nitrate leaching

→ chemical farming hard in semi-arid

→ uranium ~5% not 60%

→ REE = 17, China first not India second

→ Mundra/JNPT lead containers, not Visakhapatnam

→ even NH = N–S, odd = E–W. **Connects to:** Indian Geography (soils, Gondwana coal, rivers, ports)

→ Climatology

→ Human Geography.

Recall drill: cover and answer

Kharif, Rabi, Zaid months?	Jun–Oct · Oct/Nov–Mar/Apr · Mar–Jun	Dry, dryland, wetland farming?	<75 cm/<75 d · 75–115/75–120 · >115/>120
Dry/dryland share: maize, bajra, pulses, oilseeds?	80% · 90% · 95% · 75%	Least water-efficient crop (2021)?	Sugarcane (1,800–2,200 mm)
Who: Green, White revolution?	M.S. Swaminathan; Verghese Kurien	Copper shares: Rajasthan, MP, Jharkhand?	52.25%, 23.28%, 15.14%
Indian coal traits; why import (2012)?	non-coking, high ash, low sulphur; short supply + coking coal	Renewables, solar, wind, nuclear?	283.46 GW; 157 GW (280 by 2030); 56 GW; 8,180 → 22,480 MW
Weber: weight-losing, isotim, isodapane?	at raw material; equal transport cost; equal total cost	Gauges; NH rule?	1,676 / 1,000 / 762-610 / 1,435 mm; even N–S, odd E–W
Ports: count, newest, only riverine?	13 major + Vadhavan; Galathea Bay; Kolkata–Haldia (SPM)		